

Design of an Optical Phase Locked Loop

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Jacques Van Damme, June 2019

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Summary

In this work, the design of an optical phase locked loop (OPLL) is investigated. A SIMULINK model is built to function as simulation, and design tool for different OPLL architectures. Distributed feedback (DFB) lasers are compact, robust, and easily frequency modulated through their input current. These properties make them promising candidates for high performance optical phase locked loops. However they have an undesired phase reversal in the frequency modulation response, severely limiting the possible performance. Measurement techniques to extract the influential parameters from the DFB lasers are discussed. First order, and second order OPLL architectures are optimized for DFB type lasers. A promising double loop architecture is proposed with a considerable increase of the loop bandwidth.

Keywords

OPLL; DFB laser; Simulink.

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Abstract—In this work, the design of an optical phase locked loop (OPLL) is investigated. A SIMULINK model is built to function as simulation, and design tool for different OPLL architectures. Particular emphasis is put on semiconductor laser (SCL) based OPLL's. Distributed feedback (DFB) semiconductor lasers are compact, robust, and easily frequency modulated through their input current. These properties make them promising candidates for high performance optical phase locked loops. However, they have an undesired phase reversal in the frequency modulation response, severely limiting the possible performance. First order, and second order OPLL architectures are optimized for DFB type lasers. A promising double loop architecture is proposed with a considerable increase of the loop bandwidth.

Index Terms—OPLL; SIMULINK; DFB laser.

I. Introduction

THE upcoming 5G mobile network strives to increase the bandwidth for more users. One possibility to achieve this goal would be decreasing cell sizes. More, and smaller cells would require an increase in infrastructure, and result in a very high cost. To reduce the cost, lean antenna heads with a minimal amount of electronics could be used in the cells, while the data processing is centralized. The large amount of data traffic between the cells and the central base stations would require optical fibers to transport the received and transmitted radio frequencies (RF). One possible way to realize this RF-over-fiber is with an optical phase locked loop (OPLL). An OPLL is a system that controls the phase and frequency of a laser through feedback, usually with respect to another laser's phase and frequency. Such a system can be used in numerous and diverse applications like spectroscopy, coherent demodulation, optical carrier recovery. In this work, the design of an optical phase locked loop with a distributed feedback (DFB) SCL is researched. The DFB laser is a very popular laser architecture, as it is a compact single mode SCL that can be integrated with electronic chips using III-V-on-silicon technologies [1]. The performance of OPLL's is inherently limited by loop propagation delay, integration of the optical, and electronic parts of the system on the same chip is desired. SCL OPLL's have been designed successfully in the past [2]. The highest performance loops have been realised with distributed bragg reflector (DBR) lasers, however they are less compact, and more difficult to fabricate than DFB lasers. The problem with DFB lasers, is the extra performance limitation posed by the frequency modulation (FM) response of the DFB laser when directly modulating the laser active region with the input current. Typical DFB lasers show a phase reversal in the FM response between

0.1 MHz and 10 MHz [3], [4], which limits the achievable stable loop bandwidth.

A theoretical description of different OPLL architectures is based on a phase domain representation. The different architectures are simulated with a SIMULINK model, which could be used to design, and optimize different OPLL systems.

II. Phase Domain Description

The OPLL concept is shown in Figure 1. The combination of master laser, slave laser, and photodiode (PD) form a current controlled oscillator (CCO). The phase of the beatnote of the two lasers is referred to an external RF offset frequency with a phase detector. The phase error is fed back to the slave laser through a loop filter, closing the loop. XOR-gate phase detectors are preferred, as the intensity modulation (IM), that also results from current modulation of the DFB laser, can be disregarded.

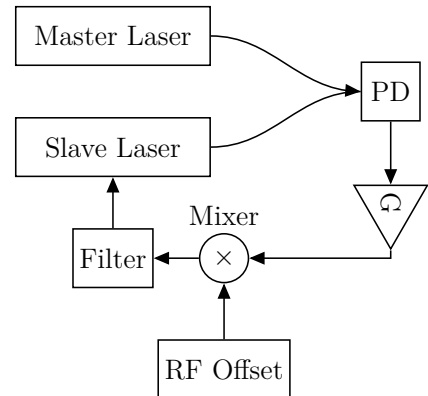


Figure 1: OPLL block diagram

If the loop is assumed to be in lock, the changes in phase difference between the CCO output and the external RF signal should remain very small. This means that the non-ideal phase detector implementation works in the linear region, and can be considered as ideal. If only the phases of the signals in Figure 1 are considered, a linear phase domain feedback system can be used to describe the OPLL behavior, Figure 2 is the block diagram implementation in the Laplace domain. A more detailed derivation of the phase domain description of phase locked loops can be found in [5].

From the block diagram in Figure 2, a few useful transfer functions can be derived (1), (2), (3), and (4). $LP(s)$ is the XOR-gate averaging filter, $F(s)$ is the loop filter, $FM(s)$ is the FM response of the slave laser, and τ_d is the loop propagation delay.

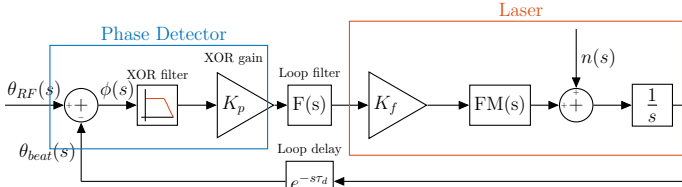


Figure 2: Phase domain OPLL block diagram

$OL(s)$ is the open-loop transfer function, $H_{\theta\theta}(s)$ is the closed-loop transfer function describing how beatnote phase θ_{beat} depends on the RF-offset phase θ_{RF} . $H_{\phi\theta}(s)$ describes how the remaining phase error ϕ depends on the input phase θ_{RF} , and $H_{\phi n}(s)$ is the transfer function that incorporates the influence of the frequency noise on the remaining phase error ϕ .

$$OL(s) = \frac{K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}}{s} \quad (1)$$

$$H_{\theta\theta}(s) = \frac{\theta_{beat}(s)}{\theta_{RF}(s)} = \frac{K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (2)$$

$$H_{\phi\theta}(s) = \frac{\phi(s)}{\theta_{RF}(s)} = \frac{s}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (3)$$

$$H_{\phi n}(s) = \frac{\phi(s)}{n(s)} = \frac{-e^{-s\tau_d}}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (4)$$

A. Stability

The loop stability criterium can be formulated in terms of the closed-loop transfer function $H_{\theta\theta}(s)$, stating that all poles should be in the left half-plane, or it can be formulated in terms of $OL(s)$.

$$\begin{cases} |OL(j\omega_{cr})|^2 = 1 \\ \arg\{OL(j\omega_{cr})\} = -\omega_{cr}\tau_d + \arg\{LP(j\omega_{cr})\} \\ \quad + \arg\{F(j\omega_{cr})\} + \arg\{FM(j\omega_{cr})\} - \frac{\pi}{2} \\ \quad = PM - \pi \end{cases} \quad (5)$$

At the critical angular frequency ω_{cr} for which the open-loop transfer function has unity gain, the phase shift due to $OL(j\omega_{cr})$ should stay clear from $-\pi$ with the desired phase margin (PM).

The loop delay τ_d , and the phase shift due to $FM(s)$, are limiting factors for the loop stability [7], [6].

B. Phase Error Variance

The OPLL performance can be expressed in terms of the residual phase error ϕ in lock, commonly referred to as the phase error variance σ_ϕ^2 , and given by equation (6).

$$\sigma_\phi^2 = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_\phi(j\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |H_{\phi n}(j\omega)|^2 S_n(j\omega) d\omega \quad (6)$$

The influence of the loop filter $F(s)$ on the residual phase error variance is discussed for a first order and a second order OPLL in [8] without the inclusion of the laser FM response. The second order loop can outperform the first order loop, provided that the propagation delay is small enough.

III. Loop Architectures

A typical DFB laser has a FM response with a phase reversal due to a shift in modulation mechanism, from thermal to electronic. This behavior is included through $FM(s)$ in equation (7), where $f_{FM} = \frac{1}{2\pi\tau_{FM}}$ is referred to as the FM crossover frequency.

$$FM(s) = \frac{1 - \frac{\tau_{FM}}{a}s}{1 + \tau_{FM}s} \quad (7)$$

K_f is the current-frequency sensitivity of the laser in $[\frac{\text{GHz}}{\text{mA}}]$. A first order low-pass filter $LP(s) = \frac{1}{1 + \tau_{LP}s}$ is used, and forms, together with the XOR-gate, the phase detector. The loop filter $F(s)$ can be designed for loop performance. The order of the loop refers to the number of perfect integrators in the open-loop transfer function $OL(s)$.

A. First, and Second Order Loop

A first order loop, is the most basic OPLL architecture where $F(s) = 1$. The only design parameter is the frequency independent gain K_p . The maximum stable gain value $K_{p,maxstable}$ can be derived from the stability criterium in equation (5). A second order loop has several advantages like increased lock range, the ability to correct frequency steps to a zero remaining phase error and, as will be shown, a superior performance in the case of pink frequency noise. The second order OPLL has a loop filter $F(s)$ given by equation (8).

$$F(s) = \frac{1 + \tau_1 s}{s} \quad (8)$$

The second order OPLL has a second design parameter, being the zero at $f_1 = \frac{1}{2\pi\tau_1}$. For both the frequency noise performance, and stability, f_1 is best chosen as small as possible. However the zero at lower frequencies will slow down the loop response, it will take longer for the phase error ϕ to become zero. To keep simulation times reasonable, the second order loop is tested with $f_1 = 20$ kHz, while the performance calculations and bode plots are made for $f_1 = 100$ Hz as it improves the loop bandwidth and noise performance.

The first, and second order loops are optimized for

minimal phase error variance σ_ϕ^2 . The integral of equation (6) is not readily solved analytically, therefore numerical integration methods were used to investigate the influence of the frequency independent gain on the residual phase error variance. In Figures 3, and 4 the numerical solution of (6) is plotted as a function of the normalized stable gain. This is done for white frequency noise, and for pink frequency noise respectively. The plots are normalized to the minimal σ_ϕ^2 . The parameter values of Table I were used during the optimization.

f_{LP}	f_{FM}	a	f_1	τ_d	K_f
1 GHz	1 MHz	1.5	100 Hz	1 ns	0.7 GHz/mA

Table I: Parameter values used for optimization of first order, and second order OPLL

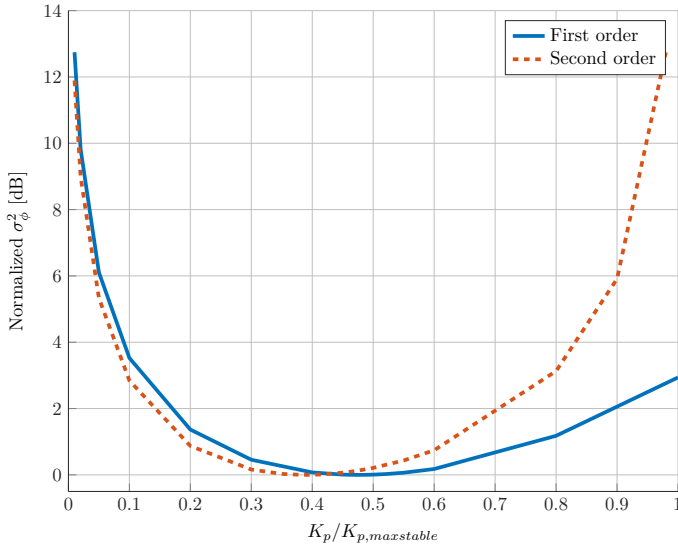


Figure 3: Phase error variance as function of stable loop gain in the case of white frequency noise

The optimal K_p value can be derived from Figures 3, and 4. In the case of white frequency noise: $K_p = 0.48 \times K_{p,maxstable}$, and $K_p = 0.39 \times K_{p,maxstable}$ for the first order, and second order OPLL respectively. In the case of pink frequency noise: $K_p = K_{p,maxstable}$, and $K_p = 0.6 \times K_{p,maxstable}$ for the first order, and second order OPLL respectively.

It is clear from the plot in Figure 4 that the second order OPLL performs better than the first order OPLL in the case of pink frequency noise. This can be explained by looking at plot of $|H_{\phi n}(j\omega)|^2$ in Figure 5. The second order loop has a band-pass effect on the frequency noise injected by the lasers, while the first order loop exhibits low-pass behavior.

B. Double Loop

The first order, and second order loop architectures have their bandwidth limited by the cross-over frequency $f_{FM} \in [0.1 \text{ MHz}; 10 \text{ MHz}]$ of the slave laser FM response. A more complicated loop architecture, called the double

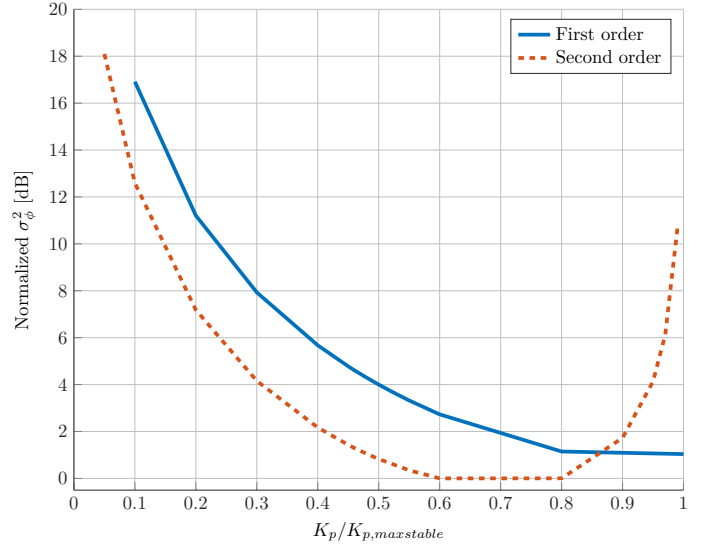


Figure 4: Phase error variance as function of stable loop gain in the case of pink frequency noise

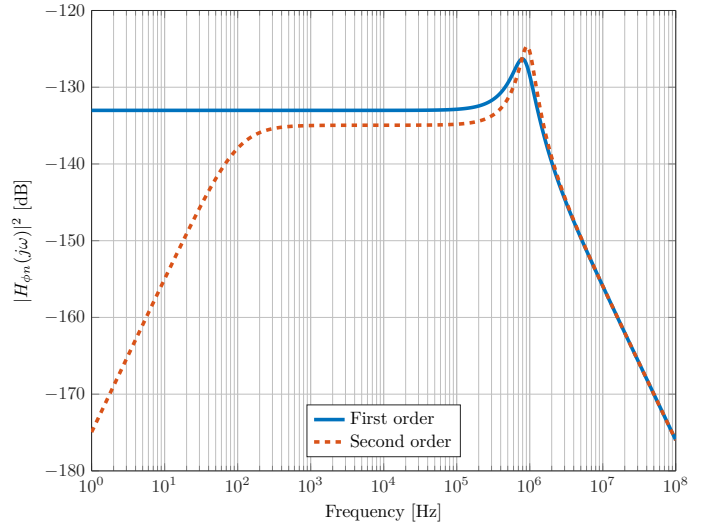


Figure 5: Phase error transfer function for first, and second order OPLL

loop OPLL can increase the loop bandwidth to frequencies far above the stability limit posed by f_{FM} .

The idea is that the feedback to the slave laser will correct the frequency difference between beatnote and RF-offset, while the parallel feedback path with an external optical phase modulator (OPM) can correct phase errors with a much larger bandwidth. The OPM can form a substitute for the slave laser, as the output can only vary over a small range of phase changes, unable to follow a frequency difference. The concept is shown in Figure 6.

There are multiple options to implement an optical phase modulator. Electro-optical modulators based on Pockels cells, and dual drive Mach-Zehnder Interferometers operated in push-push mode are two valid options. This will not be further elaborated, as it is out of the scope of

this work.

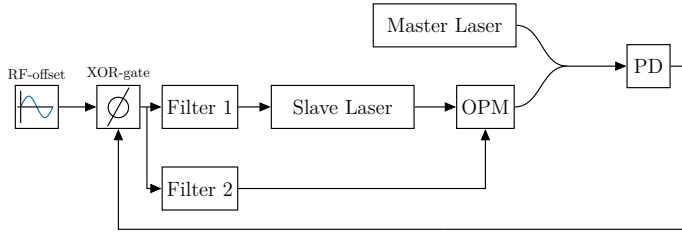


Figure 6: Conceptual block diagram of double loop OPLL architecture

The open-loop transfer function is now given by equation (9), where the output limitation of the OPM is ignored.

Each branch of the double loop has its own frequency independent gain ($G_{tot,1} = G_1 K_p K_f$ and $G_{tot,2} = G_2 K_p$), its own loop filter ($F_1(s)$ and $F_2(s)$), and its own propagation delay ($\tau_{d,1}$ and $\tau_{d,2}$).

$$\begin{aligned} OL(s) &= OL_1(s) + OL_2(s) \\ &= K_p LP(s) \left(\frac{G_1 K_f F_1(s) e^{-s\tau_{d,1}}}{s} + G_2 F_2(s) e^{-s\tau_{d,2}} \right) \end{aligned} \quad (9)$$

The stability criterium expressed in terms of the open-loop transfer function in equation (5) suddenly becomes less convenient when $OL(s)$ consists of a sum, illustrated by equation (10).

$$\begin{cases} |OL(j\omega_{cr})|^2 = |OL_1(j\omega_{cr})|^2 + |OL_2(j\omega_{cr})|^2 \\ \quad + 2\Re[OL_1(j\omega_{cr})OL_2(j\omega_{cr})] = 1 \\ \arg\{OL(j\omega_{cr})\} = \arg\{OL_1(j\omega_{cr}) + OL_2(j\omega_{cr})\} \\ \quad = PM - \pi \end{cases} \quad (10)$$

Previously, the choice of filters, and parameter values, could be deduced from the stability criterium. The double loop design proposed here, is a more ad-hoc solution based on the open-loop bode plots. $OL_1(s)$ is first designed separately as a second order loop described previously. $OL_2(s)$ is given by equation (11).

$$\begin{aligned} OL_2(s) &= G_2 K_p LP(s) F_2(s) e^{-s\tau_{d,2}} \\ &= \frac{G_2 K_p (1 + \tau_2 s) e^{-s\tau_{d,2}}}{(1 + \tau_{LP} s)(1 + \tau_3 s)} \end{aligned} \quad (11)$$

$OL_2(j\omega)$ does not have any perfect integrations, making $OL_1(j\omega)$ dominant for low frequencies. $OL_2(j\omega)$ should increase the bandwidth of the system, at a certain frequency the OPM branch should become dominant, in such a way that the stability limiting FM behavior of the slave laser can be neglected. At the region of frequencies where $OL_1(j\omega)$ and $OL_2(j\omega)$ are of the same order of magnitude, their phases should not lay drastically apart to avoid unexpected phase jumps in $OL(j\omega) =$

$OL_1(j\omega) + OL_2(j\omega)$. According to these intuitions, a stable solution was designed. The resulting bode plot is shown in Figure 7, where a loop bandwidth of $f_{cr} = 144$ MHz is achieved with a phase margin of $PM = \pi/3$ rad. The parameter values are tabulated in Table II.

$G_{tot,1}$	$\tau_{d,1}$	f_{FM}	a	f_1
1.21×10^{11}	1 ns	1 MHz	1.5	10 kHz
$G_{tot,2}$	$\tau_{d,2}$	f_{LP}	f_2	f_3
433.8	1 ns	100 MHz	20 MHz	100 kHz

Table II: Double loop OPLL parameters

With the double loop architecture, stable loop bandwidths can be achieved far beyond the slave laser FM crossover frequency f_{FM} ($f_{cr} = 144$ MHz $\gg f_{FM} = 1$ MHz), which is not possible with the single loop architectures. The performance increase are illustrated with the simulations of the SIMULINK OPLL model in the next section.

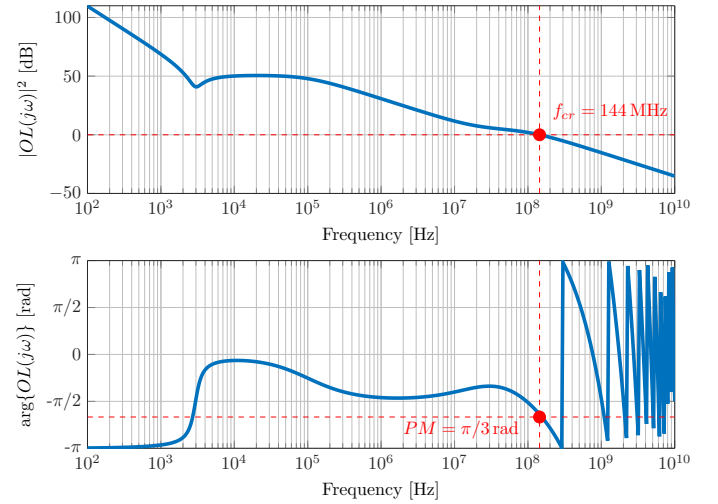


Figure 7: Bode plot of double loop $OL(s)$

IV. SIMULINK Model

Single, and double loop OPLL models are created in SIMULINK according to Figures 8, and 9. The XOR, MLSL, and PD blocks are also models made in SIMULINK to represent the XOR-gate, master laser and slave laser, and photodiode respectively. Both models are tested with free-running beatnote of 1 GHz, and a RF-offset signal of $f_{RF} = 1$ GHz + 300 kHz. White frequency noise is added in the MLSL model, and the parameters of the loops during the tests are tabulated in Tables I, and II for the single loop, and double loop models respectively.

The power spectral density (PSD) of the beatnote in lock is plotted in Figure 10 for the single loop second order OPLL architecture. The design was made for optimal phase noise performance based on Figure 3. Due to the FM crossover frequency $f_{FM} = 1$ MHz, the performance is limited.

The PSD of the beatnote in lock from the double loop architecture is plotted in Figure 11. Only a slightly cleaner

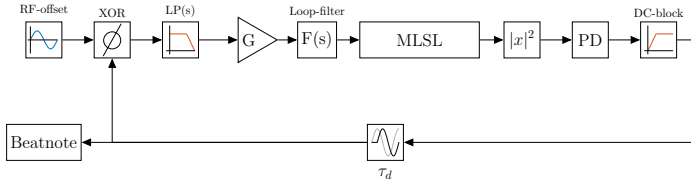


Figure 8: SIMULINK model of single loop OPLL architecture

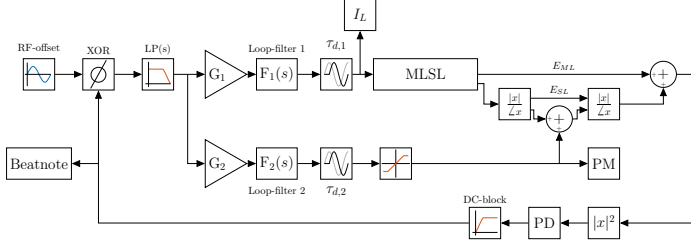


Figure 9: SIMULINK model of double loop OPLL architecture

spectrum is obtained. The PSD was calculated from the beatnote values after the transients died out $t > 60 \mu\text{s}$. This is a consequence of the OPM output limits.

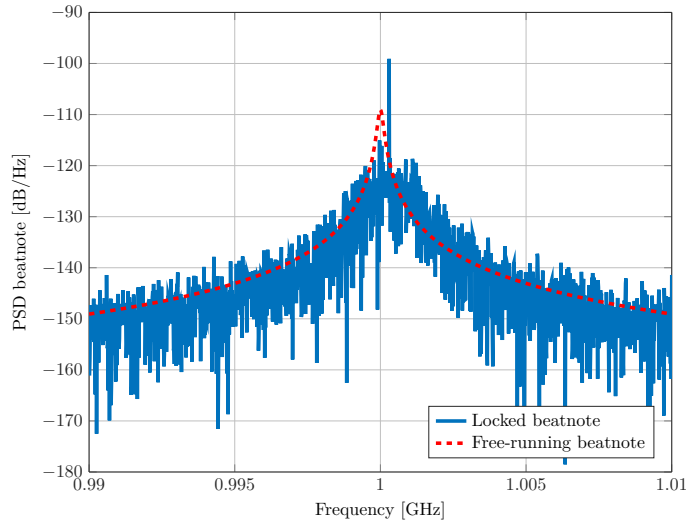


Figure 10: Simulated beatnote PSD of second order OPLL

The input current I_L of the slave laser in Figure 12. I_L achieves a steady DC-value to force the beatnote frequency to follow the RF-offset after $t = 60 \mu\text{s}$. The strange jitter that occurs randomly on top of the signal can be explained by looking at the output of the OPM branch in Figure 13. The OPM output is limited by the interval $[-4\pi \text{ rad}; 4\pi \text{ rad}]$, chosen arbitrarily to simulate the finite output range of the OPM.

The second branch corrects for the random phase fluctuations within the bandwidth, however occasionally the saturation limits will be reached. When the OPM branch saturates, it basically shuts off, and the primary branch with the slave laser has to work alone until the phase error ϕ is back within the interval $[-4\pi \text{ rad}; 4\pi \text{ rad}]$.

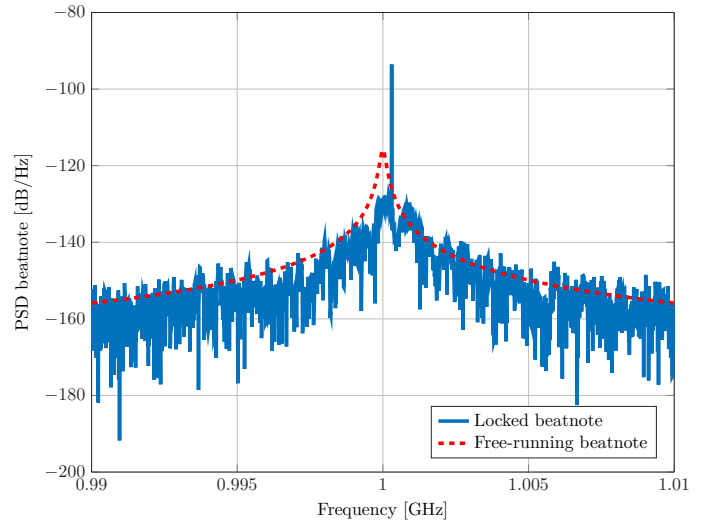


Figure 11: Simulated beatnote PSD of double OPLL, $t > 60 \mu\text{s}$

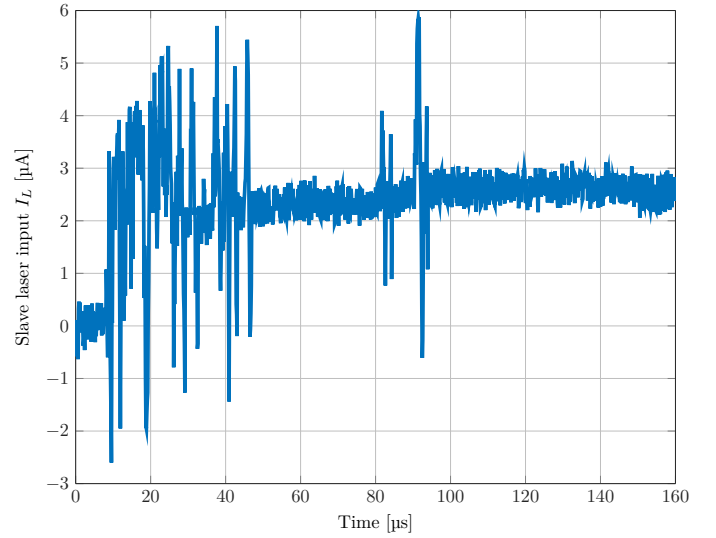


Figure 12: Simulated slave laser input

While the OPM branch is saturated, the remaining phase error ϕ increases, and translates into the peaks of the slave laser input current I_L .

The PSD of the beatnote in the interval $t \in [100 \mu\text{s}; 160 \mu\text{s}]$, where both loop branches are active, results in the very clean spectrum of Figure 14.

Keeping the OPM saturations to a minimum, results in better performance of the double loop architecture. Using lasers with fine linewidths will make sure that the amount of frequency noise injected in the loop is minimal, and random fluctuations beyond the saturation limits of the OPM will occur less often. Another course of action that would reduce the clipping, is finding a way to increasing the saturation limits of the OPM. Also, the design proposed here is not optimized in any way, the only goal was a stable system with some bandwidth enhancement. The double loop OPLL architecture looks

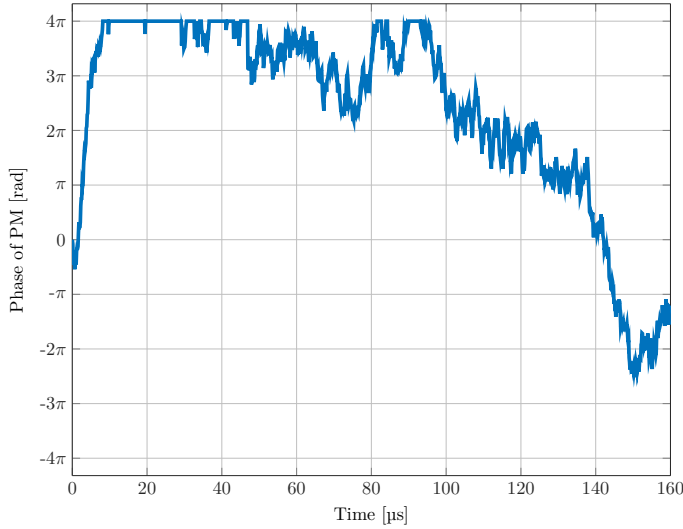


Figure 13: Simulated OPM output

very promising, and there is still a lot of room for improvement.

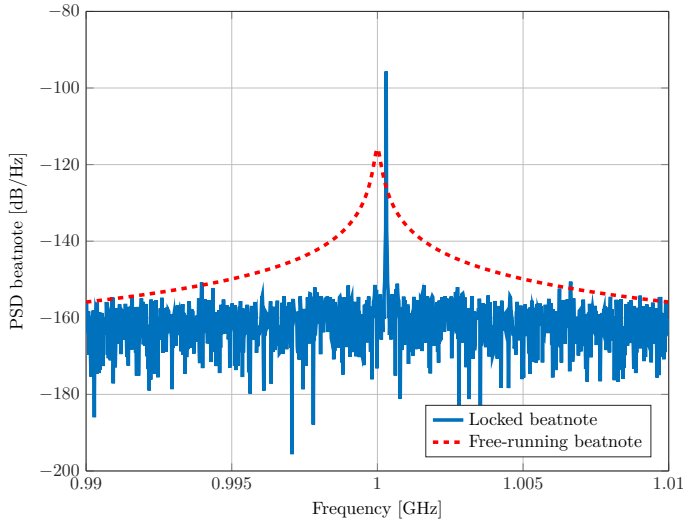


Figure 14: Simulated beatnote PSD of double OPLL, $t \in [100 \mu\text{s}; 160 \mu\text{s}]$

V. Conclusion

A careful analysis of optical phase locked loops with single-electrode DFB lasers was performed. The design process of first order, and second order OPLL systems is optimized for minimal phase error variance. A SIMULINK simulation model was created to investigate, and compare the behavior of different loop architectures.

Performant OPLL's with DFB lasers should be possible with double loop architectures. However this increase in loop complexity is in contrast with the primary reason to use DFB lasers, being the simplicity of fabrication, and simple FM with the input current. Instead of working around the phase reversal in the FM response, a better solution is using semiconductor lasers without this stability

limiting factor. DBR lasers can be frequency modulated by thermally modulating the cavity length instead of the laser active region. Special design methods can also be employed to fabricate DFB lasers without FM phase reversal [9].

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1

Introduction

Mobile communication has never been as important as today, and the increasing demand for data for even more users is not about to slow down. The next generation 5G network is almost upon us, and research into high capacity mobile networks is booming. Optical fibers are the option of choice for very high bandwidth interconnections, how can these optical links benefit the mobile network of the future?

1.1 RF-Over-Fiber

One possibility to increase the mobile network capacity is by increasing the number of remote antenna units (RAU). The amount of users can be shared over all RAU, giving each user a larger portion of the available data traffic. However, the extra infrastructure will come at a high cost. To keep this cost minimal, the complexity of the RAU could be reduced by introducing data-processing centers that would do all the processing of the received, and transmitted signals. Optical fibers can form the high bandwidth connections needed to transport all the mobile data between the base stations and the RAU. The transmitted frequencies for mobile communication are referred to as Radio-Frequencies (RF). The most popular wavelength for optical communication is $\lambda = 1550 \text{ nm}$, corresponding to a carrier frequency of 193 THz. In RF-Over-Fiber (ROF), the RF signals are put on the optical carrier frequency.

1.1.1 Problem Description

The remaining problem is how to realize ROF. In conventional optical communication systems, the modulation of choice is amplitude modulation (AM) with an external modulator. However, most mobile communication protocols employ frequency modulation (FM), as it has several benefits over AM. FM is not as sensitive to amplitude variations, which are a certain consequence of using a wireless channel. Also FM transmitters, and receivers have less stringent linearity specifications, as the amplitude of the signals can remain constant.

Semiconductor lasers (SCL) are a popular choice in optical communication, as they are robust, compact, and cheap lasers. SCL also have a very high current-frequency sensitivity, allowing very easy frequency modulation. How can we use these SCL to put RF signals with FM on the optical carrier?

1.1.2 Optical Phase Locked Loop

One possibility to transport the frequency modulated RF signals to the base stations with an optical carrier, would be with an optical phase locked loop. An optical phase locked loop (OPLL) is an extension of the well known electronic phase locked loop (PLL), where the phase, and frequency of a signal is controlled with respect to a stable oscillator. In the OPLL, the frequency of a laser is controlled with a feedback system, this concept will be explained further in Chapter 2.

The OPLL has other interesting applications like frequency synthesis, and spectroscopy.

1.1.3 Alternatives

Another possibility is achieving the ROF with conventional amplitude modulation of the optical carrier, and letting the RAU convert the AM to FM. This would again increase the base station complexity, and cost.

One could also try to directly frequency modulate the lasers without a feedback loop. This would require very narrow linewidth lasers with minimal amounts of frequency noise.

1.2 Thesis Objective

In this thesis, the design of an optical phase locked loop with a distributed feedback (DFB) SCL will be researched. The DFB laser is a very popular laser architecture, as it is a compact single mode SCL that can be integrated with electronic chips using III-V-on-silicon technologies [1]. The performance of OPLL's is inherently limited by loop propagation delay, integration of the optical, and electronic parts of the system on the same chip is desired. SCL OPLL's have been designed successfully in the past, as described by [2]. The highest performance loops have been realised with distributed bragg reflector (DBR) lasers, however they are less compact, and more difficult to fabricate than DFB lasers. The problem with DFB lasers, is the extra performance limitation posed by the frequency response of the DFB laser when directly modulating with the input current.

The design process proposed in this work is based on a SIMULINK model of the OPLL system. First the simulation tool is created in SIMULINK, and based on the design insights gathered from the model, a DFB laser based OPLL will be built.

2

The Optical Phase Locked Loop

An optical phase locked loop (OPLL) is a system that controls the phase and frequency of a laser through feedback, usually with respect to another laser's phase and frequency. Such a system can be used in numerous and diverse applications like spectroscopy, coherent demodulation, optical carrier recovery. The main envisioned application of this thesis is the OPLL as optical upconverter for RF-over-fiber.

The upcoming 5G mobile network strives to increase the bandwidth for more users. One possibility to achieve this goal would be decreasing cell sizes. More, and smaller cells would require an increase in infrastructure, and result in a very high cost. To reduce the cost, lean antenna heads with a minimal amount of electronics could be used in the cells, while the data processing is centralized. The large amount of data traffic between the cells and the central base stations would require optical fibers to transport the received and transmitted radio frequencies (RF). One possible way to realize this RF-over-fiber is with an OPLL.

2.1 Concept

When two lasers are incident on the same photodiode, their interference results in an intensity beatnote at the difference frequency. The photodiode's output current is proportional to the intensity of the incident light, for frequencies within its bandwidth, and will also contain the

beatnote, illustrated in equation (2.1).

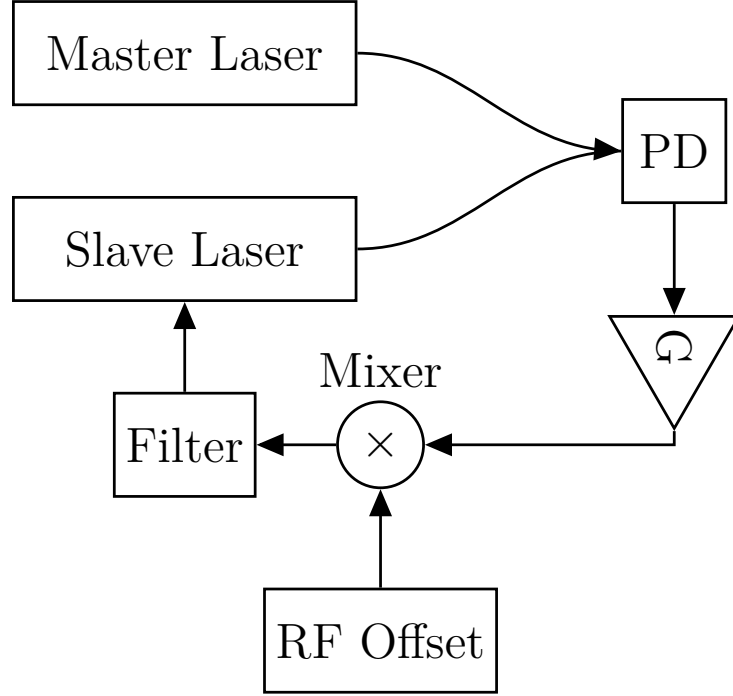


Figure 2.1: OPLL concept blockdiagram

In an OPLL, the goal is to control the offset frequency between the two lasers. The conceptual blockdiagram of Figure 2.1 shows the working principle. The beatnote is phase locked to an electrical signal at a controllable offset frequency by feeding the error signal of a phase detector back to the slave laser.

$$\begin{aligned}
 I_{PD} &\propto |E_1(t) + E_2(t)|^2 \\
 &= |E_{0,1}e^{j\omega_1 t + \phi_1(t)} + E_{0,2}e^{j\omega_2 t + \phi_2(t)}|^2 \\
 &= |E_{0,1}|^2 + |E_{0,2}|^2 + 2E_{0,1}E_{0,2} \cos((\omega_1 - \omega_2)t + \Delta\phi(t))
 \end{aligned} \tag{2.1}$$

The slave laser is forced to follow the master laser fluctuations at the RF offset frequency. For the RF-over-fiber application, the master laser and the locked slave laser optical signals are put on the same fiber. The only required component at the receiver is a photodiode with sufficient bandwidth to extract the RF signal from the optical carrier.

2.2 Components

The (O)PLL is essentially a system with three major components, a **phase detector**, a **loop filter**, and a **controllable local oscillator**. For an optical phase locked loop, the two beating lasers and photodiode together, form a current controlled oscillator. The most general schematic of Figure 2.2 is used to explain how, and why an (O)PLL works. The following theory is gathered from [3, 4].

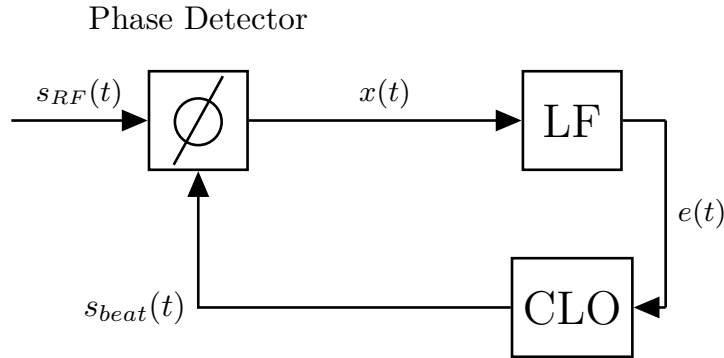
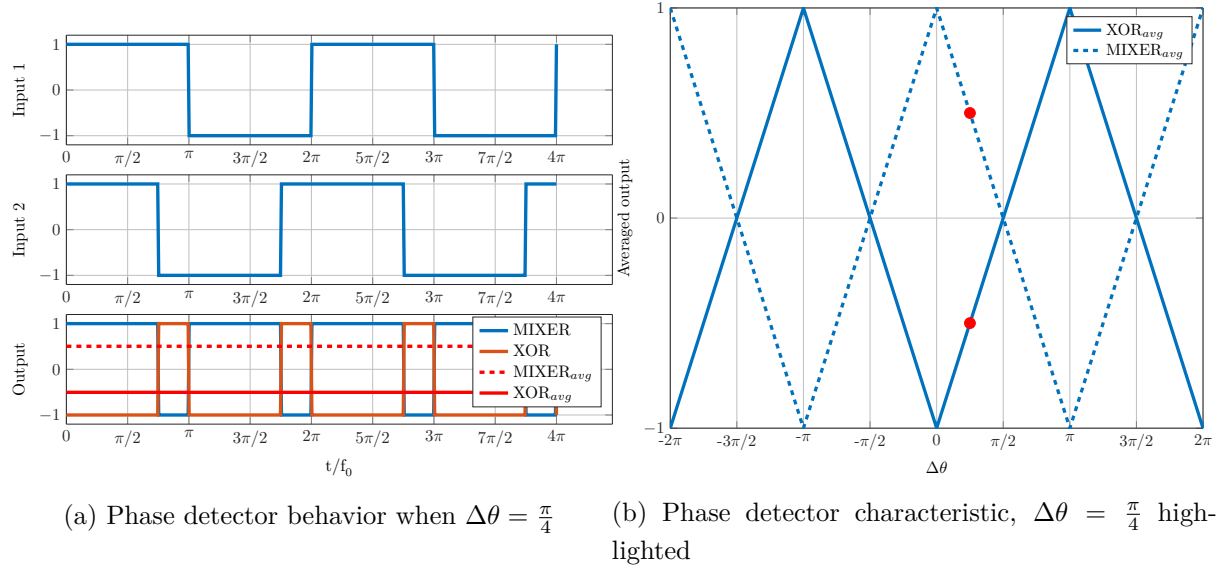


Figure 2.2: PLL general block diagram

2.2.1 Phase Detector

The **ideal linear phase detector** has as an **output** the **scaled phase difference of the two input signals**. In the block diagram of Figure 2.2, with $s_{RF}(t) = A_{RF} \cos(\theta_{RF}(t))$, and $s_{beat}(t) = A_{beat} \cos(\theta_{beat}(t))$ for example, the output would be $x(t) = K_p[\theta_{RF}(t) - \theta_{beat}(t)]$ (K_p being the phase detector gain). However, **if $s_{RF}(t)$ and $s_{beat}(t)$ have a different frequency, their phase difference would keep increasing or decreasing as a linear function of time**. This would require that the phase detector should be able to deliver outputs that can keep increasing or decreasing as time passes, which ofcourse no physical system is able to do.

Phase detectors can be build in various ways. Mixers or XOR-gates followed by a low-pass (averaging) filter are the choices considered here, but also more complex systems like phase-frequency-detectors with charge pumps exist, and could be considered to make improvements on the OPLL later on.

Figure 2.3: Phase detector implementations, $K_p = \frac{2}{\pi}$

If the input signals of the phase detector are blockwaves with the same frequency f_0 , but different phase, both the mixer and the XOR-gate output will be a blockwave with a duty-cycle dependent on the phase difference of the inputs. Using a low-pass filter to average out the output, will result in a DC-value proportional to the phase difference $\Delta\theta$, as can be seen in Figure 2.3a. Plotting the average XOR-gate or mixer output against $\Delta\theta$ results in the piecewise linear phase detector characteristic of Figure 2.3b. The value of K_p in the figures is the phase detector gain, and it is equal to the absolute value of the gradient of the slopes.

The phase detector implementations are linear over a domain of π . An advantage of the alternating positive and negative slopes of the linear pieces, is that the XOR-gate and mixer phase detectors make the loop immune to sign errors. If there would be positive feedback, the working point would run over the characteristic until the slope changes and negative feedback is achieved. For this thesis, a XOR-gate phase detector is chosen. The advantage is that the output is independent of the amplitude of the inputs, unlike with mixers. The intensity modulation (IM) of the slave laser can be disregarded for the loop design.

2.2.2 Lasers and Photodiode as Current Controlled Oscillator

The second component of the (O)PLL is a controllable local oscillator (CLO). It is the phase of the CLO output that is locked against an external oscillating signal. Based on the phase error $x(t)$, the phase (or frequency, $\omega(t) = \frac{d\theta(t)}{dt}$) of the CLO is adjusted. In the OPLL system of 2.1, the CLO is built from the master laser, the slave laser and the photodiode. The slave laser can be frequency modulated with the input current, and the beatnote of the two lasers is converted

back to electrical current by the photodiode according to equation (2.1). Together, the two lasers and the photodiode form a current controlled oscillator, with equation (2.2) describing the output $s_{beat}(t)$ as function of the input $e(t)$.

$$\begin{aligned} s_{beat}(t) &= |E_{0,1}|^2 + |E_{0,2}|^2 + 2E_{0,1}E_{0,2}\cos\left((\omega_1 - \omega_2)t + K_f \int_0^t e(u)du\right) \\ &= |E_{0,1}|^2 + |E_{0,2}|^2 + 2E_{0,1}E_{0,2}\cos(\theta_{beat}(t)) \end{aligned} \quad (2.2)$$

In equation (2.2), the frequency modulation (FM) of the slave is assumed to be frequency independent, although this is not the case for real devices. In Chapter 3, the FM response of the lasers is discussed, and in Chapter 4 the implications for the OPLL system.

2.2.3 Loop Filter

The loop dynamics can be changed with a linear loop filter (LF) inbetween the phase detector and the CLO. Equation (2.3) shows how the CLO input $e(t)$ depends on the phase detector output $x(t)$, $f(t)$ is the impulse response of the LF, and $e_0(t)$ is the transient, which will die out if the linear filter is stable. How the LF can be chosen to benefit loop behavior will be discussed in the next section.

$$e(t) = e_0(t) + \int_0^t x(t-u)f(u)du \quad (2.3)$$

2.3 Phase Domain

If the loop is assumed to be in lock, the changes in phase difference between the CLO output and the external RF signal should remain very small. This means that the non-ideal phase detector implementations work in their linear region, and can be considered as ideal. If only the phases of the signals in Figure 2.2 are considered, a linear phase domain feedback system can be used to describe the (O)PLL behavior, Figure 2.4 is the block diagram implementation in the time domain.

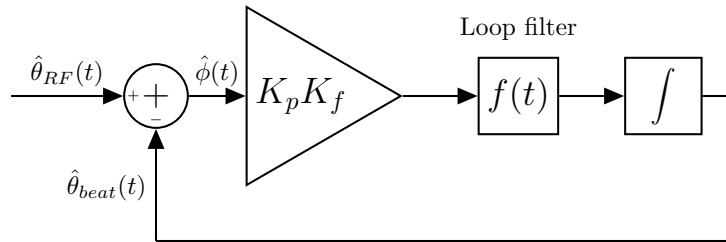


Figure 2.4: Phase domain OPLL block diagram

Linear feedback systems are most easily described in the Laplace domain, Figure 2.5 is the same system as Figure 2.4 with the addition of frequency noise n , frequency modulation response $FM(s)$, and loop delay in the feedback path.

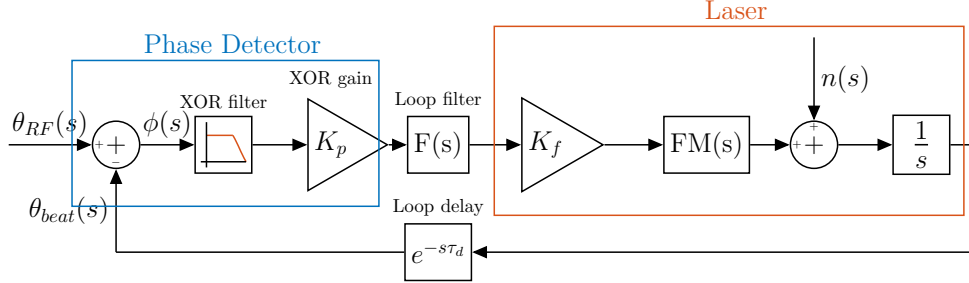


Figure 2.5: Phase domain OPLL block diagram in the Laplace domain

From the block diagram in Figure 2.5, a few useful transfer functions can be derived (2.4), (2.5), (2.6), and (2.7). The XOR averaging filter is called $LP(s)$ from now on.

$$OL(s) = \frac{K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}}{s} \quad (2.4)$$

$$H_{\theta\theta}(s) = \frac{\theta_{beat}(s)}{\theta_{RF}(s)} = \frac{OL(s)}{1 + OL(s)} = \frac{K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (2.5)$$

$$H_{\phi\theta}(s) = \frac{\phi(s)}{\theta_{RF}(s)} = \frac{1}{1 + OL(s)} = \frac{s}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (2.6)$$

$$H_{\phi n}(s) = \frac{\phi(s)}{n(s)} = \frac{-e^{-s\tau_d}}{s + sOL(s)} = \frac{-e^{-s\tau_d}}{s + K_p K_f LP(s) F(s) FM(s) e^{-s\tau_d}} \quad (2.7)$$

In general, the loop order of a control system is equal to the number of finite poles in the open-loop transfer function $OL(s)$, however in this thesis the OPLL order will refer to the number of perfect integrations (pole at $s = 0$) in $OL(s)$.

2.3.1 Stability

The loop stability criterium can be formulated in terms of the closed-loop transfer function $H_{\theta\theta}(s)$, stating that all poles should be in the left half-plane, or it can be formulated in terms of $OL(s)$.

$$\begin{cases} |OL(j\omega_{cr})|^2 = 1 \\ \arg\{OL(j\omega_{cr})\} = -\omega_{cr}\tau_d + \arg\{LP(j\omega_{cr})\} + \arg\{F(j\omega_{cr})\} + \arg\{FM(j\omega_{cr})\} - \frac{\pi}{2} = PM - \pi \end{cases} \quad (2.8)$$

At the critical angular frequency ω_{cr} for which the open-loop transfer function has unity gain, the phase shift due to $OL(j\omega_{cr})$ should stay clear from $-\pi$ with the wanted phase margin (PM). The OPLL should be designed to have ω_{cr} as high as possible, because it determines the loop bandwidth.

The loop delay τ_d , and the phase shift due to $FM(s)$, are limiting factors for the loop stability [5, 6].

2.3.2 Performance

The performance of the OPLL can be expressed in different metrics, choosing an appropriate loop filter $F(s)$ is based on these metrics.

Capture Range

In order for the (O)PLL to achieve phase lock, the RF-offset frequency must be close enough to the free-running frequency of the CLO. The interval of frequencies around the free-running frequency that can initialize phase lock, is called the capture range of the (O)PLL. An exact description of the capture range is complicated as described by [7], and out of the scope of this thesis. In [4], an upper bound on the capture range in the case of a sinusoidal multiplying phase detector is derived, resulting in equation (2.9). Intuitively, it can be argued that a higher open-loop critical frequency $f_{cr} = \frac{\omega_{cr}}{2\pi}$, should result in a higher capture range. From now on, f_{cr} will be referred to as the loop bandwidth.

$$\Delta\omega < \sqrt{\frac{K_p K_f}{\tau_{LP}}} \quad (2.9)$$

Lock Range

Once the OPLL is in lock, the range of offset frequencies it can follow without losing the lock, is defined as the lock range of the OPLL. The finite lock range arises from the non-ideal phase detector characteristic shown in Figure 2.3a, where the output range of the phase detector is limited. The maximal frequency range (centered around the free-running frequency of the beatnote) is given by equation (2.10). Note that the $FM(s)$ filter behaves more or less like a

pure phase shifter, with unity DC-gain, discussed more in detail further on.

$$\begin{aligned} \left[\frac{-\Delta\omega_{lock}}{2}; \frac{\Delta\omega_{lock}}{2} \right] &= \left[\frac{-K_p\pi}{2}; \frac{K_p\pi}{2} \right] * K_f |LP(0)F(0)FM(0)| \\ &= \left[\frac{-K_p K_f |F(0)|\pi}{2}; \frac{K_p K_f |F(0)|\pi}{2} \right] \end{aligned} \quad (2.10)$$

From this formula, it is clear that choosing an integrating loop filter $F(s)$ (with infinite DC-gain) is the best choice for the lock range metric. The integrator pole will have to be accompanied by a zero to allow for the stability criterium to be met.

Static Behavior

The static behavior of the OPLL describes the remaining phase error $\hat{\phi}(t)$ in regime due to a change in input $\hat{\theta}_{RF}(t)$. Equation (2.11) gives the remaining phase error after transients died out for a combination of a phase step and a frequency step $\hat{\theta}_{RF}(t) = \Delta\omega t + \theta_0 \leftrightarrow \theta_{RF}(s) = \frac{\Delta\omega}{s^2} + \frac{\theta_0}{s}$.

$$\begin{aligned} \lim_{t \rightarrow \infty} \hat{\phi}(t) &= \lim_{s \rightarrow 0} s\phi(s) = \lim_{s \rightarrow 0} sH_{\phi\theta}(s)\theta_{RF}(s) \\ &= \lim_{s \rightarrow 0} sH_{\phi\theta}(s) \left(\frac{\Delta\omega}{s^2} + \frac{\theta_0}{s} \right) \\ &= \frac{\Delta\omega}{K_p K_f LP(0)FM(0)F(0)} + 0 \\ &= \frac{\Delta\omega}{K_p K_f F(0)} + 0 \end{aligned} \quad (2.11)$$

Again choosing an integrating loop filter $F(s)$ is beneficial, frequency steps can be corrected to zero remaining phase error.

Phase Error Variance

When the loop is in lock, there will remain random phase error fluctuations due to the frequency noise $n(s)$ that is injected by the laser. The remaining power of the phase error, is given by σ_ϕ^2 , the phase error variance.

$$\sigma_\phi^2 = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_\phi(j\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |H_{\phi n}(j\omega)|^2 S_n(j\omega) d\omega \quad (2.12)$$

The influence of the loop filter $F(s)$ on the residual phase error variance is discussed for a first order and a second order OPLL in [8], and will be elaborated on in Chapter 4.

3

Photonic Integrated Circuit

An OPLL is an opto-electronic system, which consists of a photonic, and an electronic subsystem. A photonic integrated circuit (PIC) could act as the optical subsystem, where the input and output are electrical currents. By confining the optical part on one chip, the loop delays can be reduced to a minimum, which can benefit the performance of the OPLL. Based on the measurements of the PIC, a model of the optical subsystem can be constructed to be used for system-level simulations of the OPLL.

3.1 Chip and Technology

In Figure 3.1b the layout of a photonic integrated circuit (PIC) is visible, containing multiple subsystems of a laser, a coupler and a photodiode. Such a subsystem can form an implementation of the highlighted part of the OPLL in Figure 3.1a. The semiconductor distributed feedback (DFB) laser on the chip will function as the slave laser. An external master laser is still required, and can be coupled in via the grating coupler present in the other arm of the waveguide coupler on the chip. In this way, the optical signals from both lasers are coupled onto the photodiode also present on the chip. The input as well as the output of the PIC are electrical currents.

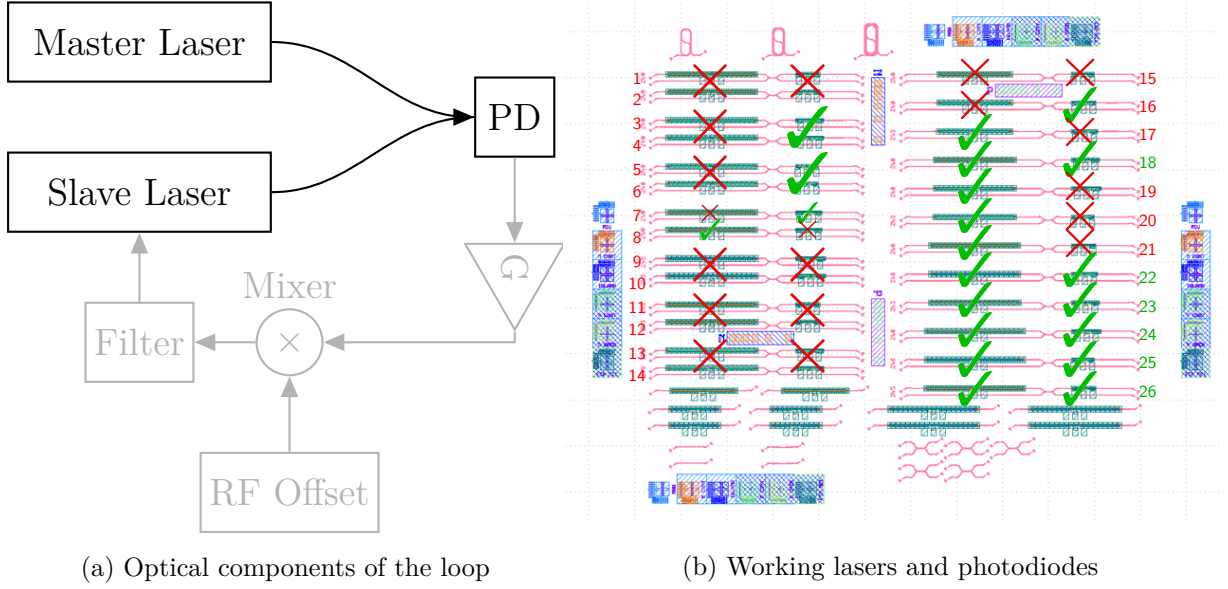


Figure 3.1: Photonic integrated chip

The PIC was designed and fabricated by my supervisor Ir. Camiel Op de Beeck from the photonics research group at Ghent University. The used technology is heterogenous III-V-on-silicon [1]. Across the subsystems on the PIC, different dimensions for parameters like the laser active region length were tried. Six {#18, #22, #23, #24, #25, #26} out of the twenty-six circuits contain both a working laser, and a working photodiode. The lasers in the left column behaved like LEDs, probably because there was not enough feedback due to narrow grating sizes and possible misalignment of the III-V-on-silicon. The exact reasons for the low yield is not investigated further because it is out of the scope of this thesis.

3.2 Photodiodes

The function of the photodiode in the loop is to detect the beatnote of the master laser and the slave laser. The optical beatnote signal can be converted into an electrical current, followed by a transimpedance amplifier (TIA) to convert the current to voltage, and is connected to the input of the phase detector of the loop, also visible on Figure 3.1a.

3.2.1 Static Measurements

A photodiode has the familiar $I(V)$ characteristic of Figure 3.2a and acts as an optical power detector when reverse biased. The output current I_{out} is linearly dependent on the incident optical power as described by equation (3.1). The static behavior of interest for loop design is

represented by the responsivity R_s of the photodiode. Figure 3.2b shows this linear relationship between the output current and the incident optical power for the photodiodes that are part of a working subsystem on the PIC.

$$I_{out} = -R_s P_{in} \quad (3.1)$$

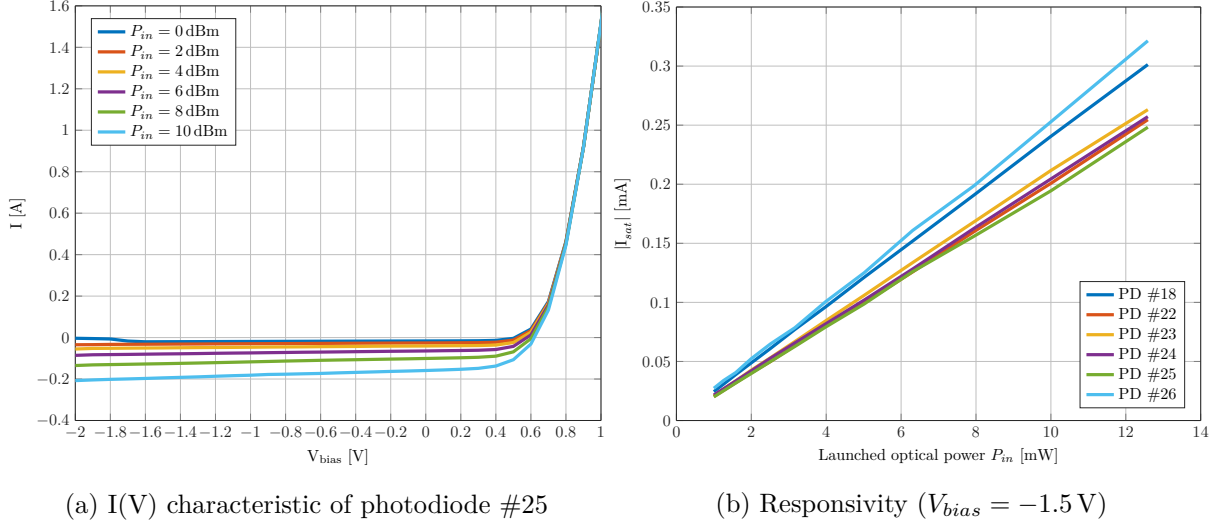


Figure 3.2: Photodiode static measurements (15 dB losses during grating coupling)

The bias point of the photodiodes $V_{bias} = -1.5$ V is chosen based on the measurement results like in Figure 3.2a. At this $V_{bias} = -1.5$ V, all working photodiodes are operating in the region of reverse saturation current with linear responsivity to the incident optical power. The optical power loss due to the grating coupling from the fibre-probe to the chip is measured to be around 15 dB. These grating losses are incorporated in the calculation of the responsivities in Table 3.1.

	PD_{18}	PD_{22}	PD_{23}	PD_{24}	PD_{25}	PD_{26}
$R_s [\frac{A}{W}]$	0.7558	0.6375	0.6639	0.642	0.6192	0.7988

Table 3.1: Photodiode responsivity ($V_{bias} = -1.5$ V)

3.2.2 Frequency Response

The frequency response of the photodiodes was measured by modulating the intensity of the incident light with a Mach-Zehnder interferometer (MZI). The modulation frequency is swept by port 1 of a vector network analyzer (VNA, the Keysight PNA-X Microwave Network Analyzer), while port 2 is connected to the high-speed GSG probe landed on the photodiode contacts. In Figure 3.3 the magnitude of the S_{21} parameter is plotted. At the chosen bias point of $V_{bias} = -1.5$ V the 3 dB bandwidth is around 4 GHz for most of the working photodiodes on the

PIC. The lower bandwidth when the photodiode is not properly reverse biased can be explained by the decrease in width of the depletion region in the p-n junction. A wider depletion region lowers the junction capacitance, and also the 3 dB bandwidth.

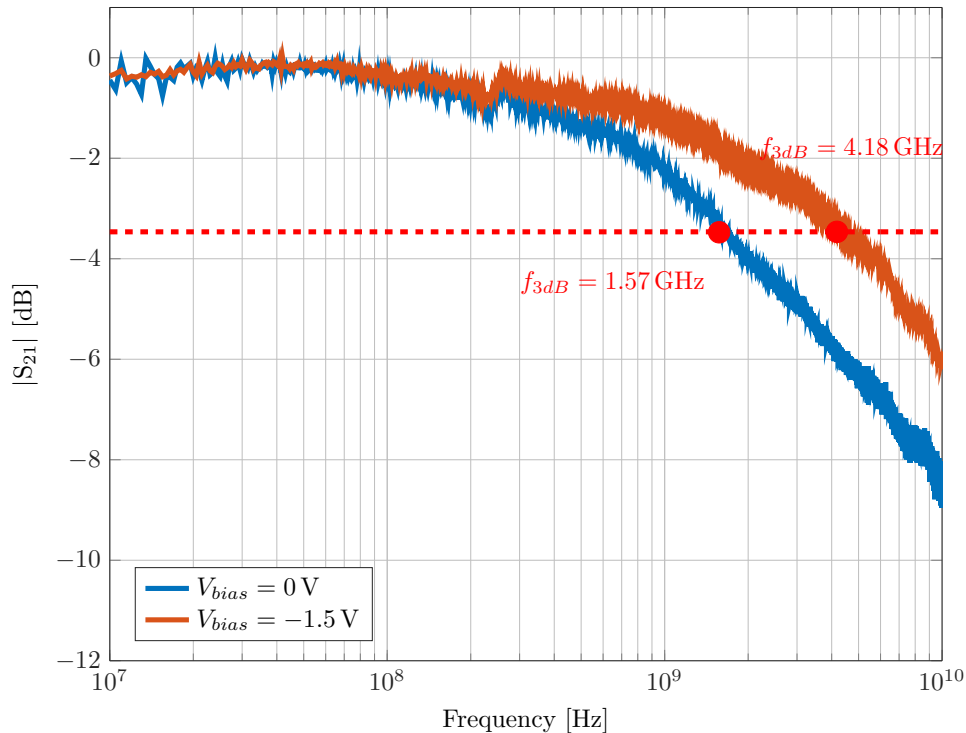


Figure 3.3: Photodiode #25 frequency response

3.3 Lasers

The semiconductor lasers on the chip would function as the slave laser in the OPLL system block diagram 3.1a. Frequency modulation (FM) will be used by the loop to force the slave laser to follow an external master laser. Single mode DFB, or distributed Bragg reflector (DBR) lasers can be frequency modulated through input current modulation. The frequency modulation is accompanied by intensity modulation (IM), which is unnecessary for the OPLL application. Three-electrode DBR lasers can be modulated with IM suppression as discussed in [9]. However, the DFB structure has the advantage of being more compact, and more easily fabricated. As discussed in Chapter 2, choosing a XOR-gate phase detector makes the loop insensitive to IM, and therefore, a DFB slave laser is chosen. The disadvantage is that isolating the FM behavior for characterization, will show to be difficult.

3.3.1 Static Measurements

The static $P(I)$ curves of the lasers that are part of a working subsystem on the PIC are plotted in Figure 3.4. The laser threshold current I_{th} is between 20 mA, and 40 mA. The optical output power of the lasers on the chip P_{out} (probing the grating couplers with an optical fibre accounts for approximately 15 dB losses) is between 0 dBm, and 10 dBm. It is verified for all working DFB lasers on the PIC, that they are monomode with more than 40 dB sidemode suppression. Figure 3.5 shows the optical spectrum analyzer (OSA) plot for lasers #18 and #25, their lasing wavelength is $\lambda = 1538$ nm which is inside the C-band [1530 nm; 1565 nm] used for optical communication applications.

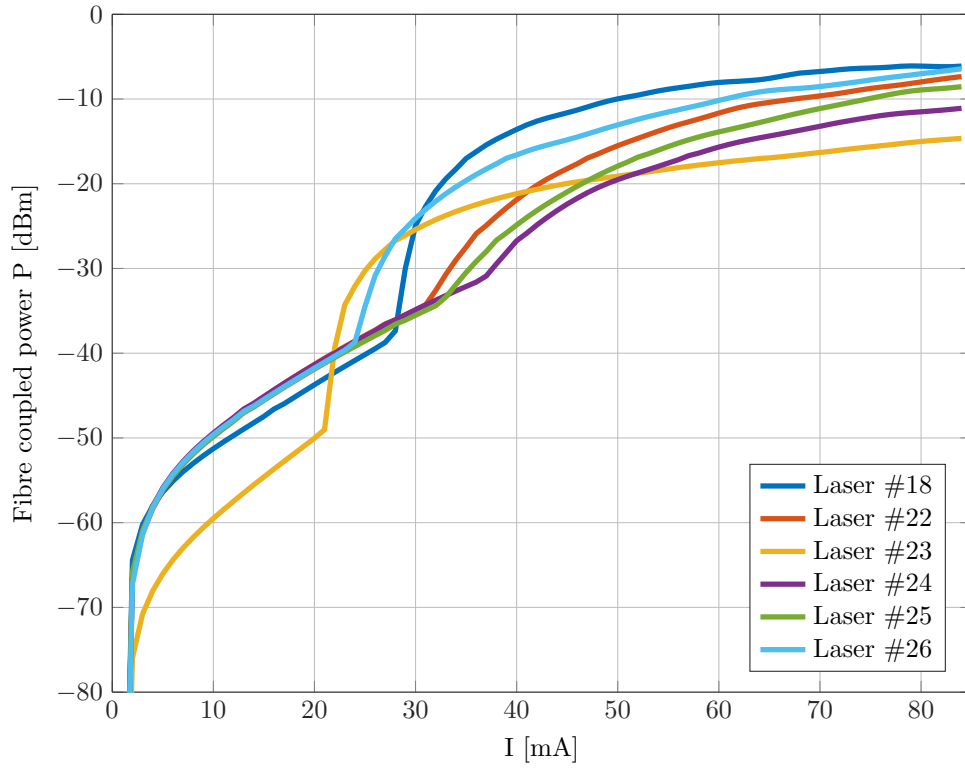


Figure 3.4: $P(I)$ characteristic of lasers (15 dB losses from grating coupler to fibre probe)

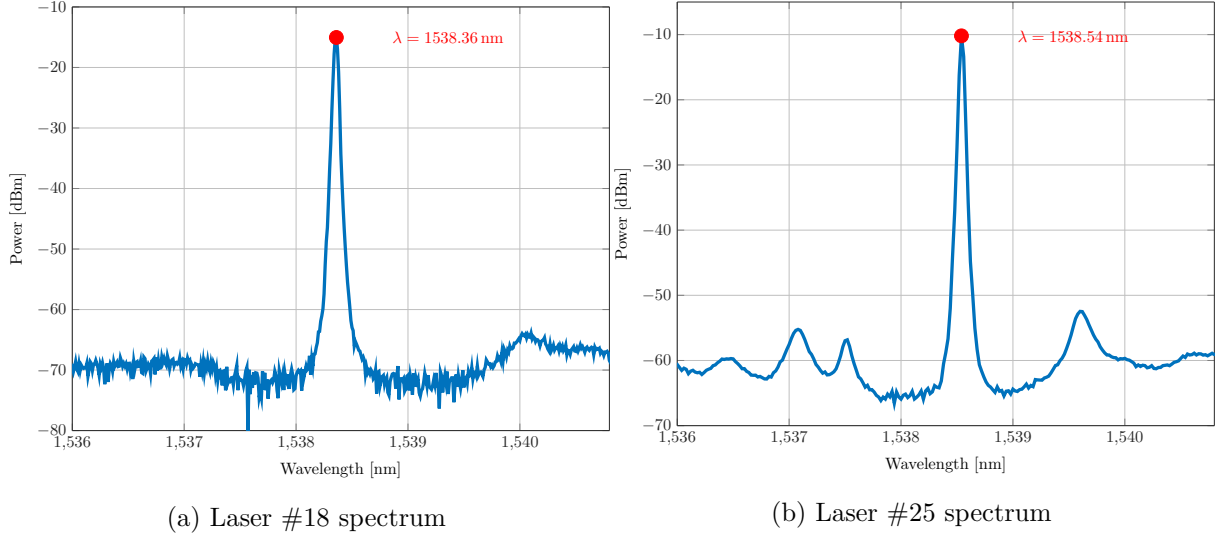


Figure 3.5: Optical spectrum analyzer measurement

3.3.2 Intensity and Frequency Modulation

The slave laser functions as current controlled oscillator. Therefore it is important to know the frequency modulation response, or chirp, of the DFB lasers on the PIC. This is however not trivial, as the direct modulation of the input current results in both intensity modulation (IM) and frequency modulation (FM). In order to separate both, the ADVANTEST Optical Chirpform Test Set (OCTS) could be used. This machine has not been used by anyone from the photonics research group at Ghent university, and some necessary information and/or software is missing to correctly process the measured data. Other measurement techniques exist, and according to literature, the expected FM response for DFB lasers is approximately flat until the frequency becomes too large for thermal contributions, then there is a dip before the electronic contribution (due to carrier injection) becomes dominant. This dip is accompanied by a phase shift of 180° , which will show to have drastic consequences for the loop stability. [10, 11, 12, 13] The ADVANTEST OCTS is essentially an unbalanced MZI that can be biased quickly at different working points ($\omega_0\tau_d = \frac{\pi}{2} + k2\pi$, $\omega_0\tau_d = \frac{3\pi}{2} + k2\pi$, $\omega_0\tau_d = k2\pi$), visualized in Figure 3.6a. A measurement method to extract the IM and FM response of a directly modulated laser is now derived by investigating the result of sending a modulated laser signal through a MZI. The laser is modulated directly with a sinusoidal current $i(t) = A \cos(\omega_m t)$ and equation (3.3) shows how the electric field at the output of a MZI looks. K_a and K_f are respectively the IM and FM sensitivity factors at modulation angular frequency ω_m . Performing a sweep over ω_m , and calculating the K_a and K_f at every frequency, will result in the IM and FM frequency responses.

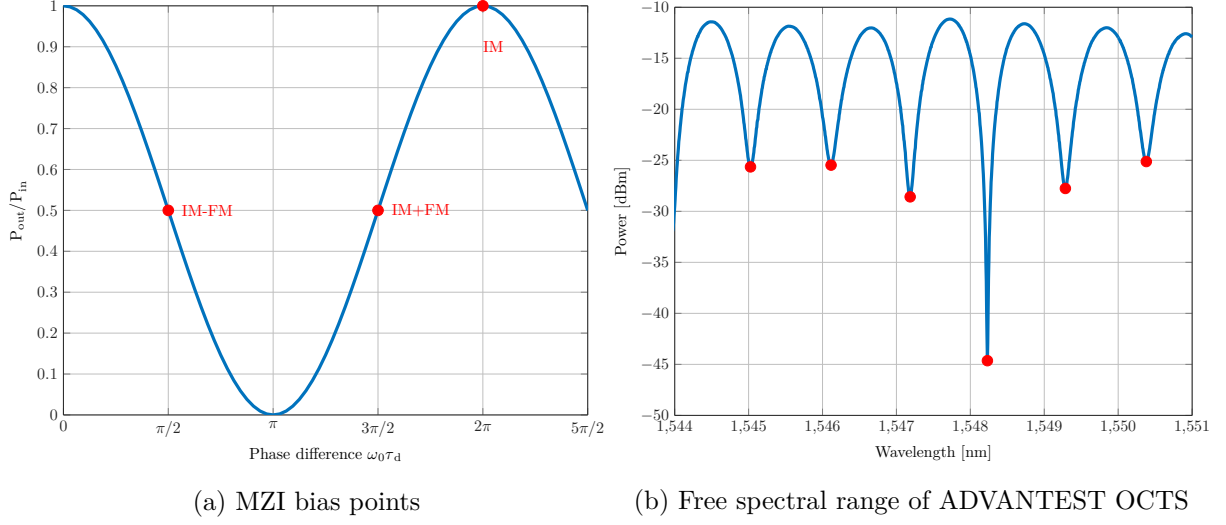


Figure 3.6

$$E(t) = \frac{\sqrt{P_{bias} + K_a i(t)}}{2} e^{j(\omega_0 t + K_f \int_0^t i(u) du + \phi(t))} \quad (3.2)$$

$$\begin{aligned}
E_{out}(t) &= \frac{E_{in}(t)}{2} + \frac{E_{in}(t - \tau_d)}{2} \\
&= \frac{\sqrt{P_{bias} + K_a i(t)}}{2} e^{j(\omega_0 t + K_f \int_0^t i(u) du + \phi(t))} + \frac{\sqrt{P_{bias} + K_a i(t - \tau_d)}}{2} e^{j(\omega_0(t - \tau_d) + K_f \int_0^{t - \tau_d} i(u) du + \phi(t - \tau_d))} \\
&= \frac{\sqrt{P_{bias} + K_a A \cos(\omega_m t)}}{2} e^{j(\omega_0 t + \frac{K_f A}{\omega_m} \sin(\omega_m t) + \phi(t))} \\
&\quad + \frac{\sqrt{P_{bias} + K_a A \cos(\omega_m(t - \tau_d))}}{2} e^{j(\omega_0(t - \tau_d) + \frac{K_f A}{\omega_m} \sin(\omega_m(t - \tau_d)) + \phi(t - \tau_d))} \\
&\approx \frac{\sqrt{P_{bias} + K_a A \cos(\omega_m t)}}{2} \left[e^{j(\omega_0 t + \frac{K_f A}{\omega_m} \sin(\omega_m t) + \phi(t))} + e^{j(\omega_0(t - \tau_d) + \frac{K_f A}{\omega_m} [\sin(\omega_m t) - \omega_m \tau_d \cos(\omega_m t)] + \phi(t - \tau_d))} \right] \quad (3.3)
\end{aligned}$$

The approximations performed in the final line of (3.3) are given by (3.4), and (3.5). These small angle approximations are justified because $\tau_d \approx 7$ ps, which was determined by measuring the free spectral range (FSR) of the ADVANTEST OCTS in Figure 3.6b ($\tau_d = \frac{\lambda^2}{c\Delta\lambda}$). The approximations hold up for modulation frequencies up to a few GHz, which is more than enough for the FM response measurement.

$$\begin{aligned}
\cos(\omega_m(t - \tau_d)) &= \cos(\omega_m t) \cos(\omega_m \tau_d) + \sin(\omega_m t) \sin(\omega_m \tau_d) \\
&\approx \cos(\omega_m t) \left(1 - \frac{\omega_m^2 \tau_d^2}{2}\right) + \omega_m \tau_d \sin(\omega_m t) \\
&\approx \cos(\omega_m t)
\end{aligned} \tag{3.4}$$

For the Taylor series approximation of (3.4), the zero order is used because (3.4) only appears as an amplitude factor in (3.3). The approximation of (3.5) appears as a phase term in the exponent of (3.3), so a first order Taylor series is used.

$$\begin{aligned}
\sin(\omega_m(t - \tau_d)) &= \sin(\omega_m t) \cos(\omega_m \tau_d) - \cos(\omega_m t) \sin(\omega_m \tau_d) \\
&\approx \sin(\omega_m t) \left(1 - \frac{\omega_m^2 \tau_d^2}{2}\right) - \omega_m \tau_d \cos(\omega_m t) \\
&\approx \sin(\omega_m t) - \omega_m \tau_d \cos(\omega_m t)
\end{aligned} \tag{3.5}$$

Equation (3.6) derives an expression for the output optical power which can be measured with a photodiode. The phase term $\Delta\phi(t) = \phi(t) - \phi(t - \tau_d)$ represents the contribution from the phase noise. By choosing the appropriate biasing points for the MZI, it is possible to isolate the IM and FM from the output optical power. This is done in (3.7), and the bias points of interest are highlighted in Figure 3.6a.

$$\begin{aligned}
P_{out}(t) &\propto |E_{out}(t)|^2 = |E_{in}(t) + E_{in}(t - \tau_d)|^2 \\
&\approx \frac{P_{bias} + K_a A \cos(\omega_m t)}{2} \left[1 + \cos(\omega_0 \tau_d + K_f A \tau_d \cos(\omega_m t) + \Delta\phi(t)) \right] \\
&= \frac{P_{in}(t)}{2} \left[1 + \cos(\omega_0 \tau_d) \cos(K_f A \tau_d \cos(\omega_m t) + \Delta\phi(t)) \right. \\
&\quad \left. - \sin(\omega_0 \tau_d) \sin(K_f A \tau_d \cos(\omega_m t) + \Delta\phi(t)) \right]
\end{aligned} \tag{3.6}$$

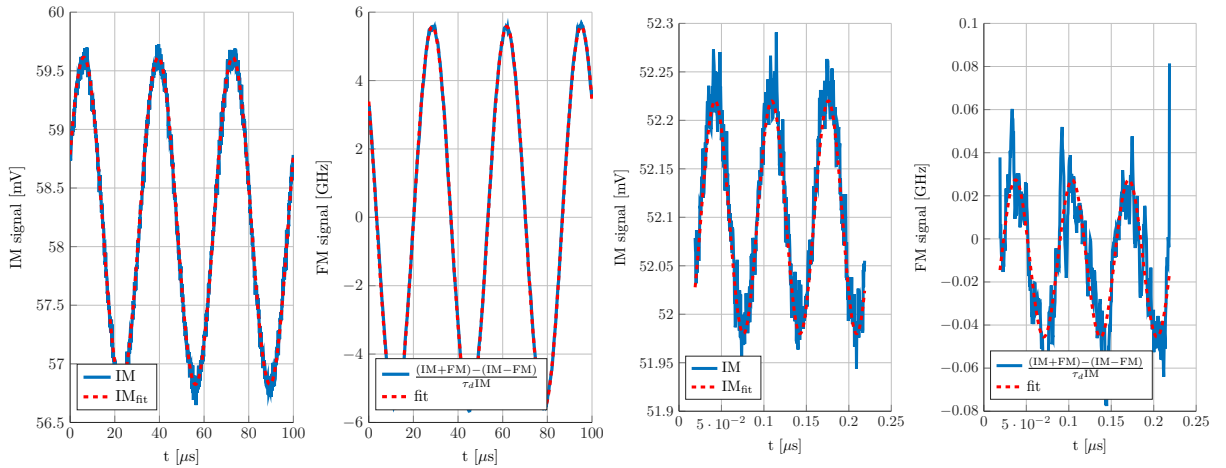
The phase noise term $\Delta\phi$ is neglected to find the final result (3.7). The IM is isolated easily by biasing at $\omega_0 \tau_d = k2\pi$, the FM can be found by subtracting the results from $\omega_0 \tau_d = \frac{3\pi}{2}$, and $\omega_0 \tau_d = \frac{\pi}{2}$. This then has to be divided by the already measured IM, and by τ_d , resulting in equation (3.8). The final approximations used in (3.7) can be justified by considering that $A \sim \text{mA}$, $K_f \sim \frac{\text{GHz}}{\text{mA}}$, and $\tau_d \sim \text{ps}$.

$$P_{out} \propto \begin{cases} \frac{P_{in}(t)}{2} [1 + \cos(K_f A \tau_d \cos(\omega_m t))] & \omega_0 \tau_d = k2\pi, k \in \mathbb{N} \\ \approx P_{in}(t) \rightarrow \mathbf{IM} \\ \frac{P_{in}(t)}{2} [1 - \sin(K_f A \tau_d \cos(\omega_m t))] & \omega_0 \tau_d = \frac{\pi}{2} + k2\pi, k \in \mathbb{N} \\ \approx \frac{P_{in}(t)}{2} [1 - K_f A \tau_d \cos(\omega_m t)] \rightarrow \mathbf{IM} - \mathbf{FM} \\ \frac{P_{in}(t)}{2} [1 + \sin(K_f A \tau_d \cos(\omega_m t))] & \omega_0 \tau_d = \frac{3\pi}{2} + k2\pi, k \in \mathbb{N} \\ \approx \frac{P_{in}(t)}{2} [1 + K_f A \tau_d \cos(\omega_m t)] \rightarrow \mathbf{IM} + \mathbf{FM} \end{cases} \quad (3.7)$$

$$\Rightarrow K_f = \frac{(\mathbf{IM} + \mathbf{FM}) - (\mathbf{IM} - \mathbf{FM})}{\mathbf{IM} \tau_d A \cos(\omega_m t)} \quad (3.8)$$

The ADVANTEST OCTS has 3 buttons called **IM**, **IM+FM**, and **IM-FM** which should correspond to the results from (3.7). This measurement of the IM and FM response of the lasers on the PIC was done by manually sweeping the modulation frequency of the laser current with a signal generator in the range $\omega_m \in [30 \text{ kHz}; 15 \text{ MHz}]$, the optical output is sent through the ADVANTEST OCTS (once with the *IM* setting, once with the *IM + FM*, and once with the *IM - FM* setting for every ω_m) to a photodiode and measured with an oscilloscope.

The output of the ADVANTEST OCTS is plotted for modulation frequencies 30 kHz, and 15 MHz in Figures 3.7a, and 3.7b respectively. The amplitude of the sinusoidal fit is then the K_a , or K_f factor for that modulation frequency. Plotting the K_a , and K_f for the entire modulation frequency sweep results in the IM and FM responses of Figure 3.8.



(a) Laser #18 modulated with 30 kHz ADVANTEST output (b) Laser #18 modulated with 15 MHz ADVANTEST output

Figure 3.7: Measurement of the ADVANTEST OCTS output for laser #18

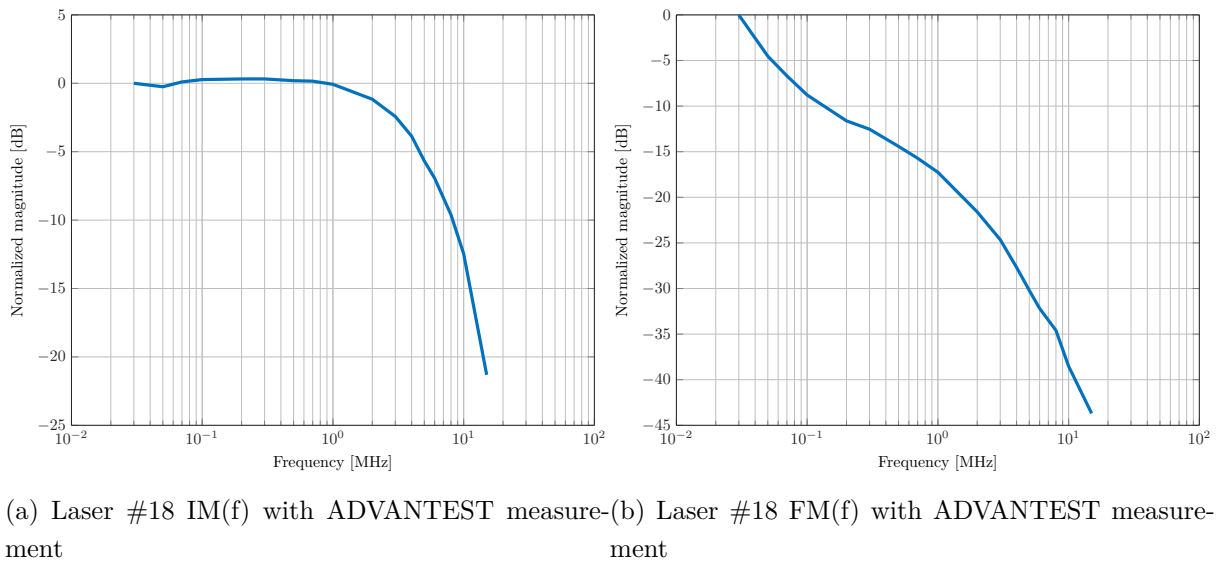
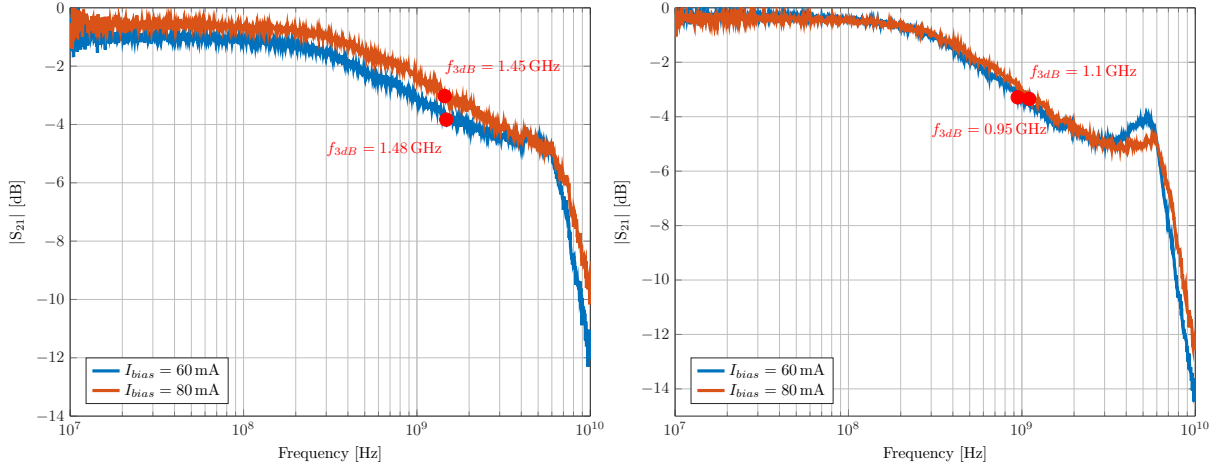


Figure 3.8: Measurement of IM and FM response of laser #18

Something obviously went wrong during the measurement. The FM response does not resemble the expected result described in literature [10, 11, 12, 13]. According to the IM response, the lasers have a modulation bandwidth below 3 MHz, which is hard to believe. Possible causes for the failed measurement:

- The used oscilloscope started to display a warning for thermal calibration halfway the frequency sweep.
- The PIC was mounted on a probing station without temperature control. The laser wavelength is highly sensitive to temperature variations. Not actively controlling the temperature during the measurement may have tempered with the results.
- No successful measurement has been performed with the (old) ADVANTEST OCTS machine at the photonics research group of Ghent University, maybe the machine is broken.

The measurement of the IM response, resulted in the strange graph of 3.8a. It looks like the DFB lasers on the PIC can only be modulated up to a few MHz, which would be unworkable for OPLL application. To check this result, the response of the lasers to direct modulation was measured with the PNA-X network analyzer. The result is visible in Figure 3.9 and it is clear that the lasers can be modulated with frequencies up to 1 GHz, confirming that something went wrong during the IM/FM-measurement.



(a) Normalized IM frequency response of laser #18 (b) Normalized IM frequency response of laser #25

Figure 3.9: IM frequency response measurement with PNA-X

3.3.3 Linewidth

The performance of the OPLL will depend on the phase noise of the master and the slave laser. Laser phase noise arises from spontaneous emission in the cavity and from carrier density fluctuations. The result of phase and frequency noise ($\omega(t) = \frac{d\phi(t)}{dt}$) is a broadened spectrum, which is typically characterized by the width of the spectral line at -3 dB of the peak power, this linewidth is called the Full Width Half Maximum (FWHM) in literature. The frequency noise power spectral density determines the shape and width of the laser field spectrum, this is described in [14], [15].

When the frequency noise is white $S_{fn} = h_0$ (double sided spectrum in $[\text{rad}^2/\text{Hz}]$), an analytical expression for the laser field spectrum exist, the laser line has a lorentzian shape with $\text{FWHM} = \frac{h_0}{2\pi}$ [15]. If the frequency noise of the lasers on the PIC is assumed to be white, measuring the linewidth will also result in the frequency noise power spectral density $S_{fn} = 2\pi\text{FWHM}$.

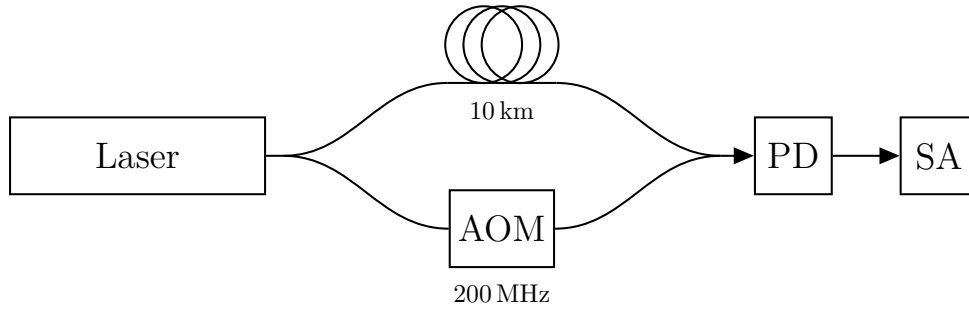


Figure 3.10: Laser lineshape measurement setup

The resolution of optical spectrum analyzers is not fine enough to measure the linewidth. In [15, 16, 17] techniques to measure the lineshape and FWHM are discussed in detail. One of those techniques is the delayed self-heterodyne measurement, schematically depicted in Figure 3.10. The laser's optical signal is split, one part is delayed through several kilometers of fiber, the other part is shifted in frequency by an acousto optical modulator (AOM). After recombination, the beatnote is detected by a photodiode, and sent to an electrical spectrum analyzer with much finer resolution. The fiber delay is required to make sure the recombined optical signals are decorrelated. 10 km fibre gives a delay of $\tau = \frac{nd}{c} \approx \frac{1.5 \cdot 10^4}{3 \cdot 10^8} \text{s} = 50 \mu\text{s}$. For white frequency noise, the correlation time of the laser electric field is given by $\tau_c = \frac{1}{\pi \text{FWHM}}$, so 10 km optical fiber should be enough to reliably measure the lineshape of lasers with a $\text{FWHM} \gg 6.366 \text{ kHz}$. Figures 3.11a and 3.11b show the output of the spectrum analyzer from the delayed self-heterodyne measurement. The δ -peak on top of the approximately lorentzian lineshape means that there is still some amount of correlation present after 10 km of fiber, the frequency noise must increase for lower frequencies (like $\frac{1}{f}$ pink noise). Increasing the fiber length even more would be an option (extra losses 0.2 dB/km) to get a more exact value for the FWHM, but for this thesis, it is enough to have an approximation of the linewidth in the same order of magnitude.

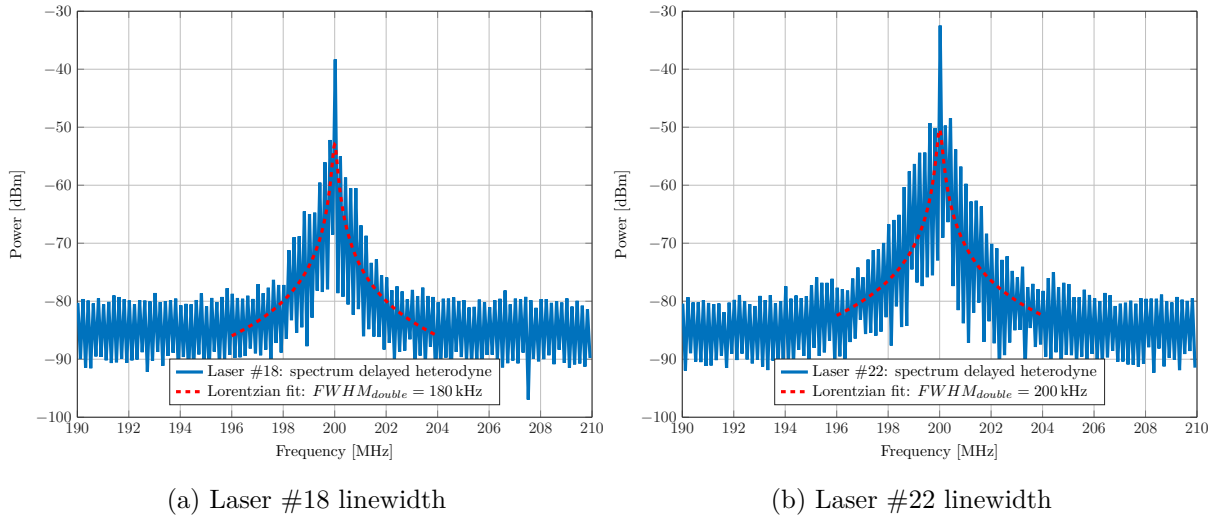


Figure 3.11: Delayed heterodyne selfbeating spectrum

The beatnote of two uncorrelated lorentzian lines will also have a lorentzian lineshape, with a FWHM that is the sum of the individual FWHM's. Therefore, the $\text{FWHM}_{\text{double}}$ used to make the lorentzian fit is twice the linewidth of the laser, resulting in $\text{FWHM}_{L\#18} \approx 90 \text{ kHz}$, and $\text{FWHM}_{L\#22} \approx 100 \text{ kHz}$.

4

OPLL System Model

A SIMULINK model for the OPLL is created, including models for the lasers and photodiodes based on the measurements of the PIC. First, all the components of the loop are modelled, then a model is built for the loop in the phase domain, and finally the components are combined into a complete OPLL system that should behave like the phase domain loop. The SIMULINK system simulations will result in the design parameters for a real life OPLL circuit.

SIMULINK is a graphical programming environment, ideally suited for the construction and simulation of systems that can be formulated in terms of a block diagram, it works well together with MATLAB because the inputs and outputs from the SIMULINK system can be easily created, and processed in MATLAB.

4.1 Photodiode Model

The mathematical function, describing a photodiode's behavior, is given by equation (4.1). Figure 4.1 is the SIMULINK block diagram implementation of this equation with the addition of a low-pass filter (in accordance to the measurement 3.3). In order for the diode to function as a photodetector, sufficiently negative bias voltages are required, in that case, only the highlighted part of the block diagram has a significant impact on the output signal.

$$I_{out}(t) = (P_{in}(t)R_d + I_{dark}) \left(e^{\frac{(V_{bias} - V_{th})q}{nkT}} - 1 \right) \quad (4.1)$$

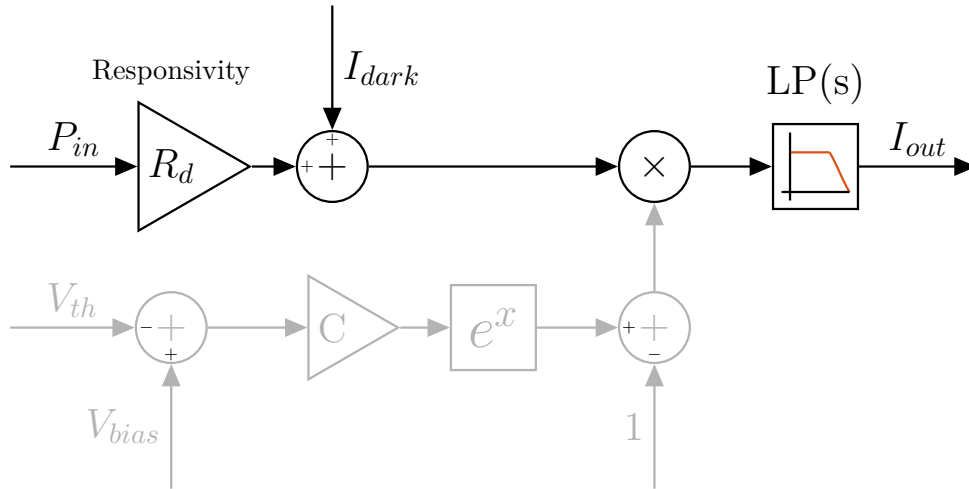


Figure 4.1: Photodiode SIMULINK block diagram

SIMULINK allows for a hierarchical buildup of a system out of subsystems. From the photodiode model 4.1, a new building block "PD" is created. To test the model's behavior, two SIMULINK test benches are created 4.2a, and 4.2b respectively. The first testbench sweeps the bias voltage V_{bias} between -2 V and 1.5 V , with a constant incident optical power $P_{in} = -5\text{ dBm}$. The second testbench sweeps P_{in} between 0 mW and 0.3981 mW (up to -4 dBm), while the bias voltage stays constant $V_{bias} = -1.5\text{ V}$. In Figures 4.3a, and 4.3b the testbench results are plotted against the measurements of the I(V) curve, and the responsivity of photodiode #22 on the PIC. For the responsivity measurement, the same data as from Figure 3.2b is used, but this time 15 dB grating losses from the grating couplers are subtracted from the laser power that came out of the fibre probe.

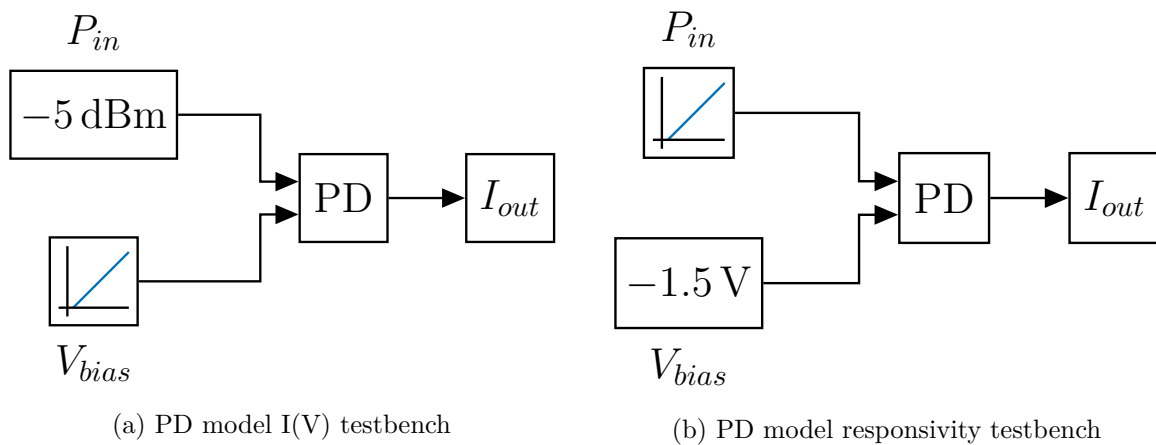


Figure 4.2: SIMULINK testbenches for the photodiode model

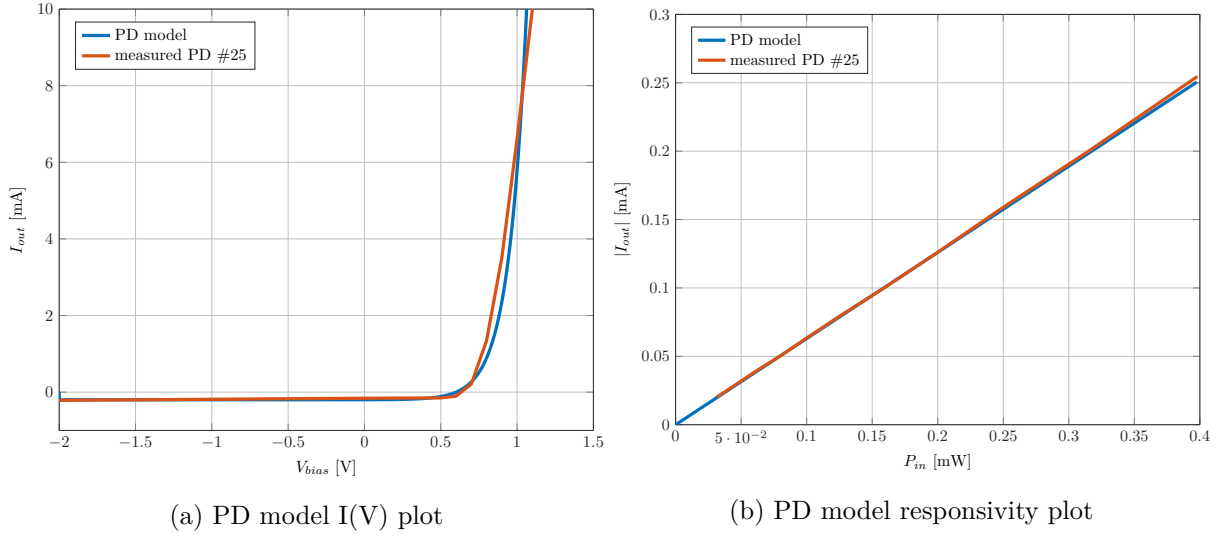


Figure 4.3: Photodiode model results

The low-pass filter $LP(s)$ used in the model is a first order low-pass filter with a -3 dB bandwidth of 4 GHz, which is an easy, and fast (in terms of simulation calculations) approximation of the measured photodiode frequency response in Figure 3.3. SIMULINK also has readily available rational filter blocks, and as long as the OPLL loop bandwidth is well below 4 GHz, there is no need for a more accurate approximation of the measured frequency response.

4.2 Laser Model

The laser's output electric field $E_{out}(t)$ is represented by equation (4.2). The SIMULINK model assumes that the laser is biased with a constant DC current that corresponds to an output power P_0 . The laser's frequency f_0 is around 193 THz, this will be inconvenient for simulations where the modulation, and filter frequencies are several orders of magnitude lower. Since the lasers are never used individually, but always as a pair beating on a photodiode, only their frequency difference matters, and the laser's central frequency f_0 can be chosen arbitrarily during simulations.

$$E_{out}(t) = \sqrt{P_0 + K_a I_{in}(t)} e^{j(2\pi f_0 t + 2\pi K_f \int_0^t I_{in}(u) du + \int_0^t n(v) dv)} \quad (4.2)$$

As discussed in Chapter 3, semiconductor DFB lasers can be directly modulated with the input current I_{in} , resulting in IM and FM. This behavior is implemented by the K_a and K_f factors in the formula. Also discussed in Chapter 3, is the lineshape, determined by the frequency

noise and included into the model as input n . Figure 4.4 is the implementation of (4.2) as a SIMULINK block diagram.

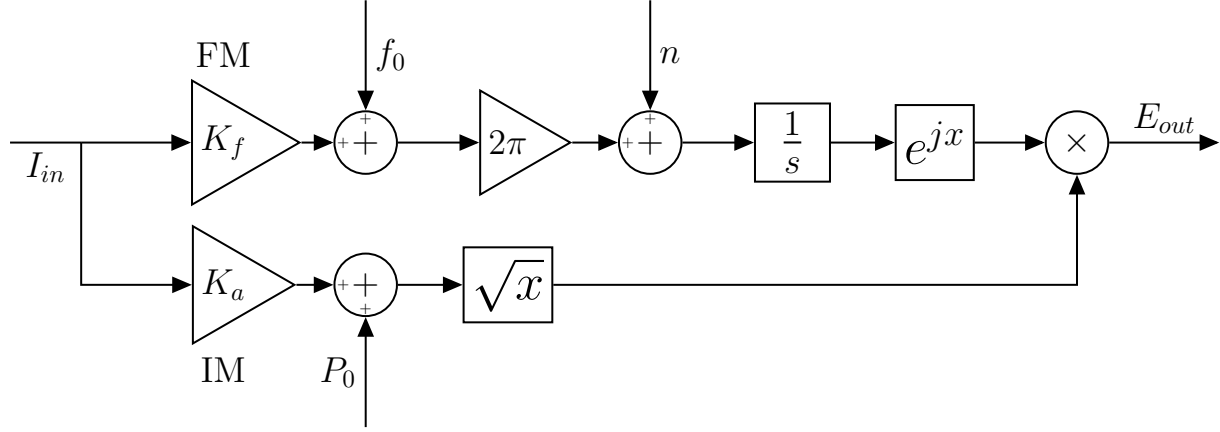


Figure 4.4: Laser SIMULINK block diagram

4.2.1 IM/FM

To visualize the intensity and frequency modulation by directly modulating the input current I_{in} , the testbench 4.5 is created. A sinusoidal input $I_{in} = A_{in} \sin(2\pi f_{in}t)$ together with the laser parameters of Table 4.1 result in the $E_{out}(t)$ plot of Figure 4.6. The IM and FM behavior of the model is clearly visible by plotting two periods of the modulation frequency.

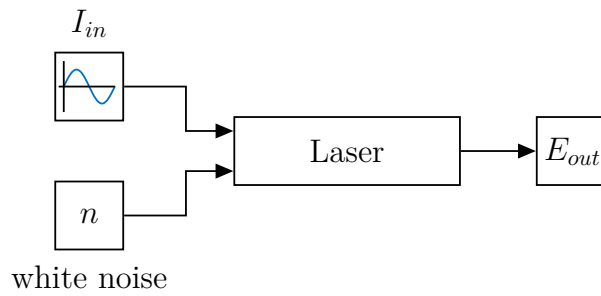


Figure 4.5: SIMULINK testbench for laser model IM and FM behavior

SIMULINK has a building block available to generate white noise, of which the double sided power spectral density (PSD) is given in units of $\left[\frac{\text{rad}^2}{\text{Hz}}\right]$. As discussed in Chapter 3 and in [15], the frequency noise double sided PSD determines the lineshape, and for white frequency noise, this is a Lorentzian shape with linewidth FWHM determined by the noise $S_n(f) = 2\pi\text{FWHM}$.

f_0	P_0	K_f	K_a	FWHM	A_{in}	f_{in}
1 GHz	2 mW	$0.7 \frac{\text{GHz}}{\text{mA}}$	$0.1 \frac{\text{mW}}{\text{mA}}$	1 MHz	5 mA	100 MHz

Table 4.1: Laser parameters for sine modulation testbench 4.5

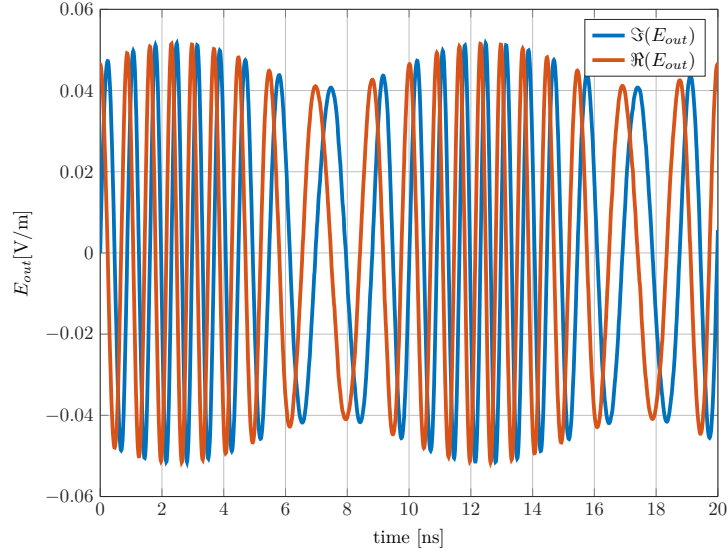


Figure 4.6: Laser model sine modulation plot

The laser model of Figure 4.4 assumes frequency independent IM and FM, however this is not accurate. Especially the frequency dependent FM(s) has a crucial influence on loop stability as the phase of FM($j\omega$) shifts 180° between 0.1 MHz and 10 MHz for typical semiconductor DFB lasers. Figure 4.7 shows the updated laser block diagram with filters FM(s), and IM(s) instead of K_f , and K_a respectively.

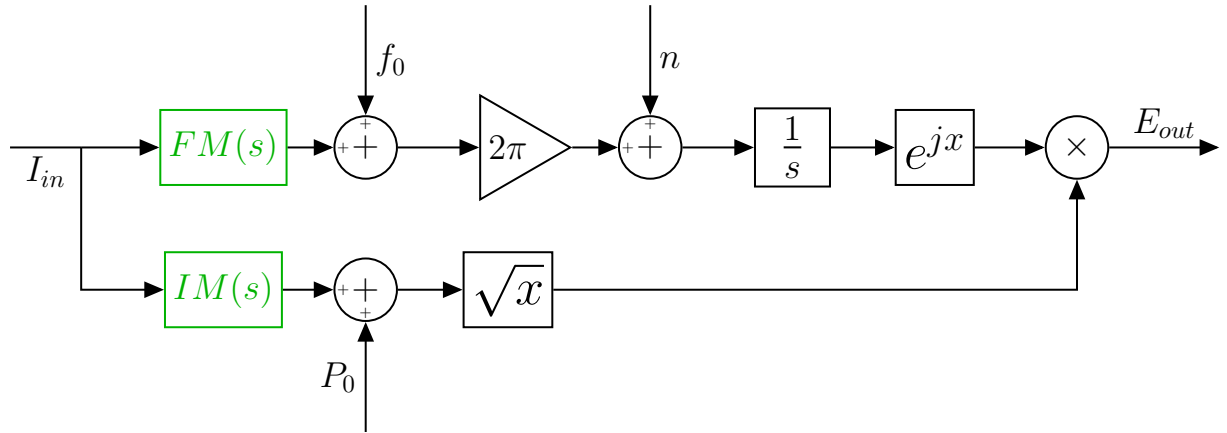


Figure 4.7: Laser SIMULINK block diagram with IM(s) and FM(s) filters

The FM frequency response measurement described in Chapter 3 did not result in useful data. Correc et. al. [13] derived, and experimentally verified, a theoretical formulation of the FM frequency response of DFB semiconductor lasers. Based on the bode plots found in literature, the filter (4.3) is chosen for the SIMULINK model of the laser, with $f_{FM} = \frac{1}{2\pi\tau_{FM}} \in [0.1 \text{ MHz}; 10 \text{ MHz}]$.

$$FM(s) = K_f \frac{1 - \frac{\tau_{FM}s}{a}}{1 + \tau_{FM}s} \quad (4.3)$$

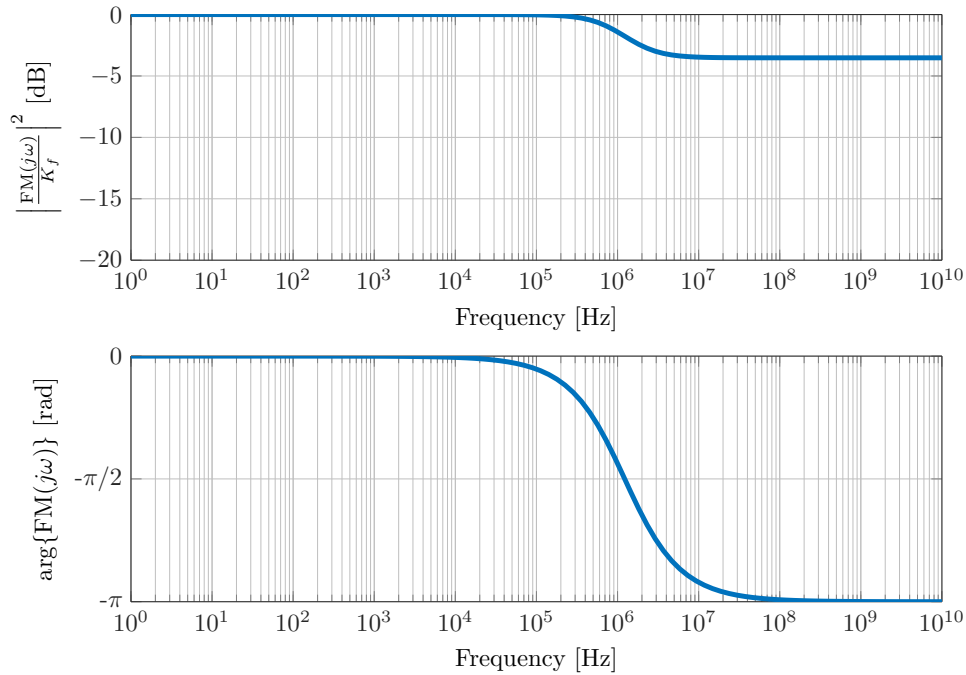


Figure 4.8: Bode plots of FM(s) filter

Based on the IM response measurement with the PNA-X in Figure 3.9, a simple first order low-pass filter is chosen as the IM(s) filter. The intensity modulation is actually of no concern for the OPLL system because a XOR-gate is used as phase detector, but it is included to keep the models more realistic. The XOR-gate output amplitude is independent of the input amplitudes.

4.2.2 Frequency Noise

To test the frequency noise behavior of the models, the testbenches of Figure 4.9 are created in SIMULINK. The lasers are not modulated, the only input is the frequency noise, making sure that the noise behavior can be verified in the spectrum of the output.

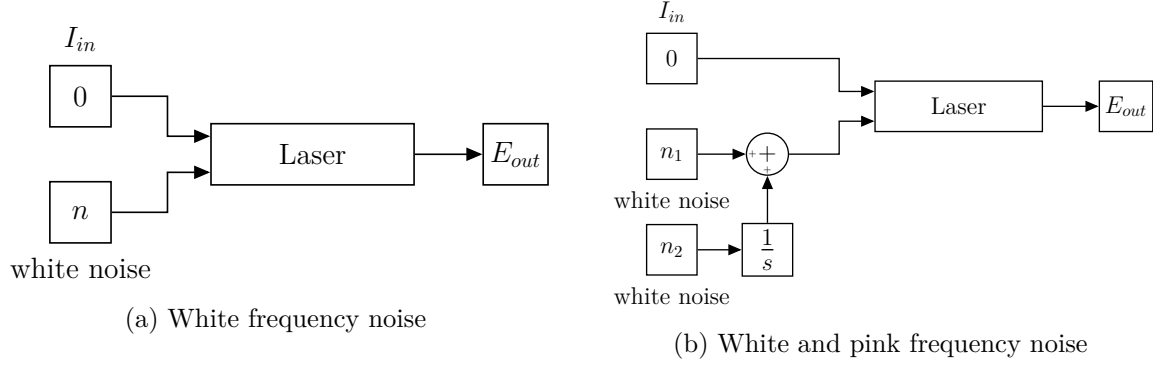


Figure 4.9: SIMULINK testbenches for the lineshape of the laser model

First, the laser models are tested with white frequency noise, because an analytical solution exists for the lineshape in the form of a lorentzian function defined by its FWHM linewidth (4.4) [14, 15]. The white noise source block of SIMULINK has a parameter h_0 , being the white noise variance in $\left[\frac{\text{rad}^2}{\text{Hz}}\right]$, the spectrum of the output from the white noise source block is plotted in Figure 4.10a (a perfectly flat spectrum for all frequencies would require an infinitely long simulation). In order to have correct frequency noise behavior, the following test is performed: when $h_0 = 2\pi\text{FWHM}$, the PSD of the electric field $E_{out}(t)$ of the laser model should have the lorentzian shape with FWHM linewidth. Figure 4.11a shows the simulated laser lineshape, and the lorentzian with FWHM, confirming the theoretical behavior in the case of white frequency noise.

$$\mathcal{L}(f) = \frac{A}{(f - f_0)^2 + \left(\frac{\text{FWHM}}{2}\right)^2} \quad (4.4)$$

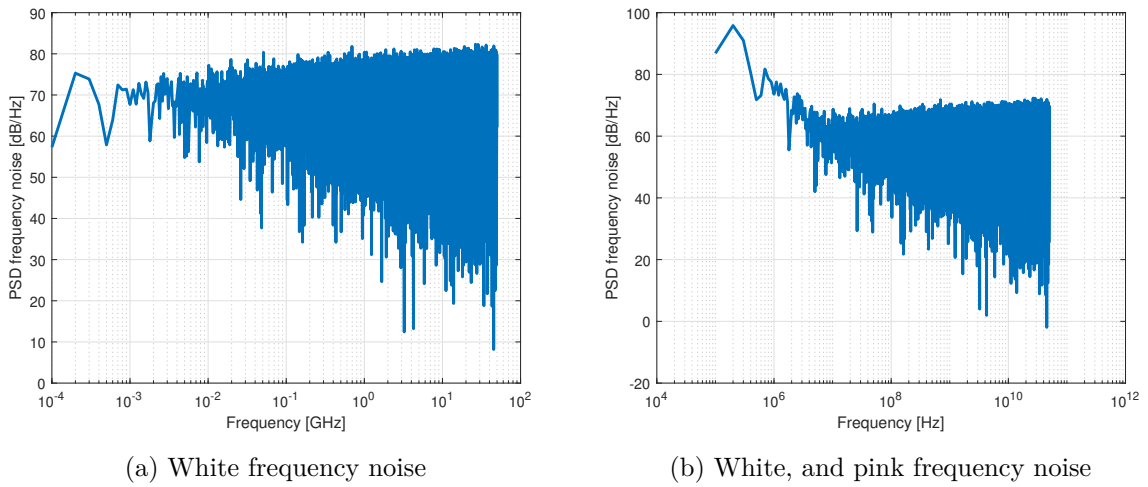


Figure 4.10: Noise spectrum

When the frequency noise has a $\frac{1}{f}$ component, there is no analytical solution for the resulting lineshape. Papers [14, 15] describe that the lineshape is approximately the convolution of a Lorentzian, and a Gaussian function, called a Voigt profile. The effect of the $\frac{1}{f}$ component is most pronounced near the peak of the spectrum, resulting in spectral line broadening. The frequency noise at higher frequencies is responsible for the tails of lineshape. SIMULINK does not have a readily available pink noise source block. The testbench 4.9b tries to create noise with a PSD $S_n(f) = h_0 + \frac{k}{f}$, and according to Figure 4.10b this works reasonably well.

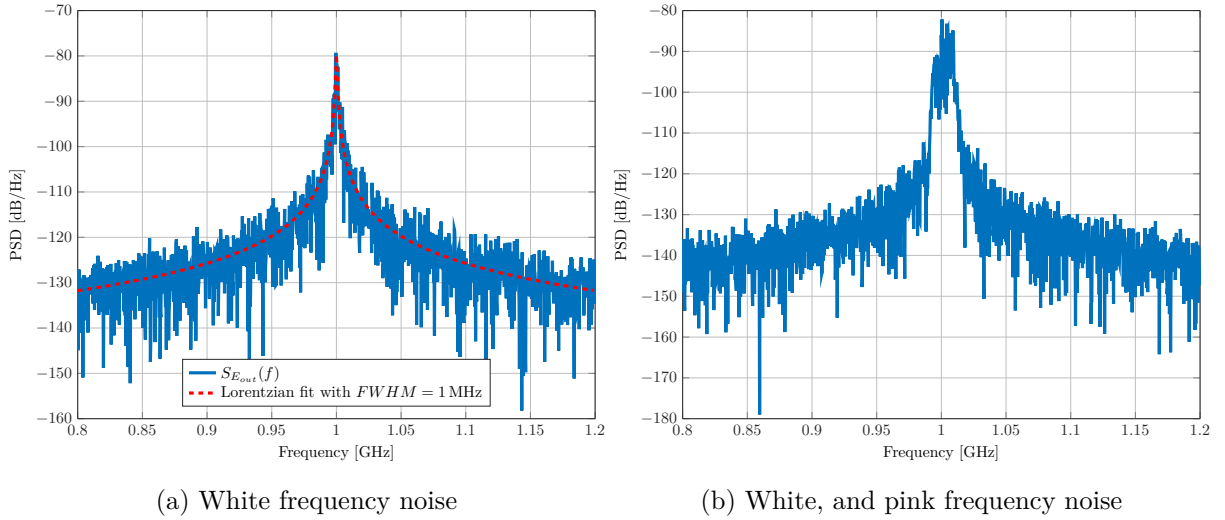


Figure 4.11: Laser model lineshape results

The lineshape resulting from testbench 4.9b is plotted in Figure 4.11b, the resulting PSD is not a Voigt profile, but there is spectral broadening. The frequency noise behavior of the laser models is according to the theory for white frequency noise, and it is possible to change/broaden the lineshape by changing the noise spectrum.

4.2.3 Master Laser and Slave Laser

The OPLL locks the frequency and phase of the beatnote of two lasers to an external electrical RF signal. A new building block is created that combines two of the laser models, the master laser, and slave laser (MLSL) block diagram of Figure 4.12. The block has two inputs to modulate the lasers, and one output being the summed electric field in exponential form $E_{out}(t) = E_{ML}(t) + E_{SL}(t)$. The individual laser frequencies are of no interest for the loop, only the difference frequency Δf matters. The MLSL block functions as the current controlled oscillator of the loop.

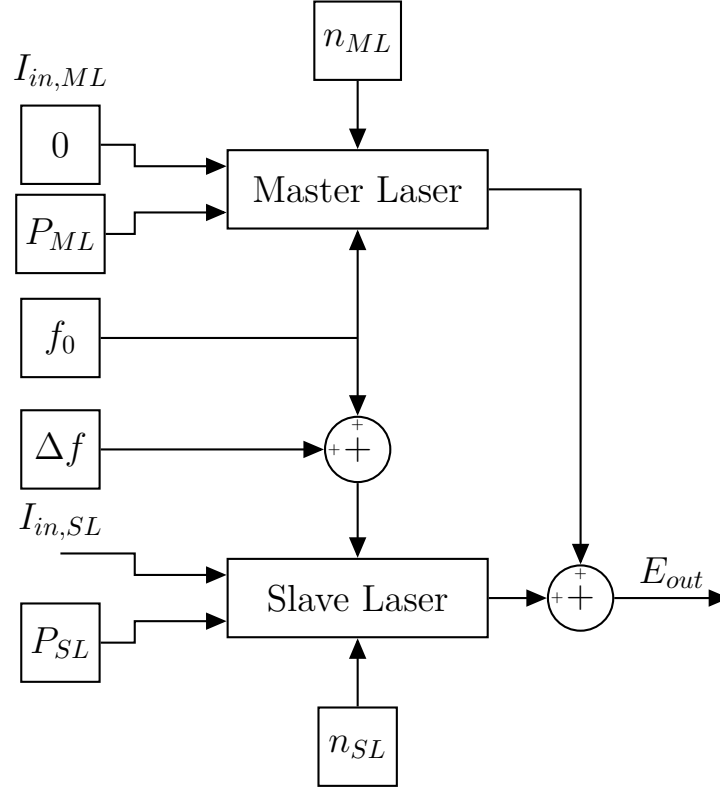


Figure 4.12: Master laser and slave laser SIMULINK block diagram

Testbench 4.13 investigates the beating behavior described by (2.1), the lasers are not modulated and their parameters for the simulation are given in Table 4.2. The PSD of the electric field should contain the two laser lines Δf apart from each other. On the photodiode the optical power is detected, the PSD of P_{out} should contain the beatnote line, with (in the case of white frequency noise) a FWHM that is the sum of the individual master, and slave laser linewidths. Figures 4.14, and 4.15b confirm the theoretical expectations.

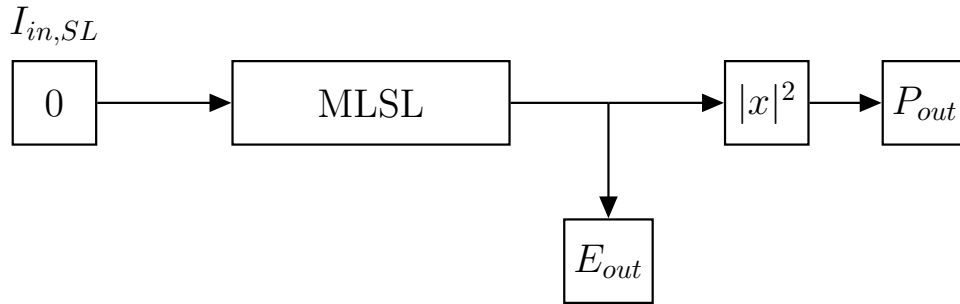


Figure 4.13: SIMULINK testbench for MLSL

f_0	Δf	P_{ML}	P_{SL}	FWHM_{ML}	FWHM_{SL}
100 MHz	100 MHz	5 mW	5 mW	1 MHz	2 MHz

Table 4.2: Laser parameters for MLSL testbench 4.13

As already discussed, the consequence of the XOR-gate phase detector is that amplitude, and IM of the beatnote should not play a role in the OPLL system. Figure 4.15a is the beatnote $P_{out}(t)$ plotted against time, the power varies between 0 mW and 20 mW = 4 · 5 mW, which is in agreement with (2.1).

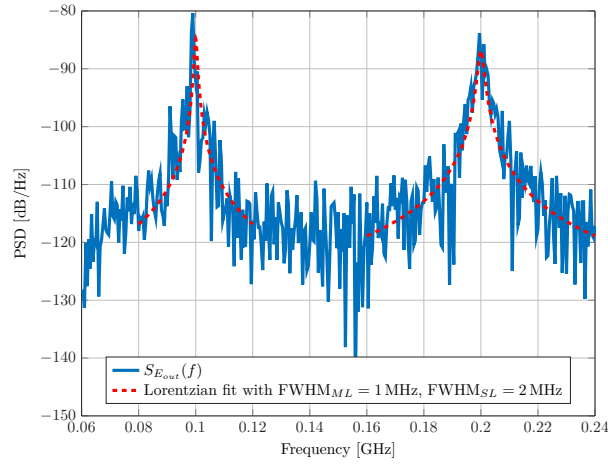
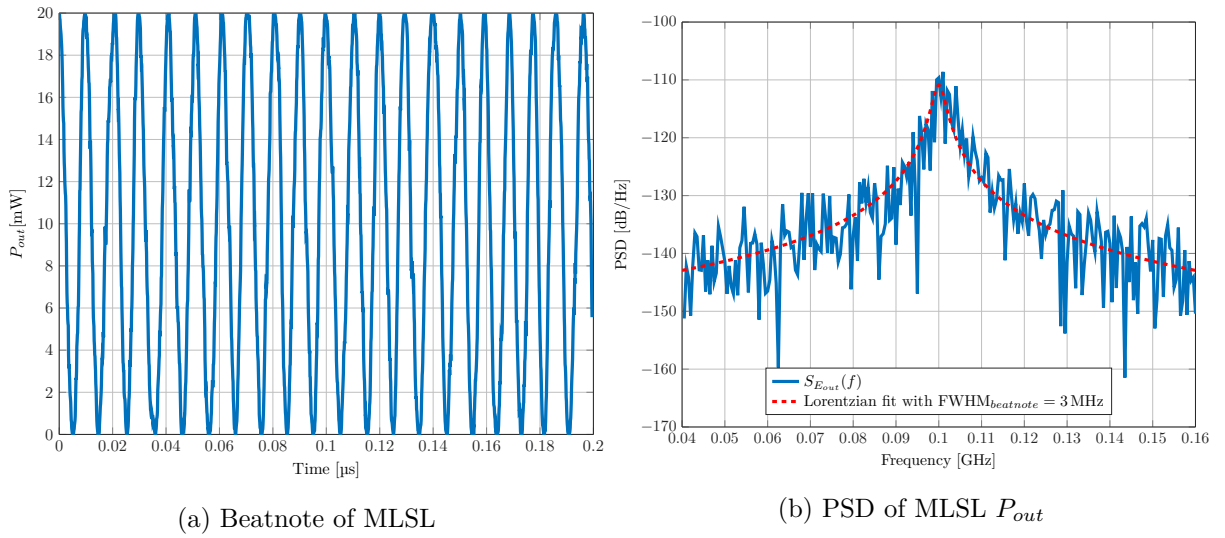
Figure 4.14: PSD of MLSL E_{out} 

Figure 4.15: MLSL model beatnote plot

4.3 Phase Domain Loop Model

Before the laser and photodiode models are incorporated into a complete OPLL system model, the loop behavior and stability are investigated with the phase domain description from Chapter 2. A SIMULINK testbench for the phase domain block diagram of Figure 2.5 is given by Figure 4.16. The behavior of the loop is tested for a frequency step input, or in other words, a linearly increasing input phase $\theta_{RF}(t) = 2\pi f_{RF}t$. The added phase noise n is assumed to be white, unless specified otherwise.

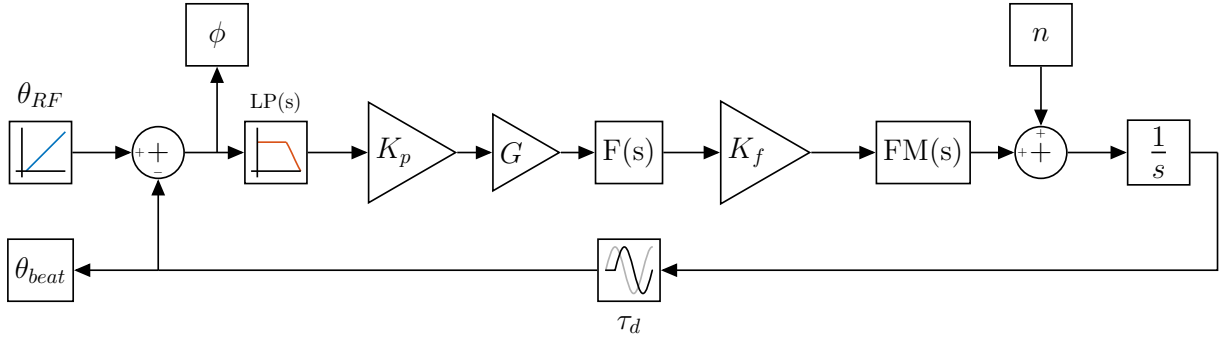


Figure 4.16: Phase domain OPLL testbench with frequency step

In the phase domain OPLL model, the filter $LP(s)$ is always a first order low-pass given by equation (4.5), the FM response $FM(s)$ of the laser is given by equation (4.6). The loop delay τ_d is determined by the loop architecture, and interconnections. The phase detector gain K_p , and the laser frequency modulation gain K_f are also fixed. The only design parameters of the loop are G , and $F(s)$. In the following subsections, a few different loop filters will be considered.

$$LP(s) = \frac{1}{1 + \tau_{LP}s} = \frac{1}{1 + \frac{s}{2\pi f_{LP}}} \quad (4.5)$$

$$FM(s) = \frac{1 - \frac{\tau_{FM}s}{a}}{1 + \tau_{FM}s} = \frac{1 - \frac{s}{a2\pi f_{FM}}}{1 + \frac{s}{2\pi f_{FM}}} \quad (4.6)$$

4.3.1 First Order

In the first order loop, the loop filter $F(s) = 1$ is used. The open-loop transfer function $OL(s)$ is given by (2.4), with the addition of an extra gain factor G . From now on, the total frequency independent gain will be referred to as $G_{tot} = GK_pK_f$. As a first step, G_{tot} is calculated from the stability criterium (2.8) to meet a target phase margin (PM). Secondly, the influence on the loop performance is investigated for $|G_{tot}| \leq |G_{tot,maxstable}|$, where the maximum stable gain

corresponds to a $PM = 0$.

The fixed parameters in Table 4.3 are chosen arbitrarily, but realistically, as the values could very well correspond to a real system. For demonstration purposes, the design parameter G_{tot} is calculated for a target phase margin of $PM = \pi/3$ rad.

Fixed				Target	Design
f_{LP}	f_{FM}	a	τ_d	PM	G_{tot}
1 GHz	1 MHz	1.5	1 ns	$\pi/3$ rad	2.07×10^6

Table 4.3: First order loop parameters

Filling in the fixed, and target parameter values into the equation for the arguments of the stability criterium in equation (2.8) results in the equation (4.7), and a value for ω_{cr} . This value for ω_{cr} is used in the second equation of the stability criterium to find the design parameter G_{tot} with equation (4.8).

$$\begin{aligned}
 -\omega_{cr}\tau_d - \arctan\left(\frac{\omega_{cr}\tau_{FM}}{a}\right) - \arctan(\omega_{cr}\tau_{FM}) - \arctan(\omega_{cr}\tau_{LP}) - \frac{\pi}{2} &= PM - \pi \\
 \Rightarrow \omega_{cr} &= 2.016 \times 10^6 \frac{\text{rad}}{\text{s}}
 \end{aligned} \tag{4.7}$$

$$G_{tot} = \sqrt{\frac{\omega_{cr}^2(1 + \omega_{cr}^2\tau_{LP}^2)(1 + \omega_{cr}^2\tau_{FM}^2)}{1 + \frac{\omega_{cr}^2\tau_{FM}^2}{a^2}}} = 2.07 \times 10^6 \tag{4.8}$$

The bode plot of the corresponding open-loop transfer function is given by Figure 4.17. It is clear that the stability limiting factor is the thermal crossover frequency f_{FM} , where a phase shift of -180° occurs due to the change from thermal to electronic FM. This f_{FM} frequency is a property of the DFB slave laser, and is typically between 0.1 MHz and 10 MHz, which severely limits the achievable loop bandwidth. There are possibilities to circumvent this problem, and these will be discussed later on. If there would be no extra phase shift between the laser's thermal and electronic contribution to the frequency modulation, the loop delay τ_d would limit the loop stability. The $-\omega_{cr}\tau_d$ term in (4.7) would start to become dominant, and is already responsible for the linear phase decrease at higher frequencies in Figure 4.17.

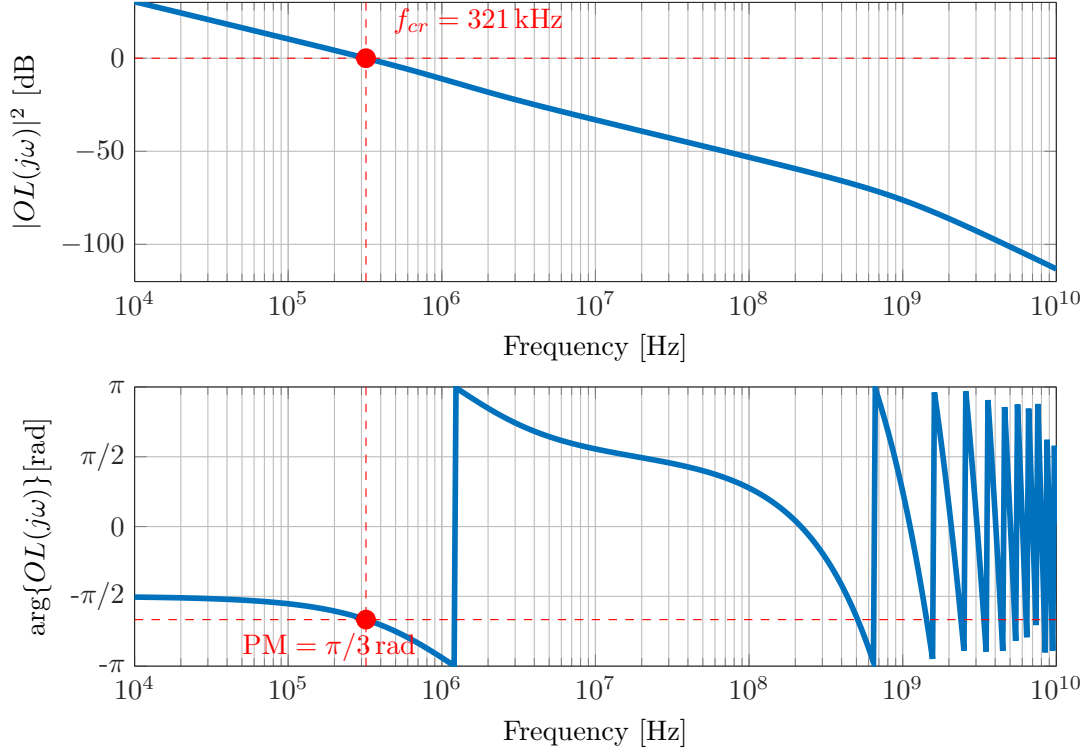


Figure 4.17: Bode plot of first order OL(s)

The maximum stable gain $G_{tot,maxstable} = 9.3 \times 10^6$ is the gain calculated for a phase margin of zero (corresponding to a loop bandwidth $f_{cr} = 1.2$ MHz). The stability calculations are verified with the testbench of Figure 4.16. An input frequency step $f_{RF} = 1$ GHz, and white frequency noise (FWHM = 0.5 MHz) was used to obtain the simulation results of Figures 4.18, 4.19, and 4.20.

Figure 4.18 gives the phase domain loop behavior for a design with $\frac{\pi}{3}$ phase margin. As discussed in Chapter 2, a first order loop is able to follow a frequency step, up to a constant phase error, given by equation (2.11). The remaining phase error in regime is $\phi_{reg}(t) = \frac{2\pi f_{RF}}{G_{tot}} = 3.0354 \times 10^3$ which agrees with the simulation result.

Figure 4.19 is the result of a loop design with zero phase margin, or $G_{tot} = G_{tot,maxstable}$. The beatnote phase θ_{beat} oscillates around θ_{RF} with a constant amplitude, confirming the loop is operating at the edge of stability. For Figure 4.20, the frequency independent gain is chosen higher than the maximum stable gain $G_{tot} = 1.05 \cdot G_{tot,maxstable}$, resulting in the clearly unstable system of Figure 4.20.

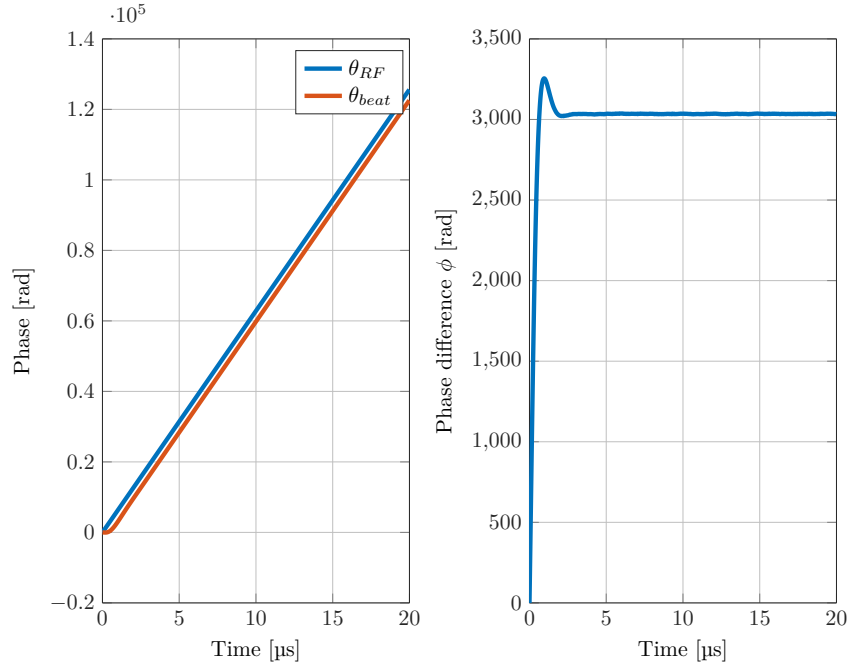


Figure 4.18: First order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $PM = \pi/3$ rad

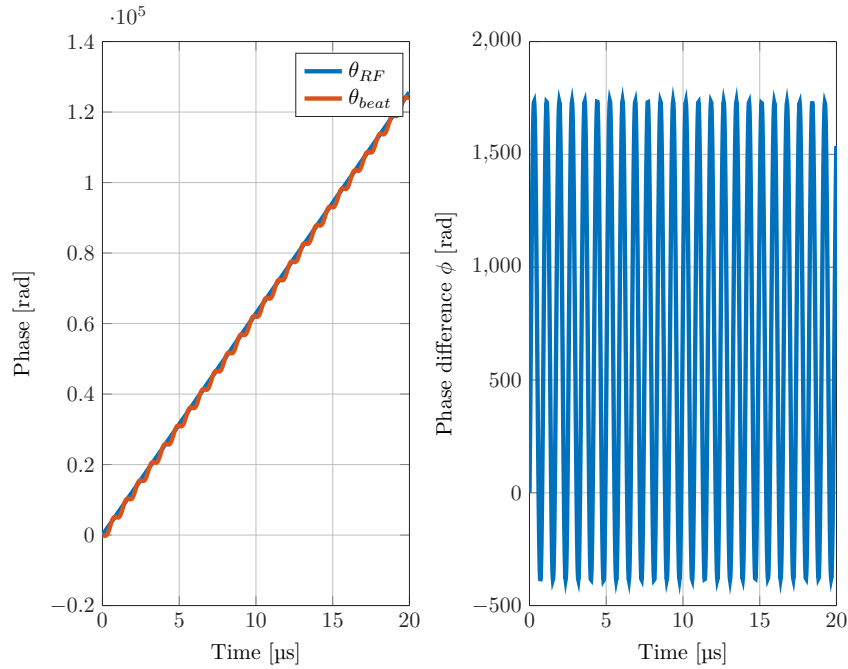


Figure 4.19: First order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $G_{tot} = G_{tot, maxstable}$

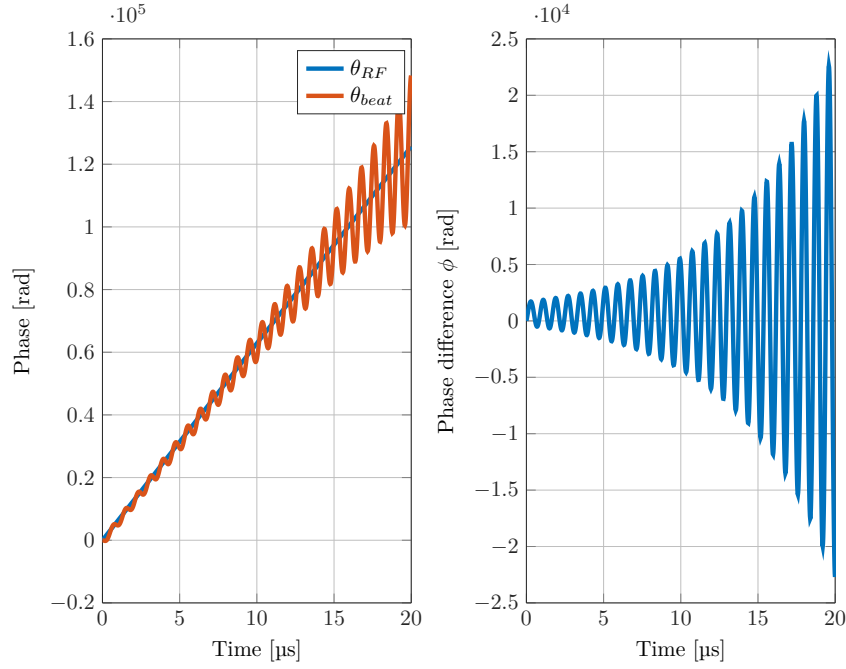


Figure 4.20: First order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $G_{tot} = 1.05 \cdot G_{tot, maxstable}$

Performance

Up to this point, the design parameter G_{tot} was chosen to meet a target phase margin. However, it would be more beneficial to optimize the frequency independent gain within the range of stable values to obtain maximal performance. The relevant performance metric depends on the application. Designing for maximal lock range, as well as minimal phase error in regime, is achieved by choosing the maximal stable gain $G_{tot} = G_{tot, maxstable}$. Designing for minimal residual phase error variance in lock, is done by minimizing equation (2.12).

The integral in (2.12), is not readily solved analytically, therefore numerical integration methods were used to investigate the influence of the frequency independent gain on the residual phase error variance. In Figures 4.21, and 4.22 the numerical solution of (2.12) is plotted as a function of the normalized stable gain. This is done for white frequency noise $S_n(\omega) = 2\pi \cdot 0.5 \times 10^6 \frac{\text{rad}^2}{\text{Hz}}$, and for pink frequency noise $S_n(\omega) = \frac{1 \times 10^{12}}{\omega} \frac{\text{rad}^3}{\text{Hz}}$ respectively.

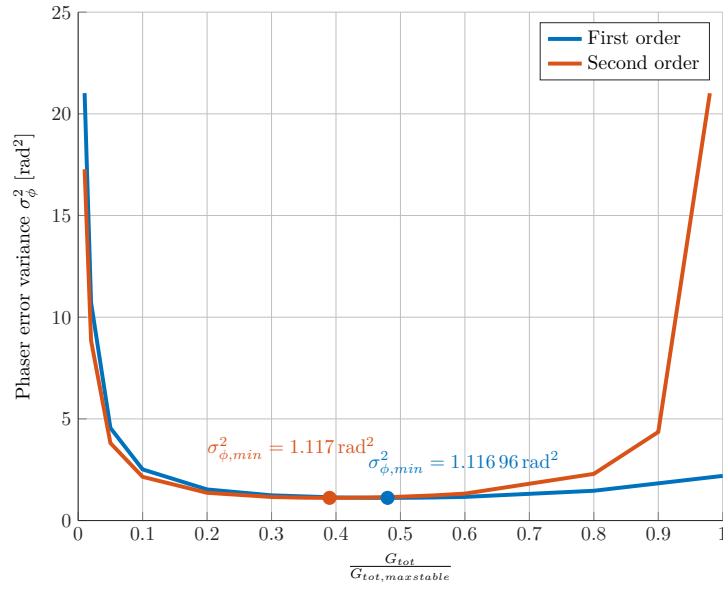


Figure 4.21: Phase error variance σ_ϕ^2 for different stable gain values G_{tot} , and white frequency noise

In the case of white frequency noise, the minimum phase error variance $\sigma_{\phi,min} = 1.117 \text{ rad}^2$ in lock is achieved for $G_{tot} = 0.48 \cdot G_{tot,maxstable} = 3.66 \times 10^6$. In the case of purely pink frequency noise, the minimum $\sigma_{\phi,min} = 0.195 \text{ rad}^2$ is achieved for $G_{tot} = G_{tot,maxstable}$.

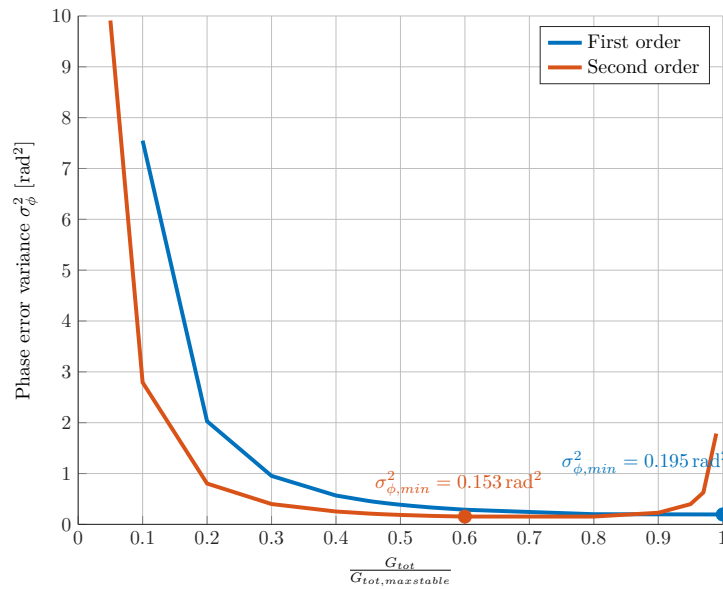


Figure 4.22: Phase error variance σ_ϕ^2 for different stable gain values G_{tot} , and purely pink frequency noise

The magnitude squared of the transfer function between the frequency noise n , injected into the loop, and the remaining phase error ϕ is plotted against frequency in Figure 4.23 ($G_{tot} = 0.48 \cdot G_{tot, maxstable}$). For the first order loop, $H_{\phi n}(j\omega)$ acts like a first order low pass filter with a resonance peak.

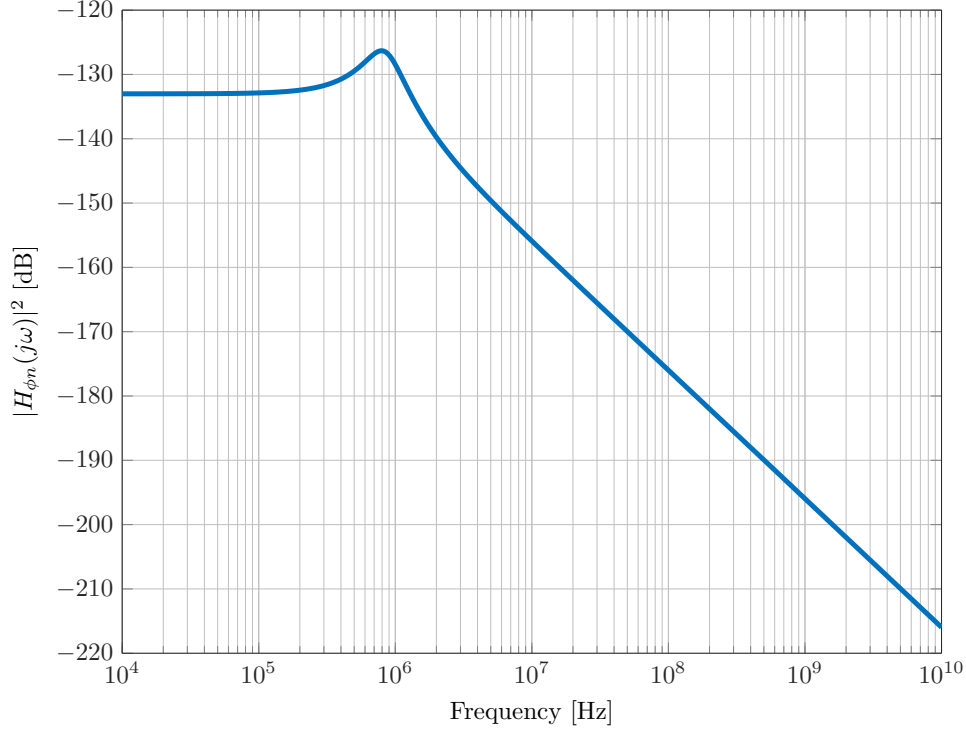


Figure 4.23: Phase error transfer function of first order loop

4.3.2 Second Order

The same approach is used to design a second order OPLL in the phase domain. A second integrator is introduced into the open-loop transfer function $OL(s)$ by the loop filter of equation (4.9). The integrator has to be accompanied by a zero to compensate for the extra $\frac{-\pi}{2}$ phase shift. This introduces a second design parameter f_1 . For stability reasons it is clear that $2\pi f_1 < \omega_{cr}$. For both frequency noise performance, and stability, f_1 is best chosen as small as possible. However the zero at lower frequencies will slow down the loop response, it will take longer for the phase error ϕ to become zero. To keep simulation times reasonable, the second order loop is tested with $f_1 = 20$ kHz, and the performance calculations and bode plots are made for the tabulated $f_1 = 100$ Hz as it improves the loop bandwidth and noise performance.

$$F(s) = \frac{1 + \tau_1 s}{s} = \frac{1 + \frac{s}{2\pi f_1}}{s} \quad (4.9)$$

The second order loop is tested with the same testbench of Figure 4.16, and with the same fixed parameters as with the first order loop. The fixed, target, and design parameters are given in Table 4.4. The design parameters are calculated from the target phase margin, and the fixed parameters via equations (4.10), and (4.11).

Fixed				Target	Design	
f_{LP}	f_{FM}	a	τ_d	PM	G_{tot}	f_1
1 GHz	1 MHz	1.5	1 ns	$\pi/3$ rad	1.3×10^9	100 Hz

Table 4.4: Second order loop parameters

$$\begin{aligned}
 -\omega_{cr}\tau_d - \arctan\left(\frac{\omega_{cr}\tau_{FM}}{a}\right) - \arctan(\omega_{cr}\tau_{FM}) - \arctan(\omega_{cr}\tau_{LP}) - \pi + \arctan(\omega_{cr}\tau_1) &= PM - \pi \\
 \Rightarrow \omega_{cr} &= 2.015 \times 10^6 \frac{\text{rad}}{\text{s}}
 \end{aligned}
 \tag{4.10}$$

$$G_{tot} = \sqrt{\frac{\omega_{cr}^4 (1 + \omega_{cr}^2 \tau_{LP}^2) (1 + \omega_{cr}^2 \tau_{FM}^2)}{(1 + \frac{\omega_{cr}^2 \tau_{FM}^2}{a^2}) (1 + \omega_{cr}^2 \tau_1^2)}} = 1.3 \times 10^9
 \tag{4.11}$$

The corresponding open-loop transfer function for the second order OPLL is represented by the bode plot of Figure 4.24. Again the stability is limited by the phase shift of $FM(j\omega)$ from the DFB laser. In Figures 4.25, 4.26, and 4.27 the stability calculations are verified with the SIMULINK testbench. When the phase margin is $\pi/3$ rad, the phase error in regime $\phi_{reg}(t)$ is equal to zero thanks to the second integrator introduced by $F(s)$. At the maximal stable gain, a marginally stable system is observed, and for $G_{tot} = 1.05 \cdot G_{tot,maxstable}$ the OPLL is clearly unstable.

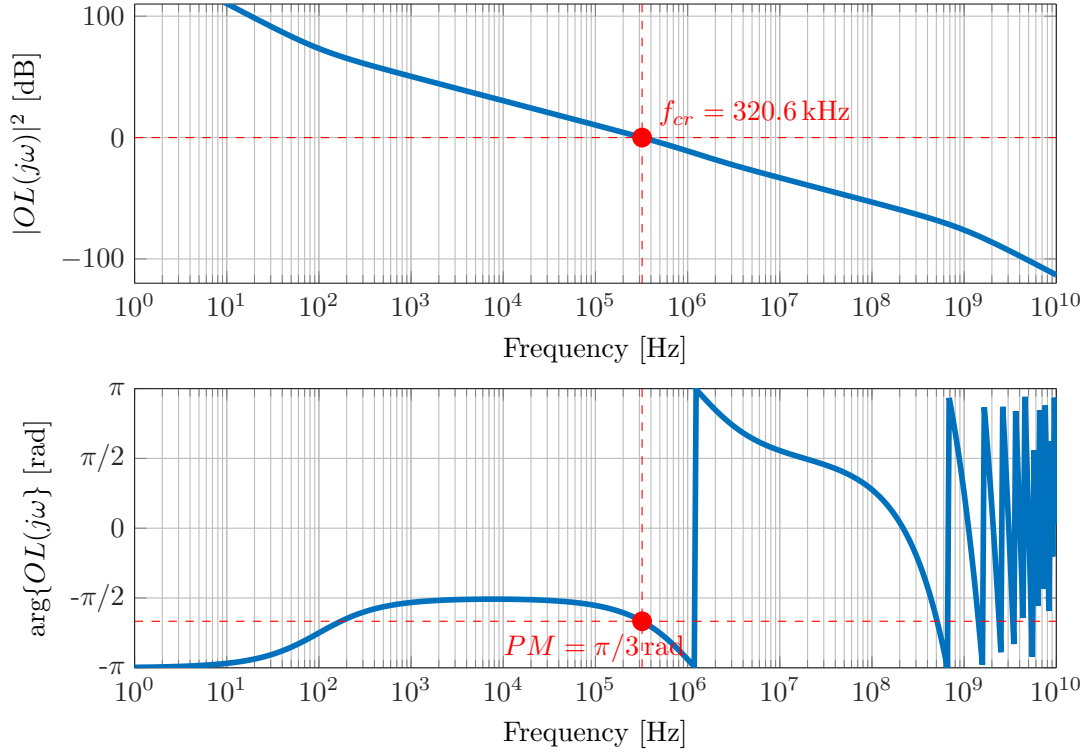
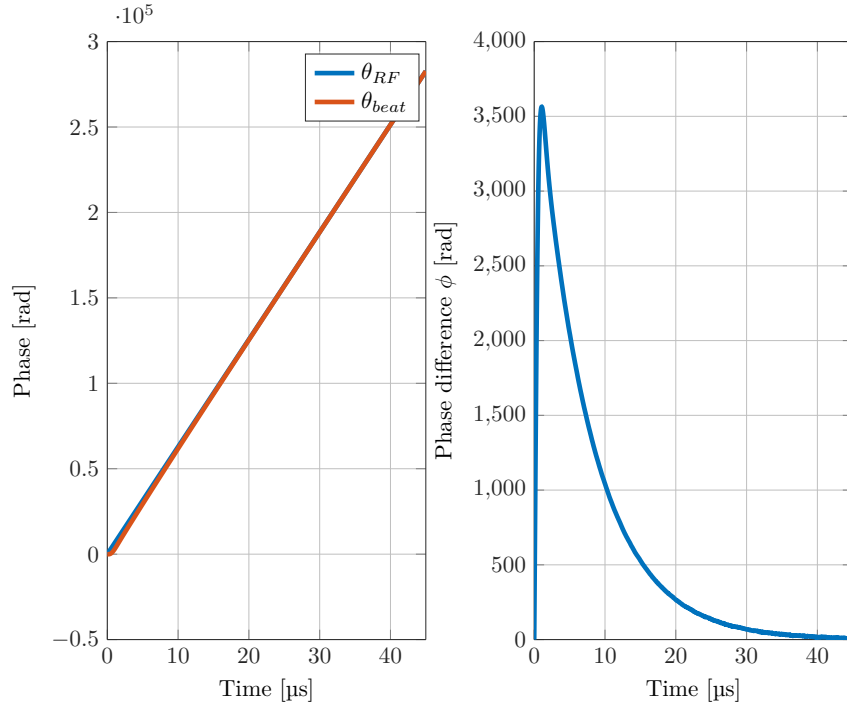


Figure 4.24: Bode plot of second order OL(s)

Figure 4.25: Second order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $PM = \pi/3$ rad

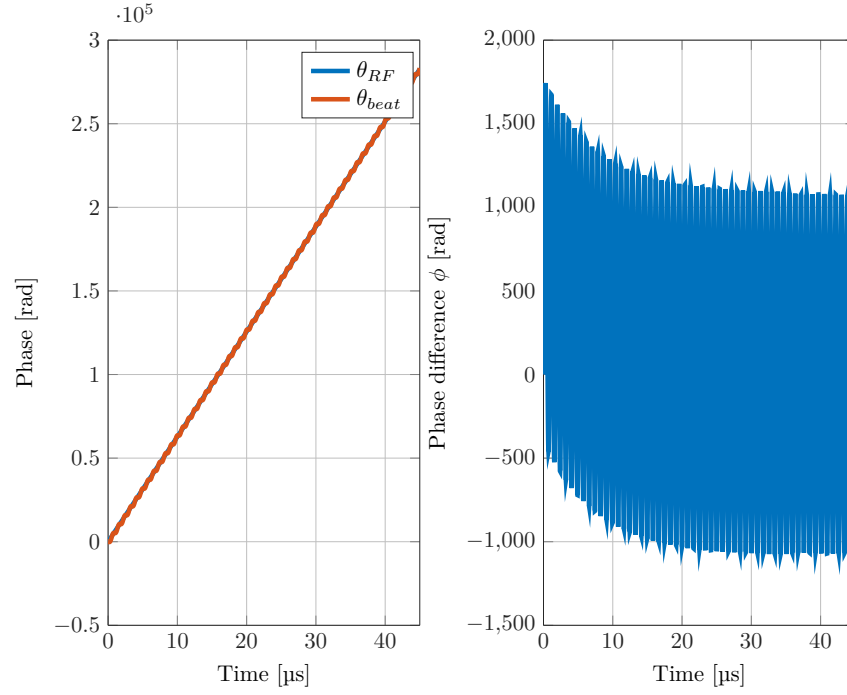


Figure 4.26: Second order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $G_{tot} = G_{tot,maxstable}$

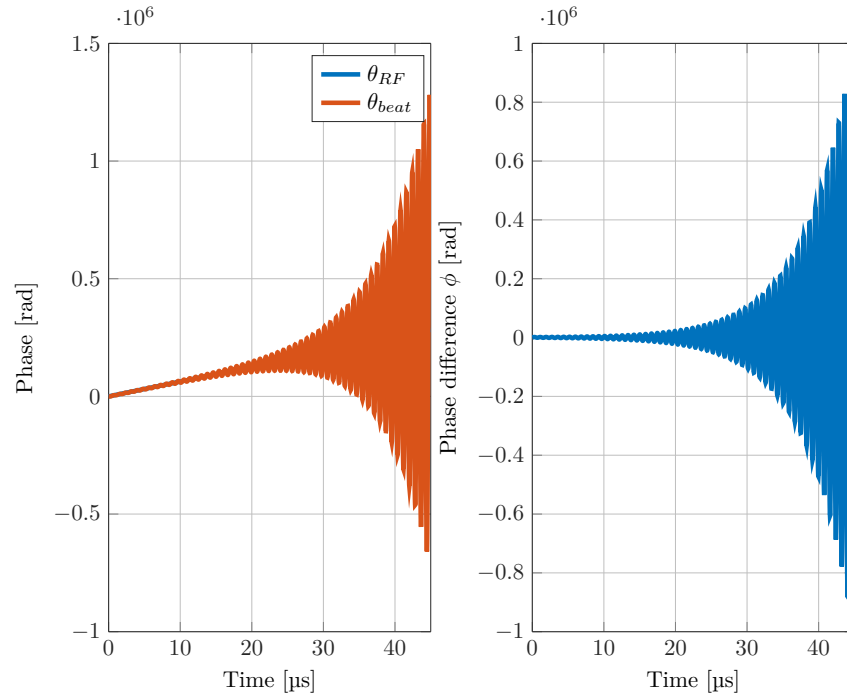


Figure 4.27: Second order phase domain OPLL responding to frequency step $\Delta f = 1$ GHz, with $G_{tot} = 1.05 \cdot G_{tot,maxstable}$

Performance

The phase error variance is calculated numerically in the same manner as for the first order loop. A sweep is performed over the stable gain values, while $f_1 = 100$ Hz. This is done for white frequency noise $S_n(\omega) = 2\pi \cdot 0.5 \times 10^6 \frac{\text{rad}^2}{\text{Hz}}$, and for pink frequency noise $S_n(\omega) = \frac{1 \times 10^{12}}{\omega} \frac{\text{rad}^3}{\text{Hz}}$ in Figures 4.21, and 4.22 respectively. In the case of white frequency noise, the second order OPLL design performs very similarly to the first order loop, and the minimal phase error variance $\sigma_{\phi, \min} = 1.117 \text{ rad}^2$ obtained for $G_{\text{tot}} = 0.39 \cdot G_{\text{tot, maxstable}}$ is the same as for the first order loop. However it is clear from Figure 4.22 that in the case of pink frequency noise, the second order loop outperforms the first order in terms of remaining phase error variance. A minimum $\sigma_{\phi, \min} = 0.153 \text{ rad}^2$ is obtained for $G_{\text{tot}} = 0.6 \cdot G_{\text{tot, maxstable}}$. The better performance for pink noise comes as no surprise when looking at the magnitude squared of the transfer function between n and ϕ given in Figure 4.28. The highpass behavior of $|H_{\phi n}(j\omega)|^2$ suppresses the $\frac{1}{f}$ frequency noise.

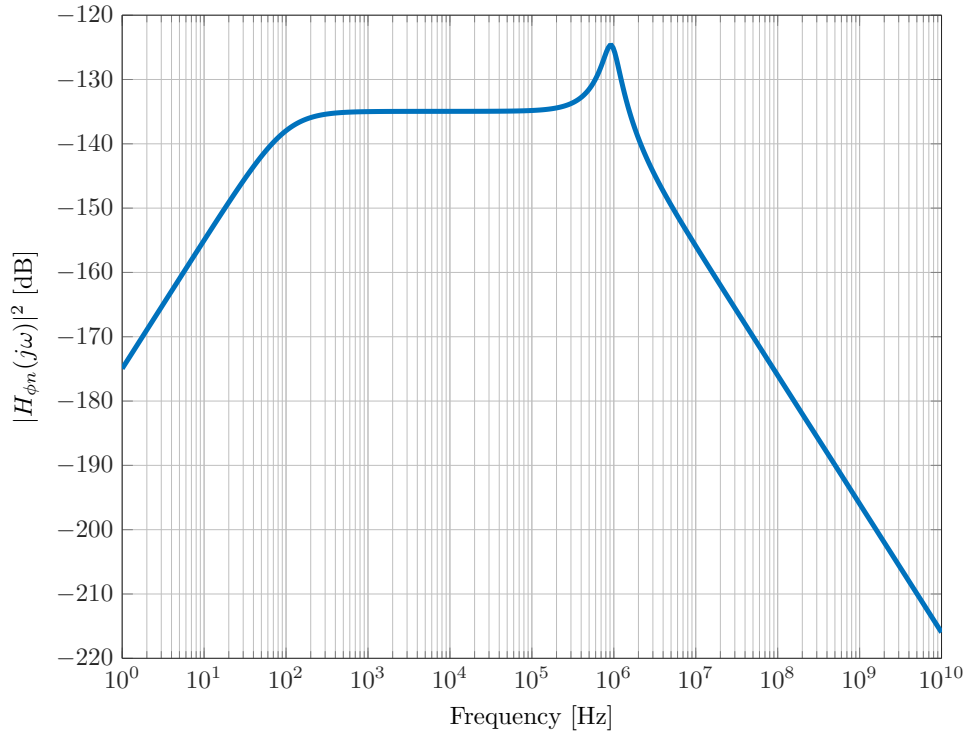


Figure 4.28: Phase error transfer function of second order loop

4.3.3 Compensation Filter

The loop bandwidth of the OPLL is limited by the FM response of the DFB slave laser. At f_{FM} , a phase shift of π rad is introduced by the change of FM mechanism (electronic instead

of thermal). One possibility to increase the loop bandwidth is with an extra compensation filter $C(s)$ given by equation (4.12). By choosing $f_a < f_{FM}$, and $f_b > f_{FM}$, a phase-bump can shift the $-\pi$ crossover to frequencies beyond f_{FM} [18].

$$C(s) = \frac{1 + \tau_a s}{1 + \tau_b s} = \frac{1 + \frac{s}{2\pi f_a}}{1 + \frac{s}{2\pi f_b}} \quad (4.12)$$

The bandwidth enhancement is illustrated in the open-loop bode plot of Figure 4.29, created with the parameter values of Table 4.5.

Fixed				Target	Design			
f_{LP}	f_{FM}	a	τ_d	PM	G_{tot}	f_1	f_a	f_b
1 GHz	1 MHz	1.5	1 ns	$\pi/3$ rad	1.3×10^9	100 Hz	0.5 MHz	1 GHz

Table 4.5: Second order loop with compensation filter parameters

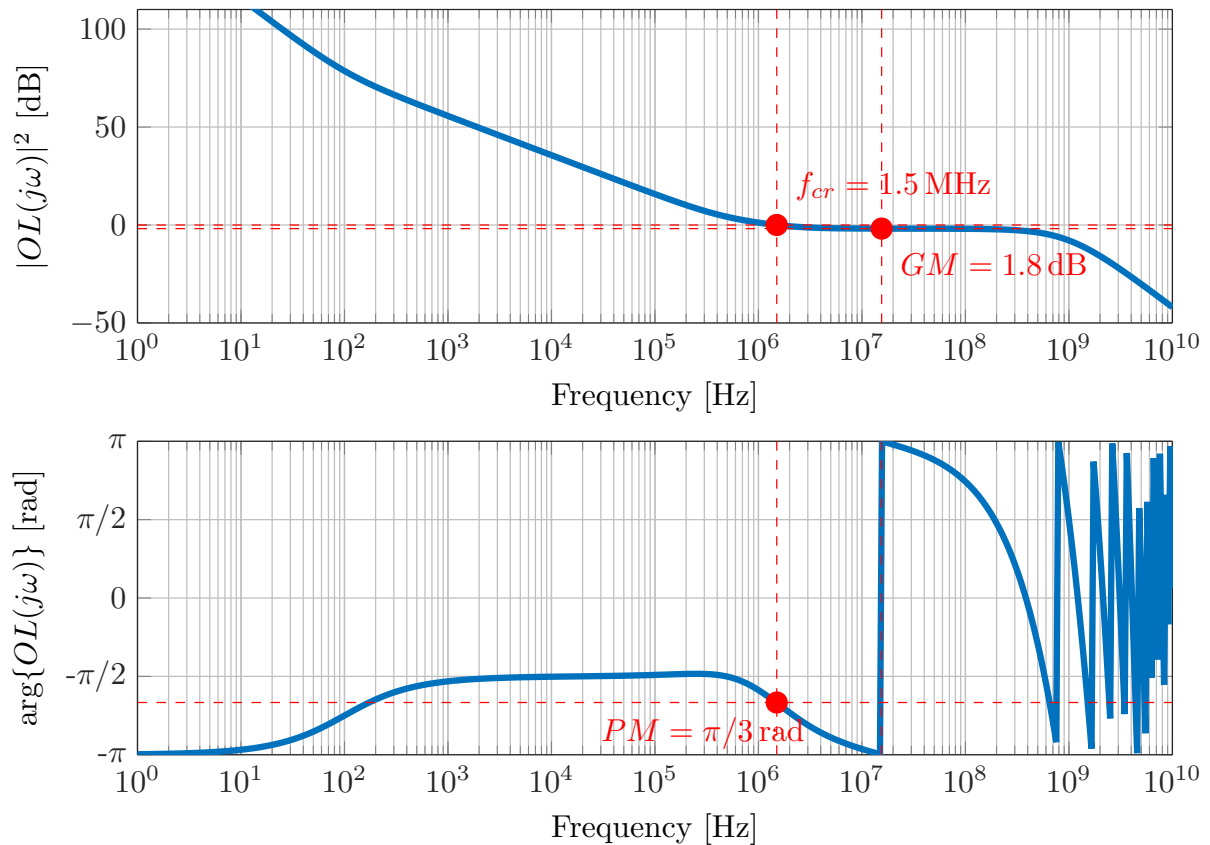


Figure 4.29: Bode plot of second order OL(s) with compensation filter

The critical frequency can be increased to $f_{cr} = 1.5$ MHz, however the zero at f_a that "bumps" the phase also cancels the slope of the magnitude plot. The consequence of increasing the

bandwidth with a compensation filter is a reduced gain margin ($GM = 1.8$ dB for this design), which is definitely a drawback as will become clear in the simulations of the complete system in section 4.4.

4.3.4 Double Loop

A more promising approach to increase the OPLL bandwidth, and increase the performance, is using a double loop architecture. To implement the second parallel loop, an external optical phase modulator (OPM) is added to the system. The idea is that the feedback to the slave laser will correct the frequency difference between beatnote and RF-offset, while the parallel feedback path with the external phase modulator can correct phase errors (within a finite interval) with a much larger bandwidth. The concept is shown in Figure 4.30.

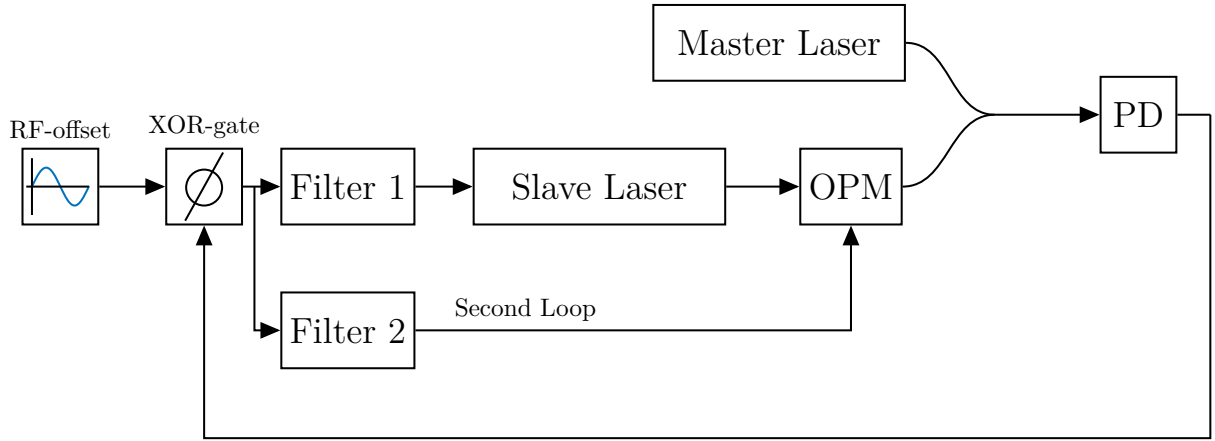


Figure 4.30: Conceptual double loop OPLL architecture

There are multiple options to implement an optical phase modulator. Electro-optical modulators based on Pockels cells, and dual drive Mach-Zehnder Interferometers operated in push-push mode are two valid options. This will not be further elaborated, as it is out of the scope of this work.

In a physical system, the OPM modulates the field of the slave laser before beating with the master laser. The OPM input is a filtered version of the phase error ϕ between RF-offset and beatnote. The double loop OPLL system in the phase domain can be described by the block diagram of Figure 4.31. The upper branch is the conventional OPLL without OPM, the lower branch models the OPM. Both branches have their own loop filters ($F_1(s)$ and $F_2(s)$), separate gain values (G_1 and G_2), and different loop delays ($\tau_{d,1}$ and $\tau_{d,2}$). The physical optical phase modulators can only change the phase over a finite interval, this is modelled by a saturation block in SIMULINK.

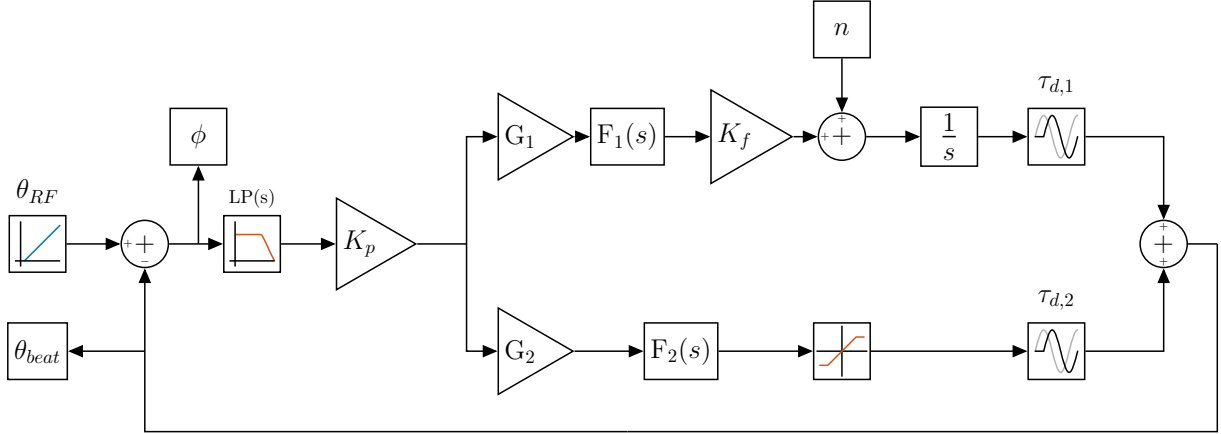


Figure 4.31: Phase domain double loop OPLL model testbench with frequency step

The open-loop transfer function is now given by equation (4.13), where the saturation is ignored. The double loop OPLL is a complicated system, ignoring the saturation effect during the design is not the only approximation or assumption that will be made in order to achieve a stable solution.

$$OL(s) = OL_1(s) + OL_2(s) = K_p LP(s) \left(\frac{G_1 K_f F_1(s) e^{-s\tau_{d,1}}}{s} + G_2 F_2(s) e^{-s\tau_{d,2}} \right) \quad (4.13)$$

The stability criterium expressed in terms of the open-loop transfer function in equation (2.8) suddenly becomes a lot less convenient when $OL(s)$ consists of a sum, illustrated by equation (4.14).

$$\begin{cases} |OL(j\omega_{cr})|^2 = |OL_1(j\omega_{cr})|^2 + |OL_2(j\omega_{cr})|^2 + 2\Re \left[\overline{OL_1(j\omega_{cr})} OL_2(j\omega_{cr}) \right] = 1 \\ \arg\{OL(j\omega_{cr})\} = \arg\{OL_1(j\omega_{cr}) + OL_2(j\omega_{cr})\} = PM - \pi \end{cases} \quad (4.14)$$

Previously, the choice of filters, and parameter values, could be deduced from the stability criterium. The double loop design proposed here, is a more ad-hoc solution based on the open-loop bode plots. $OL_1(s)$ is first designed separately as a second order loop described previously. $OL_2(s)$ is given by equation (4.15).

$$OL_2(s) = G_2 K_p LP(s) F_2(s) e^{-s\tau_{d,2}} = \frac{G_2 K_p (1 + \tau_2 s) e^{-s\tau_{d,2}}}{(1 + \tau_{LP} s)(1 + \tau_3 s)} \quad (4.15)$$

$OL_2(j\omega)$ does not have any perfect integrations, making $OL_1(j\omega)$ dominant for low frequencies. $OL_2(j\omega)$ should increase the bandwidth of the system, at a certain frequency the lower branch

of Figure 4.31 should become dominant, in such a way that the stability limiting FM behavior of the slave laser can be neglected. At the region of frequencies where $OL_1(j\omega)$ and $OL_2(j\omega)$ are of the same order of magnitude, their phases should not lay drastically apart to avoid unexpected phase jumps in $OL(j\omega) = OL_1(j\omega) + OL_2(j\omega)$. According to these intuitions, a stable solution was designed. The resulting bode plot is shown in Figure 4.32, where a loop bandwidth of $f_{cr} = 144$ MHz is achieved with a phase margin of $PM = \pi/3$ rad. The parameter values are tabulated in Table 4.6, where $G_{tot,1} = K_p K_f G_1$, and $G_{tot,2} = K_p G_2$.

f_{LP}	f_{FM}	a	$\tau_{d,1}$	$\tau_{d,2}$	$G_{tot,1}$	$G_{tot,2}$	f_1	f_2	f_3
100 MHz	1 MHz	1.5	1 ns	1 ns	1.21×10^{11}	433.8	10 kHz	20 MHz	100 kHz

Table 4.6: Double loop OPLL parameters

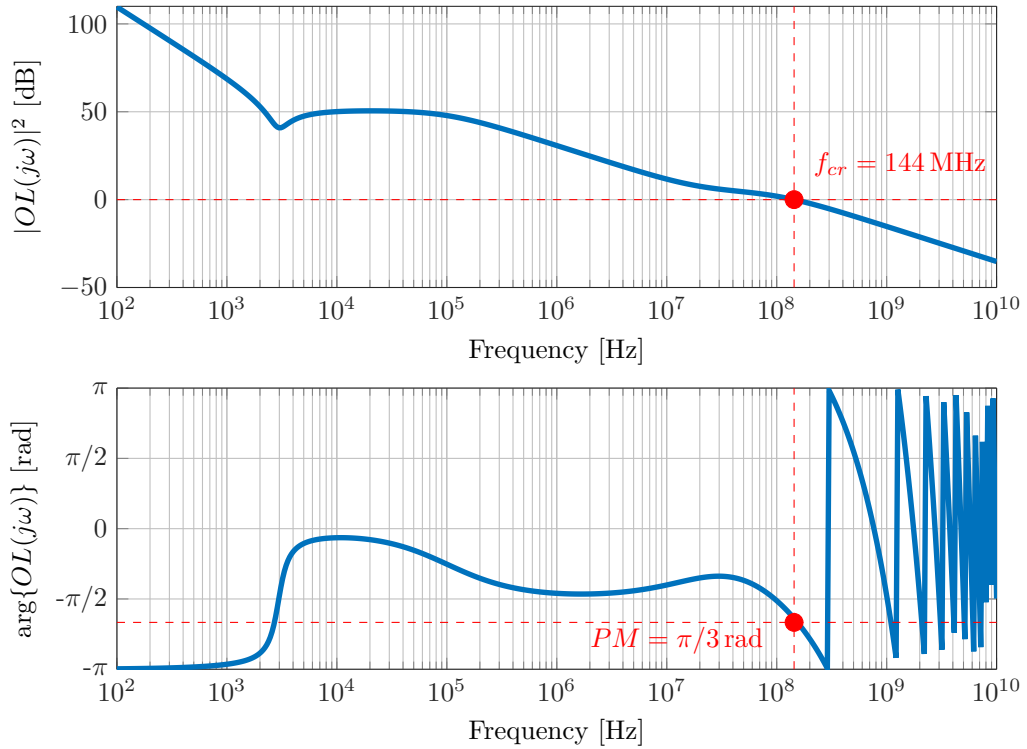


Figure 4.32: Bode plot of double loop OL(s)

The design is tested with the phase domain testbench of Figure 4.31. The result is plotted in Figure 4.33, confirming that the system is stable.

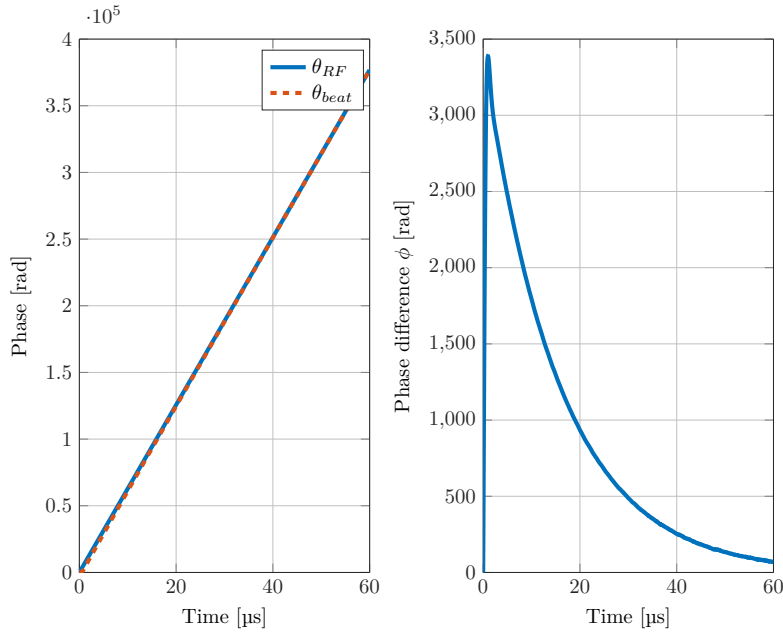


Figure 4.33: Double loop phase domain OPLL responding to frequency step, with $PM = \pi/3$ rad

4.4 OPLL System Model

All the components of the OPLL system have been modelled, and loop design is possible through the phase domain model. The complete system model is implemented in SIMULINK according to the block diagram of Figure 4.34. The four different loop architectures discussed in section 4.3 are tested in this complete system model.

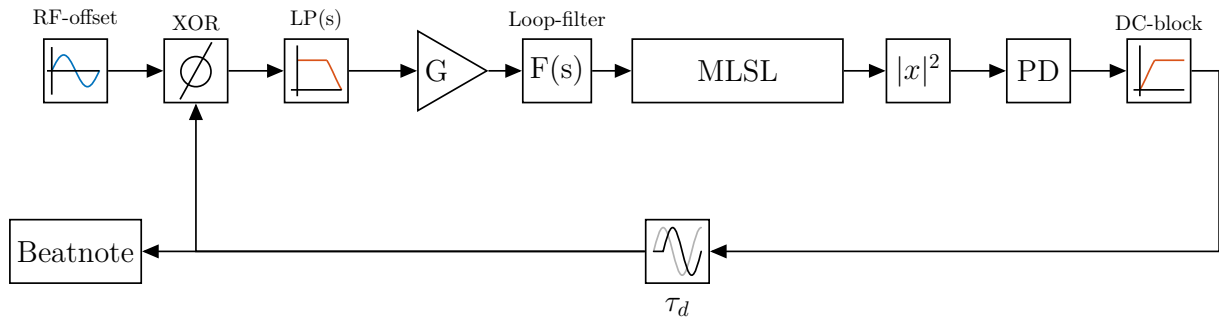


Figure 4.34: OPLL model testbench

The different designs discussed in the phase domain section are tried in the complete model to verify the match between theory and model. All the following simulations are performed with the same parameters from Table 4.7. Only the gain G , and the loop filter $F(s)$ differ between

the following designs.

MLSL					PD	XOR	RF-offset	Other
f_0	P_{ML}	K_f	FWHM_{ML}	f_{FM}	R_d	K_p	f_{RF}	τ_d
0 Hz	5 dBm	$-0.8 \frac{\text{GHz}}{\text{mA}}$	100 kHz	1 MHz	$0.63 \frac{\text{A}}{\text{W}}$	1	1 GHz + 300 kHz	1 ns
Δf	P_{SL}	K_a	FWHM_{SL}	a	f_c	f_{LP}	A_{RF}	f_{HP}
1 GHz	5 dBm	$0.1 \frac{\text{W}}{\text{A}}$	100 kHz	1.5	4 GHz	1 GHz	10 mA	50 kHz

Table 4.7: Fixed parameters of the OPLL system model simulations

4.4.1 First Order

The first order loop has $F(s) = 1$, and the design parameter $G = \frac{G_{tot}}{K_p K_f}$ is chosen, once to meet a phase margin of $PM = \pi/3$ rad: $G_{tot} = 2.07 \times 10^6$, and once for optimal frequency noise performance: $G_{tot} = 0.48 \cdot G_{tot, maxstable} = 3.66 \times 10^6$. The resulting power spectral densities of the beatnotes are plotted in Figures 4.35a, and 4.35b respectively.

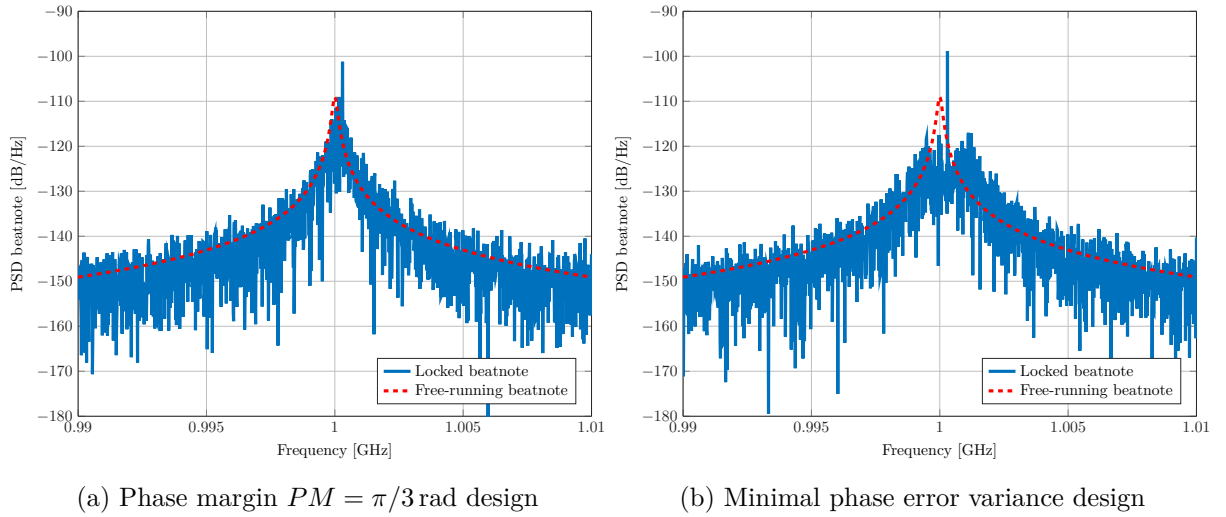


Figure 4.35: First order OPLL model simulation result

In both cases the beatnote locks to the RF-offset frequency f_{RF} which is 100 kHz higher than the free-running beatnote of 1 GHz. Designing for minimal phase error variance $\sigma_{\phi, min}^2$ also results in the slightly cleaner spectrum of Figure 4.35b.

4.4.2 Second Order

The second order loop is also tested for both the design of $PM = \pi/3$ rad, and the design for $\sigma_{\phi,min}^2$, resulting in Figures 4.36a, and 4.36b respectively. The plots are almost identical to the first order loop, as predicted by the white frequency noise performance of Figure 4.21.

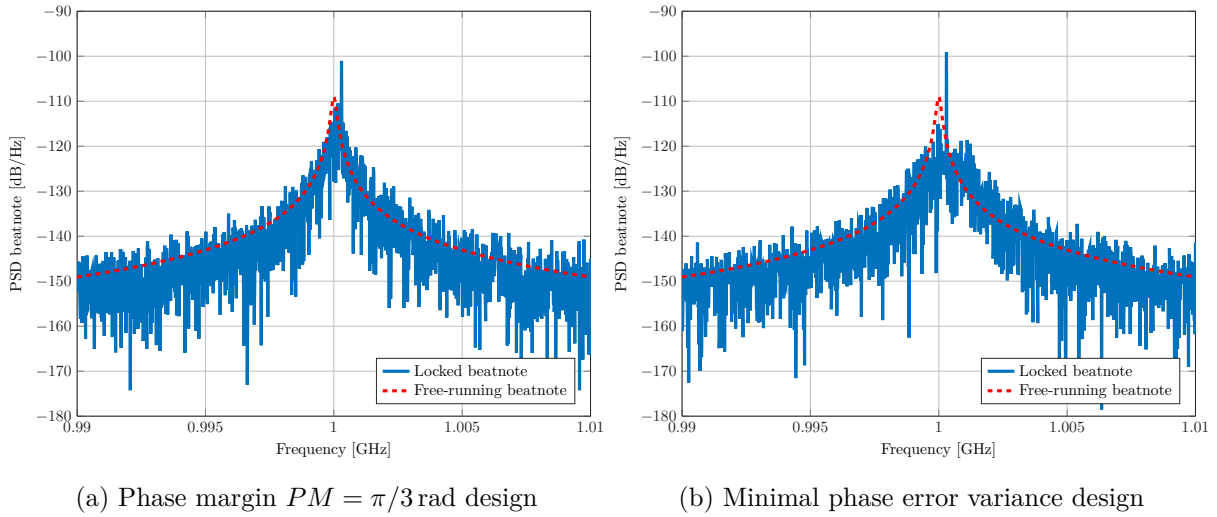


Figure 4.36: Second order OPLL model simulation result

4.4.3 Compensation Filter

In order to increase the loop bandwidth, and performance of the OPLL, a design with a compensation filter $C(s)$ given by equation (4.12) was proposed in the phase domain section. The critical frequency could be increased to $f_{cr} = 1.5$ MHz, but at the expense of gain margin. The phase domain design is tried with the complete OPLL system resulting in the beatnote PSD of Figure 4.37. The frequency noise is indeed reduced around the locked beatnote frequency, however the power in the tails of the originally Lorentzian lineshape is increased. This is a consequence of the nearly instable system. It is possible to use a compensation filter design with a higher gain margin of $GM = 3.8$ dB, and $f_{cr} = 1$ MHz for example, but even in that case building a stable system with real components will be challenging.

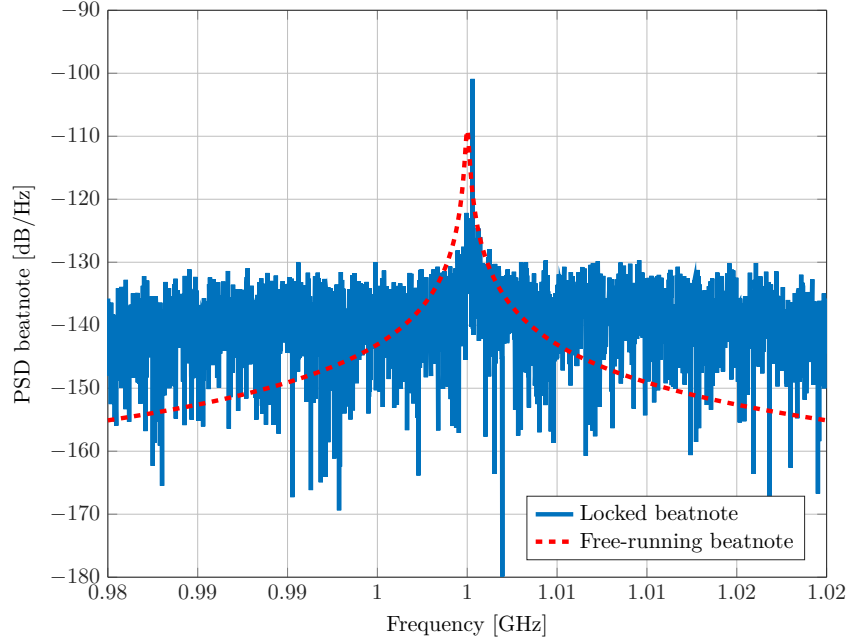


Figure 4.37: Second order OPLL model with compensation filter simulation result

4.4.4 Double Loop

The double loop OPLL architecture is implemented as a SIMULINK model according to the blockdiagram of Figure 4.38. The MLSL subsystem had to be slightly adjusted so the electric field of the slave laser can be modulated separately by an external OPM. The slave laser E-field E_{SL} is split into its magnitude and phase by a building block readily available in SIMULINK. The OPM output is added to the E_{SL} phase before recombination.

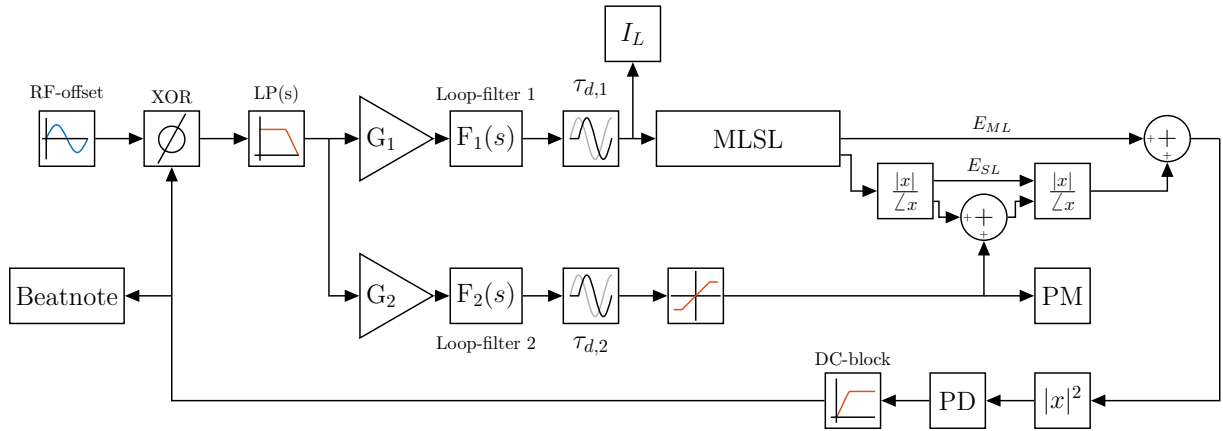


Figure 4.38: OPLL double loop model testbench

The used simulation parameters for the testbench of Figure 4.38 are given in Table 4.6, and Table 4.7. The OPM phase is allowed to vary in the interval $[-4\pi \text{ rad}; 4\pi \text{ rad}]$ by the saturation block. The saturation limits are chosen arbitrarily, they represent the finite range of phase modulations that is possible with the OPM. The saturation was ignored in the phase domain design, however it has some interesting consequences for loop behavior. The 144 MHz design is run in the OPLL simulation, and a stable lock is achieved. Figure 4.39a shows the input current of the slave laser, I_L achieves a steady DC-value to force the beatnote frequency to follow the RF-offset. The strange jitter that occurs randomly on top of the signal can be explained by looking at the output of the OPM branch in Figure 4.39b.

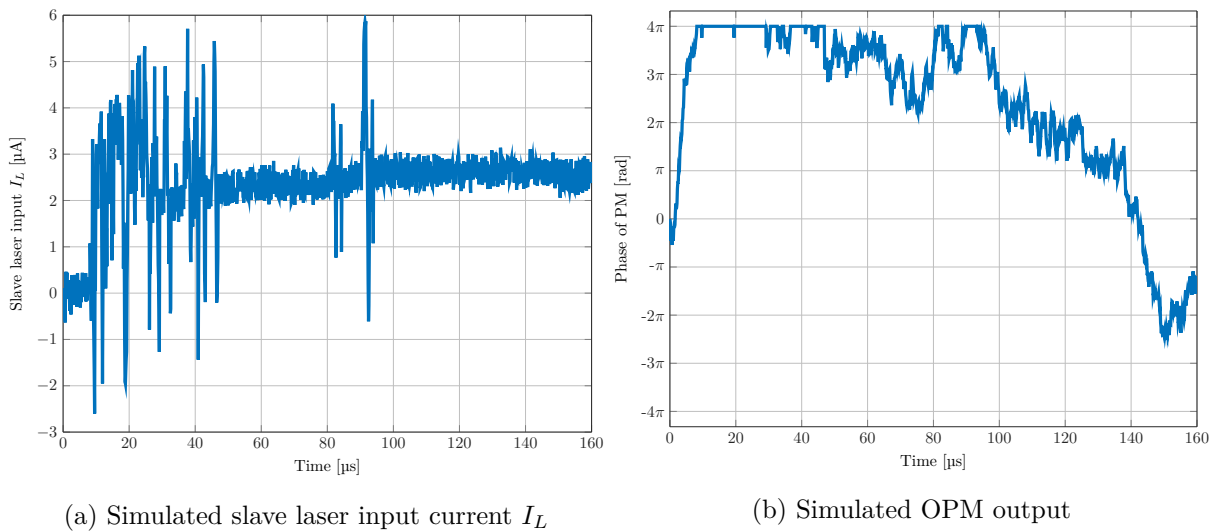


Figure 4.39: Double loop OPLL simulation signals

The second branch corrects for the random phase fluctuations within the bandwidth, however occasionally the saturation limits will be reached. When the OPM branch saturates, it basically shuts off, and the primary branch with the slave laser has to work alone until the phase error ϕ is back within the interval $[-4\pi \text{ rad}; 4\pi \text{ rad}]$. While the OPM branch is saturated, the remaining phase error ϕ increases, and translates into the peaks of the slave laser input current I_L . Figure 4.40 shows how this behavior translates in the beatnote spectrum. Figure 4.40a is the power spectral density calculated from the beatnote once the system is in stable lock. According to the I_L plot of Figure 4.39a this happens around $t = 60 \mu\text{s}$. The spectrum is already a lot cleaner than all the previous designs. However, in a time frame where no saturation occurs ($t \in [100 \mu\text{s}; 160 \mu\text{s}]$ for example), and both branches are simultaneously active, the beatnote spectrum of Figure 4.40b would be the result.

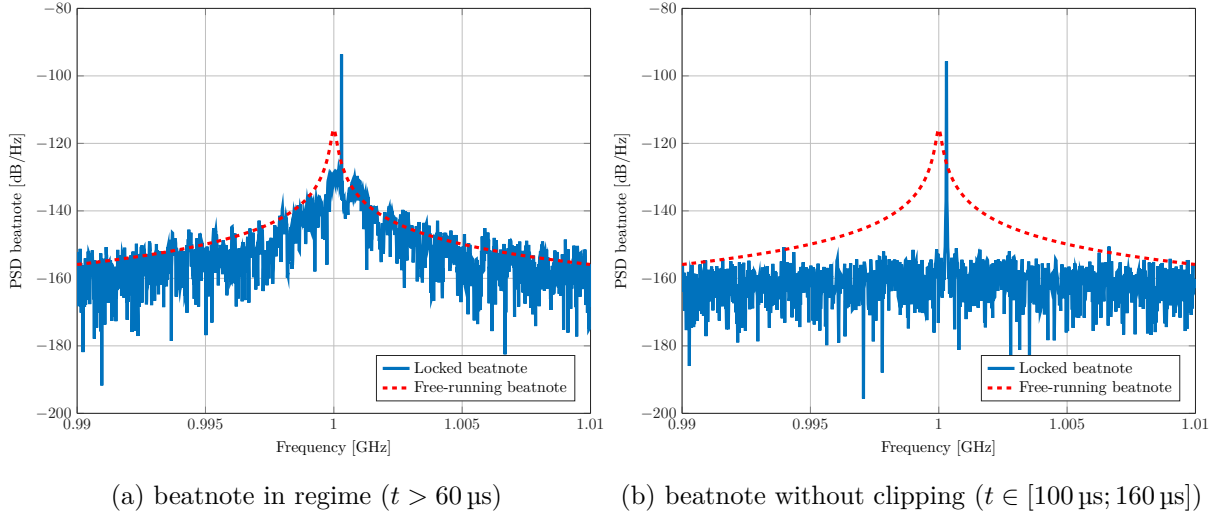


Figure 4.40: Double loop OPLL simulation beatnote PSD

Keeping the OPM saturations to a minimum, results in better performance of the double loop architecture. Using lasers with fine linewidths will make sure that the amount of phase noise injected in the loop is minimal, and random fluctuations beyond the saturation limits of the OPM will occur less often. Another course of action that would reduce the clipping, is finding a way to increasing the saturation limits of the OPM. Also, the design proposed here is not optimized in any way, the only goal was a stable system with some bandwidth enhancement. The double loop OPLL architecture looks very promising, and there is still a lot of room for improvement.

5

Building the System

With the help of the OPLL simulation tool created in SIMULINK, a real system will be designed, and built. To test the concept and the simulation tool, a simple second order loop with discrete components is tried first. Some parameters of the loop are determined by the available components in the lab. These constraints will dictate the design of the remaining parts of the loop.

5.1 Available Components

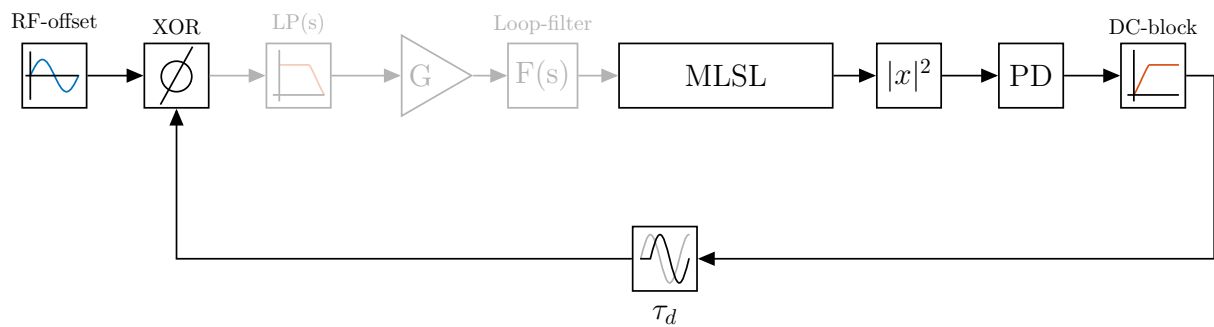


Figure 5.1: OPLL model available components

The highlighted blocks in Figure 5.1 are available as discrete components in the research lab, and were chosen out of convenience. Other options are possible, and potentially better.

5.1.1 Master Laser and Slave Laser

It was shown in Chapter 4 that a typical semiconductor DFB laser poses limitations on the achievable performance with simple first or second order loops, due to the FM response. However, the simplicity of frequency modulating with the input current, and the very compact implementation on a PIC, makes the DFB laser based OPLL a worthwhile subject of research.

Slave Laser

The ExOptronics EX-DY7005 High Speed Laser Module is an already packaged DFB laser, and will perform the role of slave laser in the OPLL system. Unfortunately the available datasheet is very limited, and no information regarding the FM response, nor the frequency noise is present. The K_f parameter in $[\frac{\text{GHz}}{\text{mA}}]$ determines the frequency sensitivity to the input current. It is essential for loop design, and is estimated as $K_f = -0.8 \frac{\text{GHz}}{\text{mA}}$ by performing a static current sweep, and looking at the resulting frequency drift. The cross-over frequency from thermal to electronic FM (accompanied by the π rad phase shift) f_{FM} is assumed to be 1 MHz, as that would be a typical value for a DFB laser. This is by no means the correct way to measure the FM behavior, but as time was limited, the decision was made to proceed with the OPLL system design without detailed knowledge of K_f , nor f_{FM} . A tunable gain in the circuit should make sure that a stable lock can be achieved.

Master Laser

A master laser is preferably tunable, and should have a low linewidth. A few master lasers can be considered, the most convenient choice is the tunable Santec C-band laser. Other tunable lasers, and a second EX-DY7005 DFB laser are also available to use as master laser.

5.1.2 Photodiode and TIA

The optical to electrical conversion will be performed by the high-speed labuddy with integrated transimpedance amplifier: DSCR409-LW from Discovery Semiconductors, Inc. The bandwidth of this photodetector is 40 GHz, and the conversion gain from optical input power to output voltage is $159 \frac{\text{V}}{\text{W}}$, this corresponds to the photodiode responsivity in the model.

5.1.3 XOR-gate Phase Detector

The HMC851LC3C 28Gbps XOR-gate from Analog Devices, is readily available on a PCB with differential inputs and output. The output of this XOR-gate is a block wave between $[-560\text{ mV}; -15\text{ mV}]$, this corresponds to $K_p = \frac{0.545}{\pi} = 0.1735$ in the model. K_p is gain associated with the slope of the linear parts of the XOR phase detector.

5.2 Designed Components

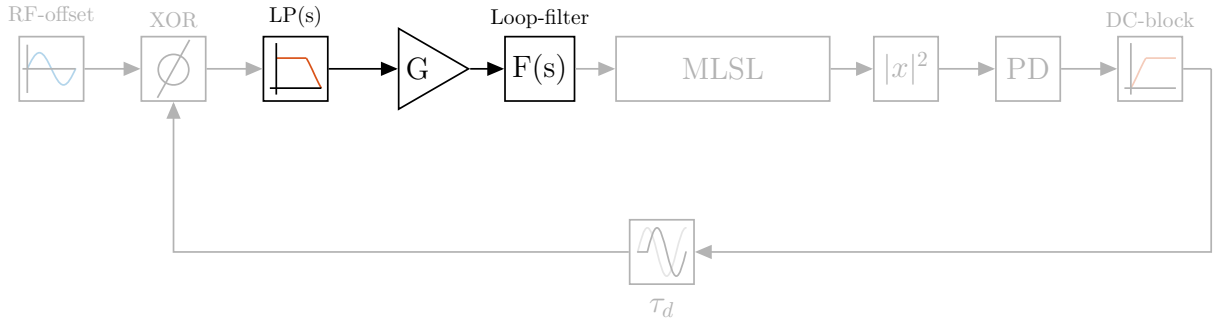


Figure 5.2: OPLL model components to design

To complete the loop highlighted in Figure 5.2, the XOR output voltage V_{XOR} has to be converted to the slave laser input current I_L according to the transfer function (5.1). An electronic circuit is designed to perform this task.

$$\frac{I_L}{V_{XOR}} = G \cdot LP(s) \cdot F(s) \quad (5.1)$$

5.2.1 System 1

In Chapter 4 it was shown that a second order loop (with an integrating loop filter $F(s)$) has some performance advantages in comparison with a first order loop. The static phase error in lock is zero, it offers better phase noise suppression for low frequency components, and a larger lock range. Because of these advantages, the first system that is designed, is a second order OPLL.

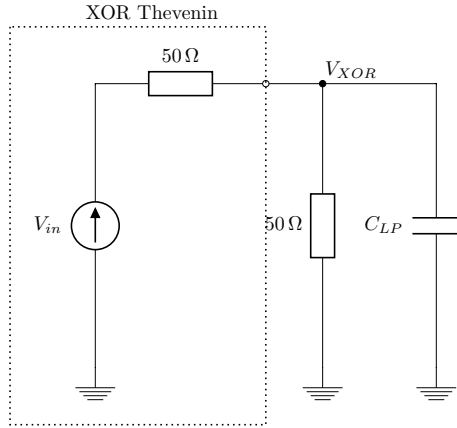
Function and Schematic

In the case of a second order loop, equation (5.1) can be written as equation (5.2). An electronic circuit that implements this transfer function is given by the schematic in Figure 5.5, it is built from three stages. The low-pass filter, the integrating loop filter, and the (tunable) gain in combination with an output stage that acts as a current source.

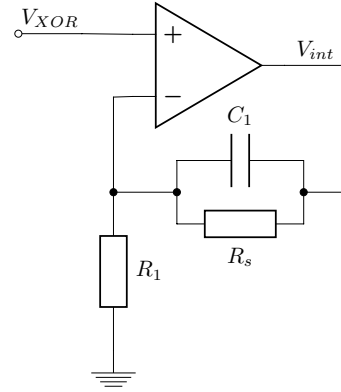
$$\frac{I_L}{V_{XOR}} = G \cdot LP(s) \cdot F(s) = \frac{G(1 + \tau_1 s)}{s(1 + \tau_{LP} s)} \quad (5.2)$$

The first stage is the first order low-pass filter. The XOR-gate has an output impedance of 50Ω , and the output voltage V_{XOR} in the datasheet assumes a 50Ω load. Figure 5.3a shows the schematic of the first stage, where the XOR-gate is represented by its Thévenin equivalent. Equation (5.3) shows the correct behavior, where the extra factor $\frac{1}{2}$ can be ignored, as the datasheet values of the output voltage of the XOR-gate corresponds to $\frac{V_{in}}{2}$.

$$\frac{V_{XOR}}{V_{in}} = \frac{1}{2} \frac{1}{(1 + 25\Omega C_{LP} s)} = \frac{1}{2} \frac{1}{(1 + \tau_{LP} s)} \quad (5.3)$$



(a) First stage $LP(s)$ schematic



(b) Second stage $F(s)$ schematic

Figure 5.3: System 1 circuit stages

The second stage implements the loop filter $F(s)$, the schematic is given by Figure 5.3b. The corresponding transfer function is given by equation (5.4), where the operational amplifier inputs are assumed to work as a nullator. The ideal integrator (with a pole at $s = 0$) is not possible, as every operational amplifier would saturate for very low frequencies and DC (the average input voltage $-15\text{ mV} \geq V_{XOR,avg} \geq -560\text{ mV}$). To ensure that the circuit can stay linear, the resistor

R_s is added in parallel to C_1 shifting the "integrator" pole to $s = \frac{1}{R_s C_1}$. The DC-gain $|F(0)|$ is no longer infinite, but still a lot higher than what is possible with a stable first order system.

$$\frac{V_{int}}{V_{XOR}} = \frac{R_1 + R_s}{R_1} \frac{1 + \frac{R_s R_1}{R_s + R_s} C_1 s}{1 + R_s C_1 s} \approx A \frac{1 + \tau_1 s}{s} \quad (5.4)$$

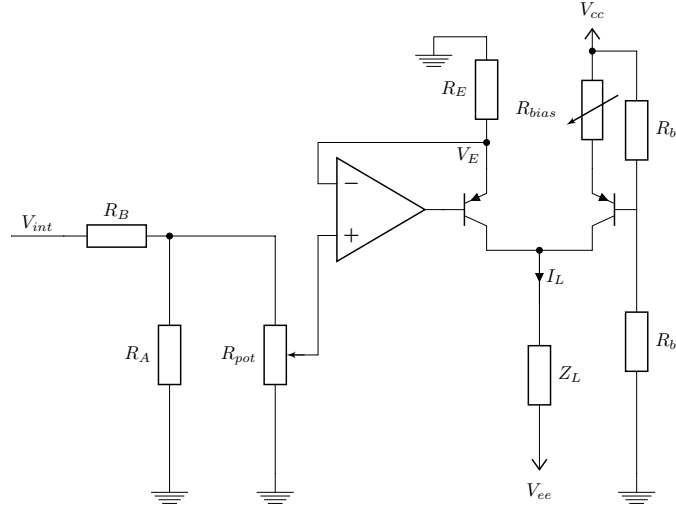


Figure 5.4: Output stage schematic

The final stage implements a tunable gain (which is actually an attenuation), and the voltage to current conversion. There is also an extra current source in parallel to set the bias current for the slave laser. The available slave laser has a dedicated bias port included in the package, so an external current source can be used to bias the laser. However, the functionality is still included in the circuit as it could allow the use of other slave lasers without the extra DC-bias input.

$$\frac{I_L}{V_{int}} = \frac{I_L}{V_E} \frac{V_E}{V_{int}} = \frac{-1}{R_E} \frac{R_A R_{p2}}{R_B (R_{p1} + R_{p2} + R_A) + R_A (R_{p1} + R_{p2})} \quad (5.5)$$

The output stage schematic is visible in Figure 5.4, the tunable gain is realised by potentiometer $R_{pot} = R_{p1} + R_{p2}$. By using the screw on the potentiometer, the resistance R_{pot} can be divided between R_{p1} and R_{p2} , with the limits $(R_{p1}, R_{p2}) = (0, R_{pot})$ and $(R_{p1}, R_{p2}) = (R_{pot}, 0)$. The operational amplifier works as a voltage follower, the pnp bipolar transistor performs the task of current source with high output impedance. The transfer function given by equation (5.5) does not include the bias current, which can just be added as both parallel paths have a very high output impedance.

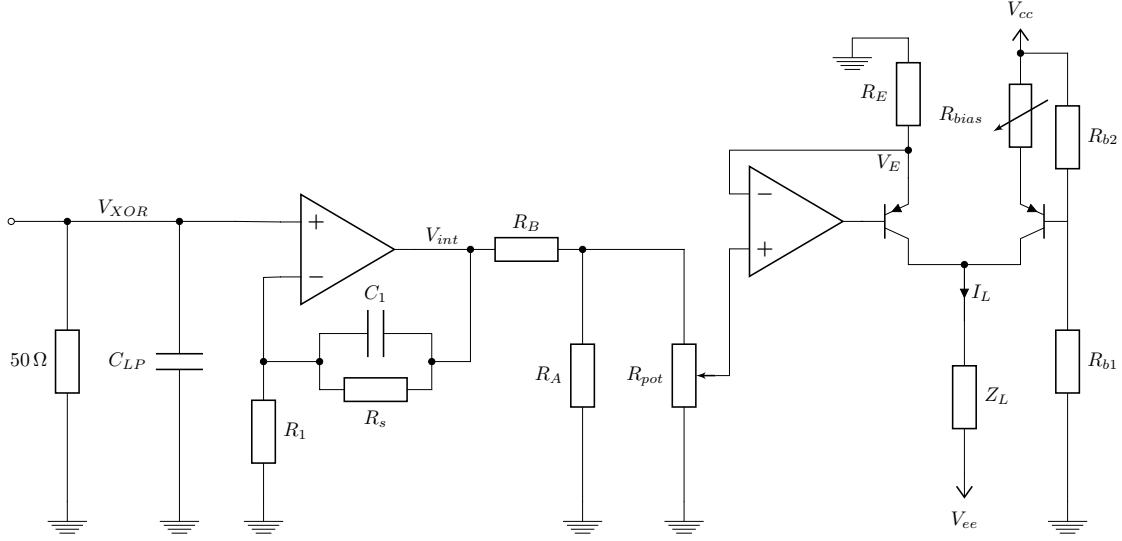


Figure 5.5: System 1 schematic

The total circuit is the concatenation of the tree stages. This can be done without a problem, as the second stage input is very high impedance, and the second stage output behaves like a voltage source with very low output impedance. The total transfer function is given by equation (5.6), which is the same as the wanted transfer function of equation (5.2), but with the "integrator" pole at a non-zero frequency $s = \frac{1}{\tau_s}$.

$$\begin{aligned} \frac{2I_L}{V_{in}} &= \frac{-R_A R_{p2} (R_1 + R_s)}{R_1 R_E (R_B (R_{p1} + R_{p2} + R_A) + R_A (R_{p1} + R_{p2}))} \frac{(1 + \frac{R_s R_1}{R_s + R_1} C_1 s)}{(1 + R_s C_1 s)(1 + 25 \Omega C_{LP} s)} \\ &= G \frac{1 + \tau_1 s}{(1 + \tau_s s)(1 + \tau_{LP} s)} \approx G \frac{1 + \tau_1 s}{s(1 + \tau_{LP} s)} \end{aligned} \quad (5.6)$$

ADS Circuit Design

The system 1 circuit is ready for design, the Advanced Design Systems (ADS) simulation tool is used in combination with the OPLL SIMULINK simulation tool of Chapter 4. The transfer function of equation (5.6) is implemented in the SIMULINK testbench of Figure 4.34 to verify that the chosen circuit elements result in a working OPLL. In ADS, the circuit should behave like the ideal schematic, and every op-amp feedback circuit has to be stable.

The starting point of the design is with the parameters determined by the available loop components given in Table 5.1. The loop delay $\tau_d = 15$ ns accounts for up to 3 m fiber propagation which will be the dominant source of delay in the OPLL consisting of discrete components. The FM response parameters f_{FM} and K_f are guessed, this uncertainty is covered by a tunable gain

with the potentiometer R_{pot} .

K_p	K_f	f_{FM}	τ_d
$0.1735 \frac{\text{V}}{\text{rad}}$	$0.8 \frac{\text{GHz}}{\text{mA}}$	1 MHz	15 ns

Table 5.1: Loop parameters from the available components

Based on the phase domain description, the remaining loop parameters are calculated to achieve a phase margin $PM = \pi/6 \text{ rad}$. The resulting parameters are given in Table 5.2, and the corresponding open-loop bode plot is given in Figure 5.6.

τ_{LP}	τ_1	τ_s	G_{tot}
15.9 ns	5.26 μs	0.47 μs	3.69×10^{-7}

Table 5.2: Calculated parameters for system 1

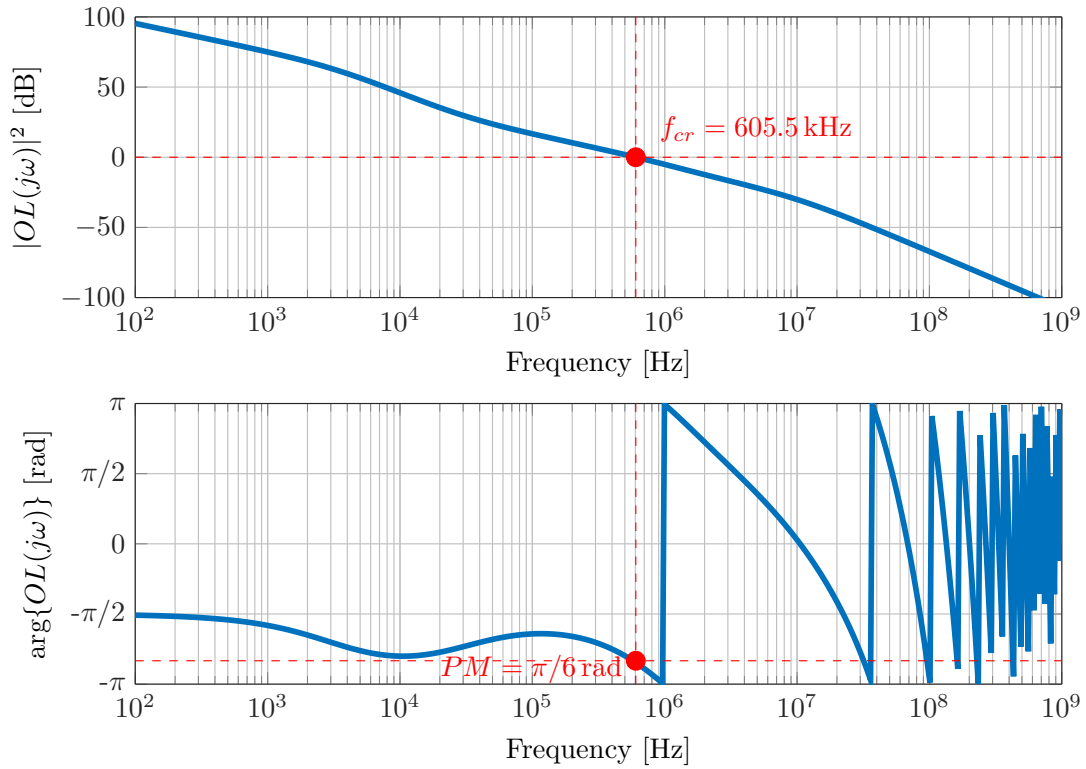


Figure 5.6: Bode plot of system 1 OL(s)

The parameters of Table 5.2 can be approximated with E24-series resistors and capacitors according to the formulas derived from the circuit schematic. The gain interval of G should be centered around $G = \frac{G_{tot}}{K_p K_f} = 3.04 \times 10^{-4}$. The resistor and capacitor values listed in Table 5.3

are chosen. The available gain values are in the interval $[0; 5.86 \times 10^{-4}] = [0; 1.77 \times G]$. The values of Tables 5.1 and 5.3 are implemented into the SIMULINK simulation. The resulting beatnote PSD in lock is plotted in Figure 5.7. The chosen design is able to phase lock in the simulation environment.

C_{LP}	C_1	R_1	R_s	R_A	R_B	R_{pot}	R_E
680 pF	470 pF	12 k Ω	100 k Ω	100 Ω	4.3 k Ω	1 k Ω	330 Ω

Table 5.3: Resistor and capacitor values of system 1

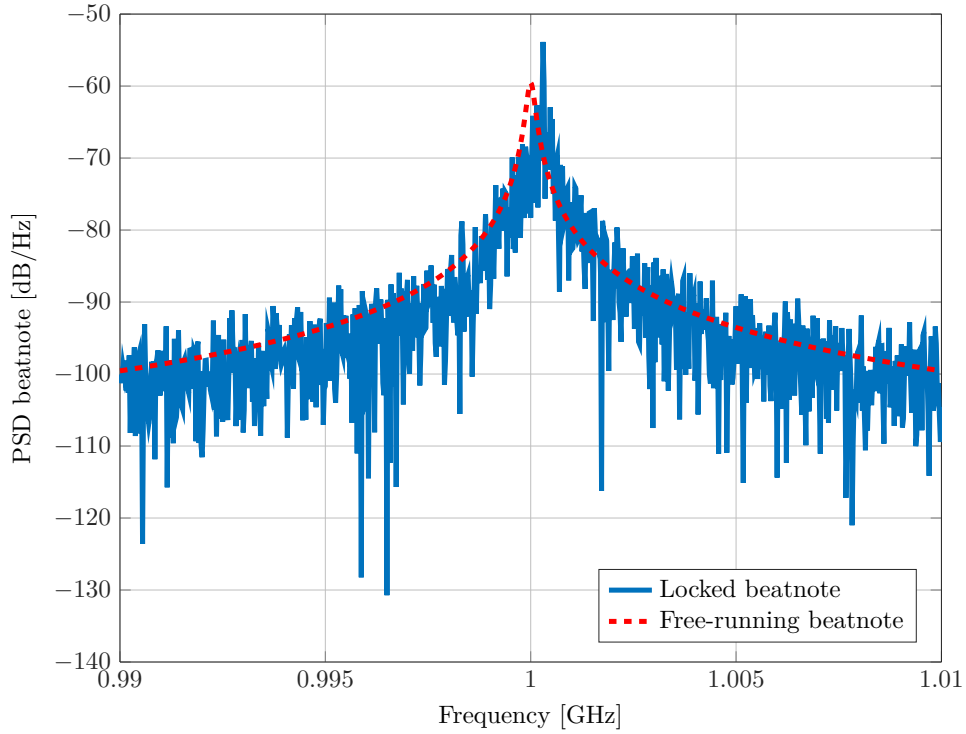
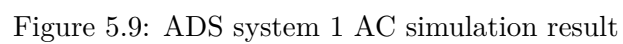
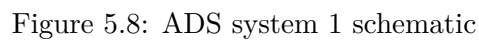
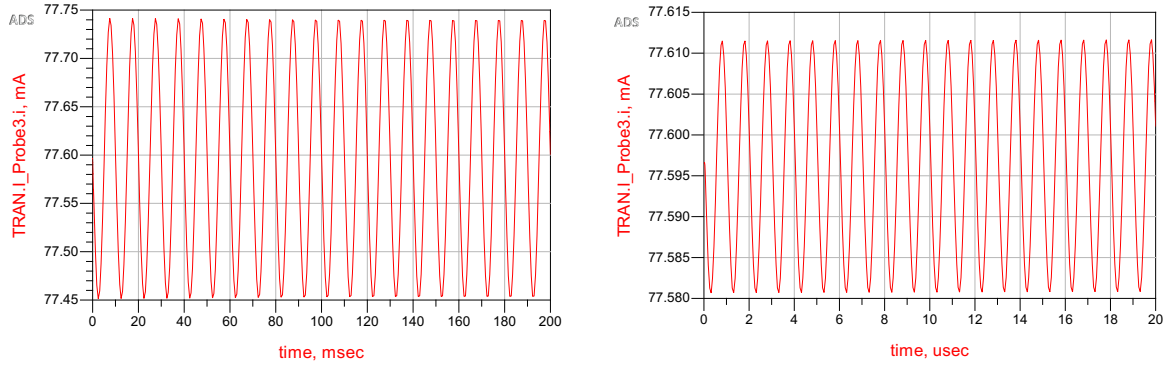


Figure 5.7: System 1 OPLL simulation result

The same values are implemented into the ADS schematic of Figure 5.8. Both operational amplifiers in the circuit are the AD845 Precision 16 MHz CBFET. These were chosen out of a database of available components, because they performed best in terms of stability. The pnp transistors are the BCX71 general purpose transistors. The AC-simulation results in Figure 5.9 confirm that the load current frequency response matches with formula (5.2). The marker on the plot of the magnitude of the transfer function $TF = \frac{2 \cdot I_L}{V_{in}}$ shows that $G = 3.024 \times 10^{-4}$ when the potentiometer is halfway ($R_{p1} = R_{p2}$). $V_{in}/2$ corresponds to the datasheet values of the XOR-gate output which assumes a 50 Ω load.

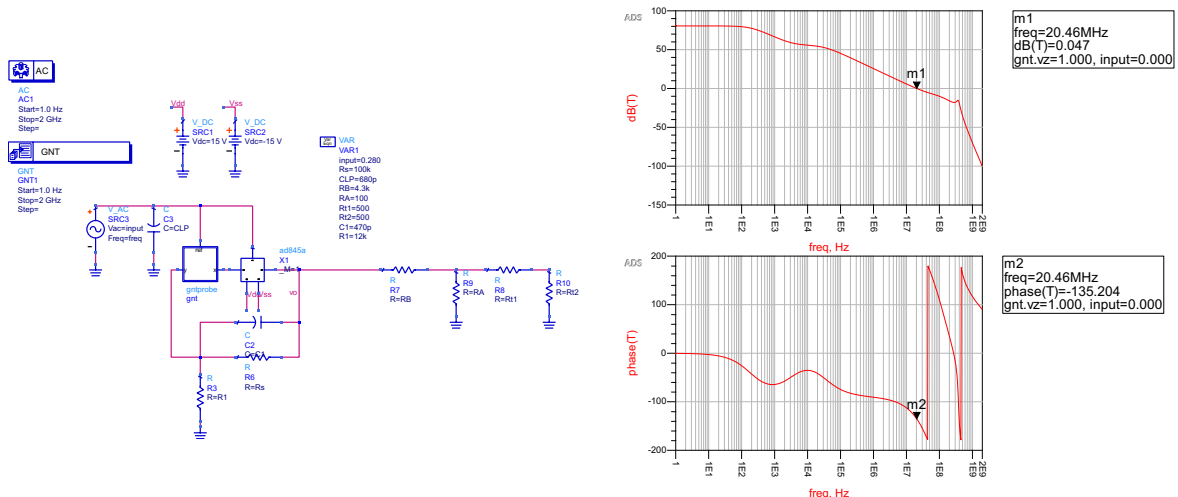




(a) ADS system 1 transient simulation at $f = 100\text{ Hz}$ (b) ADS system 1 transient simulation at $f = 1\text{ MHz}$

Figure 5.10: ADS system 1 transient simulation result

The result from the transient simulation for two different sinusoidal inputs is given by Figure 5.10a, and 5.10b at $f = 100\text{ Hz}$, and $f = 1\text{ MHz}$ respectively. The circuit is very linear for the complete frequency range. The two op-amps with feedback in the circuit are checked for stability according to the general network theorem (GNT) [19, 20] with the help of a GNT-probe in ADS. Figures 5.11a, and 5.11b show that the second stage op-amp feedback system is stable with a phase margin of 45° . Figures 5.12a, and 5.12b show that the op-amp feedback circuit of the output stage is stable with a phase margin of 62° .



(a) ADS system 1 stage 2 GNT schematic

(b) ADS system 1 stage 2 GNT plot

Figure 5.11: ADS system 1 stage 2 op-amp stability check

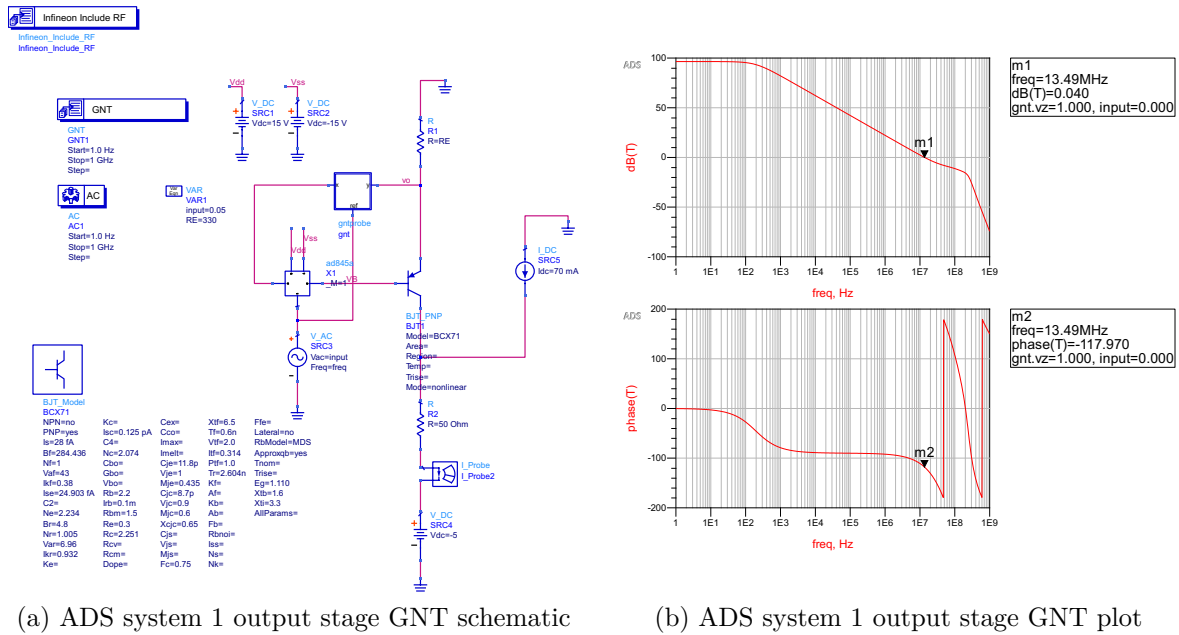


Figure 5.12: ADS system 1 output stage op-amp stability check

Printed Circuit Board

The circuit is implemented on a printed circuit board (PCB) with the lay-out software Eagle. The resulting lay-out is given in Figure 5.13. The ground plane is at the top side, at the bottom side there is a 15 V power plane, and a -15 V power plane for the op-amp supply. There is also a small plane for V_{ee} , this negative voltage makes sure that the pnp bipolar transistor can operate in the active region. In the output stage schematic of Figure 5.4, the collector voltage is determined by $V_{ee} + I_L Z_L$ and should stay below the negative base voltage. If the bias current of $I_{bias} = 70$ mA is delivered by the PCB, $V_{ee} = -5$ V is ok. If the laser is biased with an external source, $V_{ee} = -1$ V is already enough.

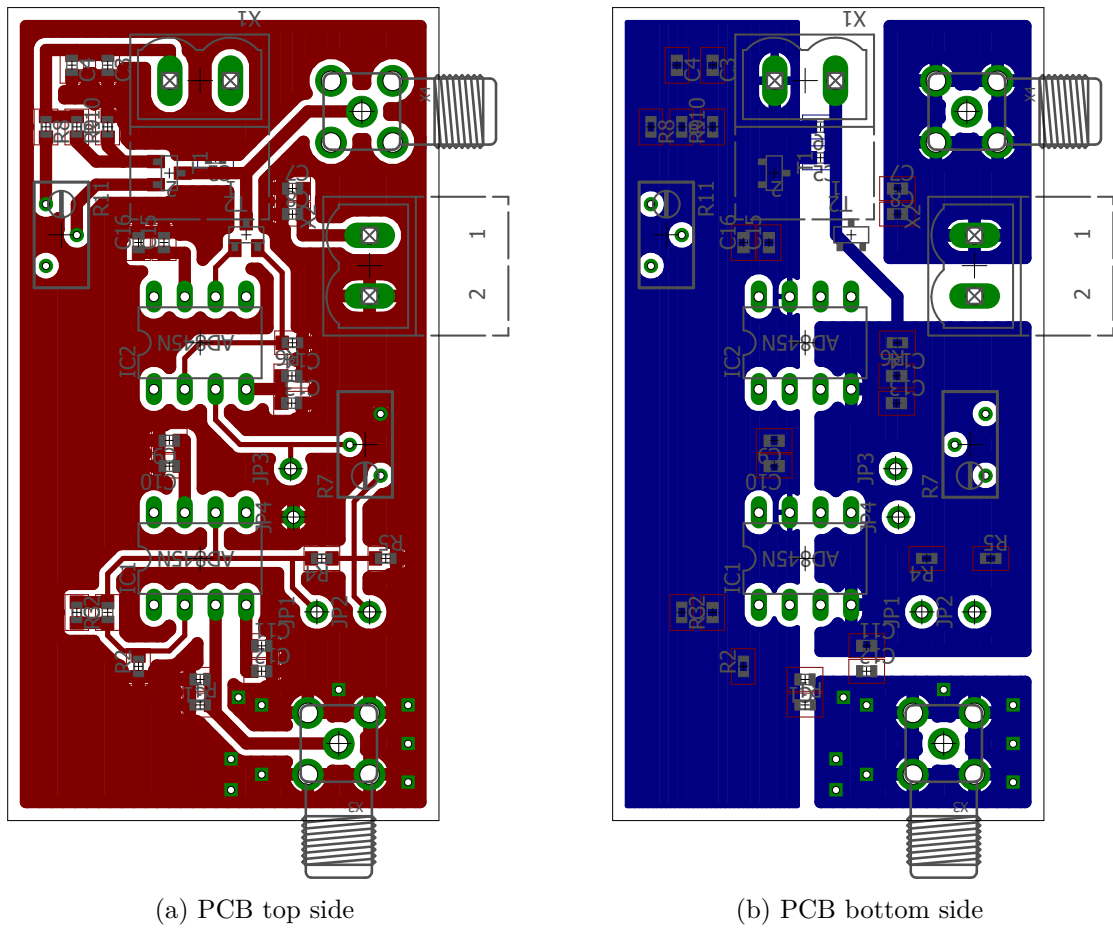
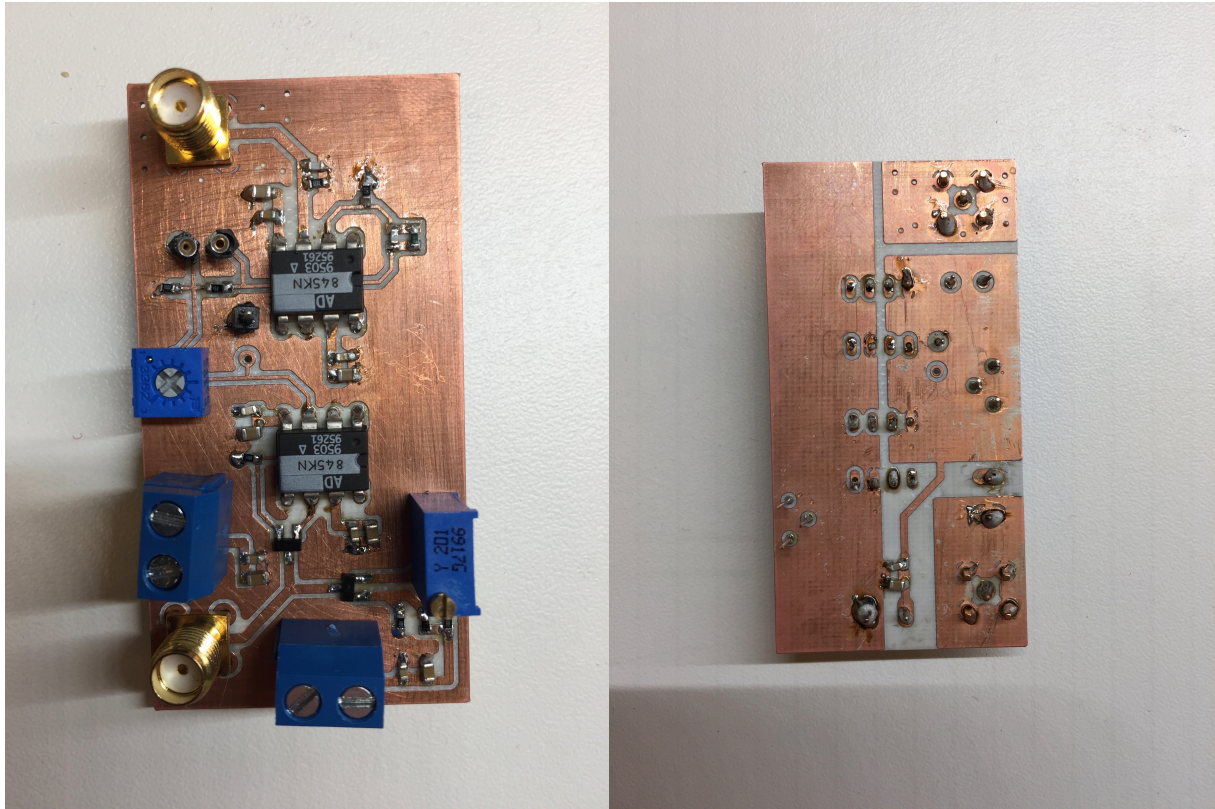


Figure 5.13: Printed circuit board lay-out of system 1



(a) PCB top side

(b) PCB bottom side

Figure 5.14: Printed circuit board of system 1

The lay-out is fabricated, and all the components are soldered in place, pictures of the resulting finished PCB are visible in Figure 5.14.

PCB Measurements

To test the PCB, the measurement setup from Figure 5.15 was built. Two signal generators with frequencies f_{RF1} and f_{RF2} are connected to the input ports of the XOR-gate, the XOR output is connected to the input of the PCB, and the PCB output is terminated with a 50Ω load (The slave laser input impedance is also 50Ω according to the datasheet). The voltage over the load V_L is measured with an oscilloscope. The output current is equal to $I_L = \frac{V_L}{50\Omega}$ and is plotted in Figure 5.16.

The measured triangle waves verify that the XOR phase detector with low-pass filter works as anticipated in Figure 2.3b. Figure 5.16a confirms that the tunable gain with the potentiometer works. The Figures 5.16b, 5.16c, and 5.16d are measurements at different frequencies, before the "integrator" pole $s = \frac{1}{\tau_s}$, halfway the "integration", and after the zero $s = \frac{1}{\tau_1}$ respectively. The amplitudes behave according to the frequency response of the AC simulation in Figure 5.9.

Note that currents are very small, this is a consequence of extremely large frequency modulation factor K_f of the slave laser $K_f = -0.8 \frac{\text{GHz}}{\text{mA}}$, making frequency modulation very sensitive.

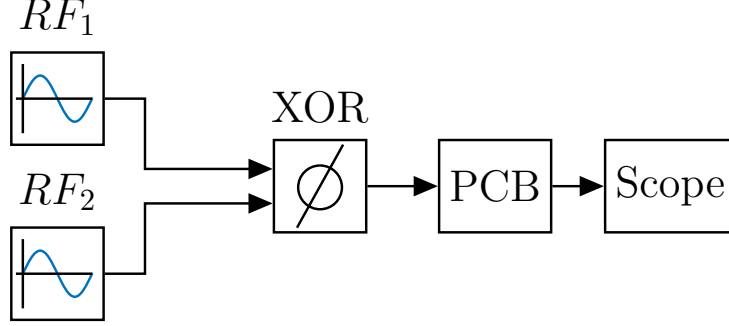
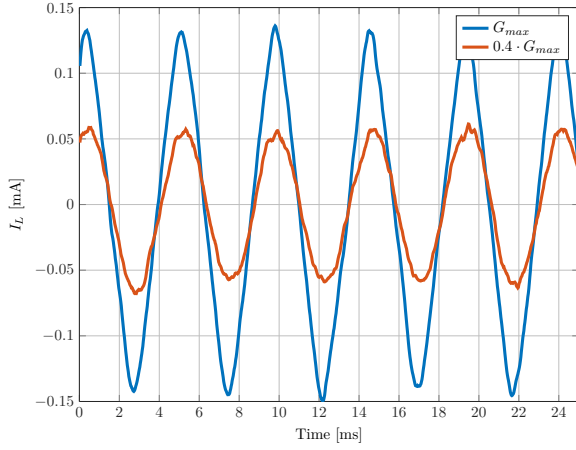
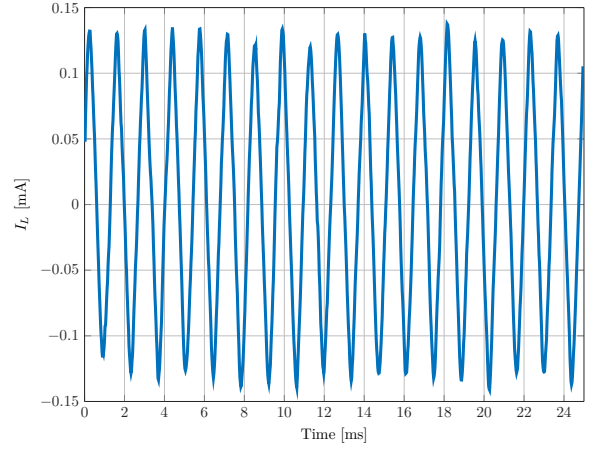


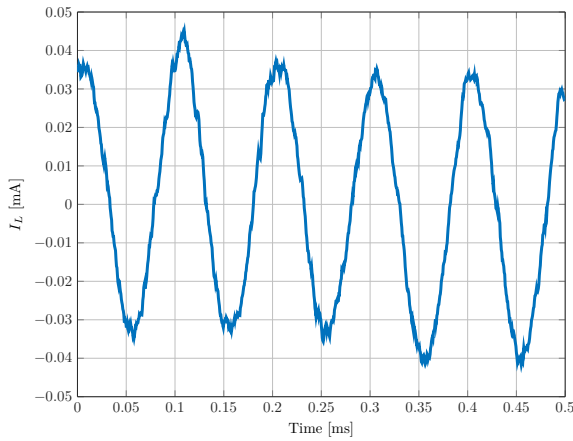
Figure 5.15: Measurement setup to test PCB behavior



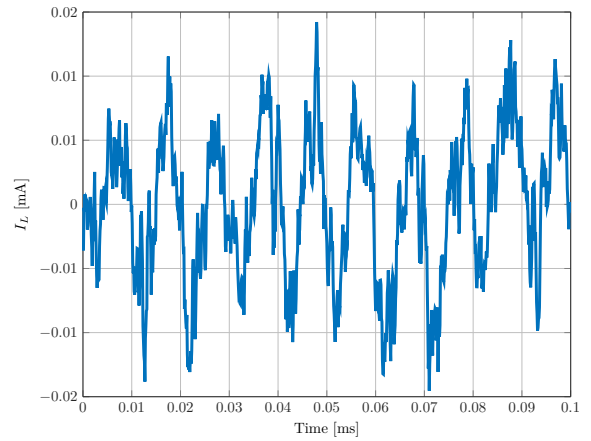
(a) $|f_{RF1} - f_{RF2}| = 215 \text{ Hz}$, 2 different gain values



(b) $|f_{RF1} - f_{RF2}| = 741 \text{ Hz}$, G_{max}



(c) $|f_{RF1} - f_{RF2}| = 10 \text{ kHz}$, G_{max}



(d) $|f_{RF1} - f_{RF2}| = 100 \text{ kHz}$, G_{max}

Figure 5.16: Measurement results from the setup of Figure 5.15

5.2.2 System 2

Unfortunately, the OPLL system 1 did not work. While the reasons for the failure were still unclear, a first order system was designed. This time, the possibility for higher gain, and a larger loop bandwidth is used. The FM response of the slave laser is still unknown, if a configuration with higher loop bandwidth could be stable, it would make phase locking more efficient. This system 2 follows the same design process, and the same PCB lay-out is used, as there was a double manufactured. The second stage is replaced by an op-amp voltage follower instead of the integrating loop filter.

Function and Schematic

The low-pass filter, the tunable attenuation, and the voltage to current conversion is realised by a circuit with the schematic of Figure 5.17. The schematic is very similar to the schematic of the second order system 1, the corresponding transfer function is given by equation (5.7).

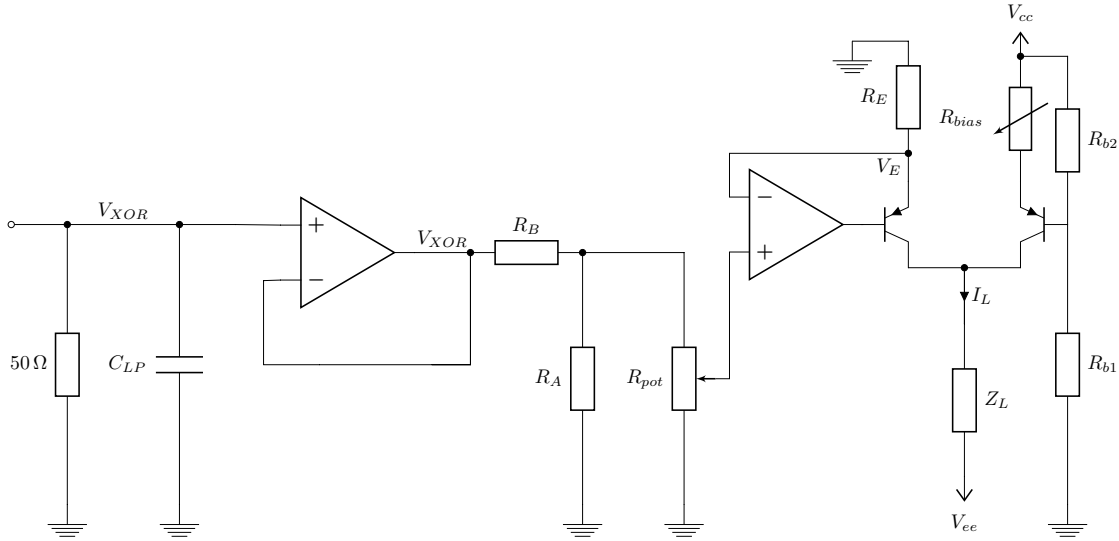


Figure 5.17: System 2 schematic

$$\begin{aligned}
 \frac{2I_L}{V_{in}} &= \frac{-R_A R_{p2}}{R_E (R_B (R_{p1} + R_{p2} + R_A) + R_A (R_{p1} + R_{p2}))} \frac{1}{(1 + 25 \Omega C_{LP} s)} \\
 &= G \frac{1}{(1 + \tau_{LPS} s)}
 \end{aligned} \tag{5.7}$$

ADS Circuit Design

This time, the tunable gain is designed to cover a much larger range of values. Enough gain is available to get a loop bandwidth of $f_{cr} = 3.42$ MHz in the case that the unknown cross-over frequency from thermal to electronic FM of the slave laser is $f_{FM} = 10$ MHz, instead of $f_{FM} = 1$ MHz. The gain is tunable between a maximum value G_{max} , and zero. This way, if stability allows it, a better loop can be made. And if stability does not allow it, the gain can always be reduced appropriately. Figure 5.18 shows the open-loop bode plot for an OPLL design with $f_{FM} = 10$ MHz, which is the maximal value for typical DFB lasers described in literature. Table 5.4 lists the corresponding OPLL parameters. In this case, the loop delay $\tau_d = 15$ ns has a non-negligible influence on the stability criterium. Keeping interconnections as short as possible is advised.

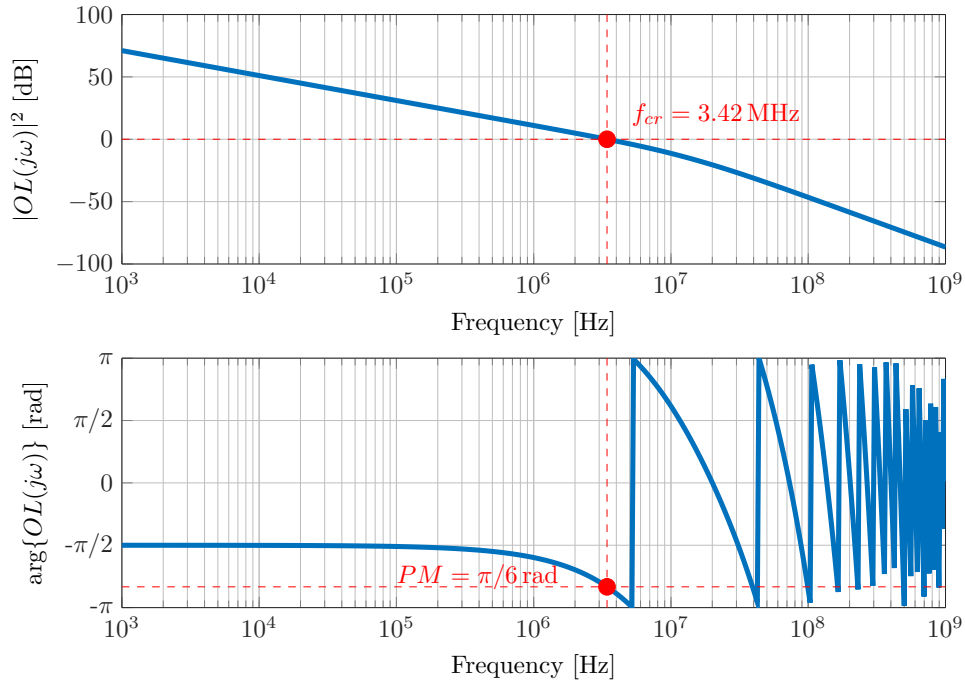


Figure 5.18: Bode plot of system 2 OL(s), $f_{FM} = 10$ MHz

K_p	K_f	f_{FM}	τ_d	f_{LP}	G_{tot}
$0.1735 \frac{\text{V}}{\text{rad}}$	$0.8 \frac{\text{GHz}}{\text{mA}}$	10 MHz	15 ns	20 MHz	2.25×10^7

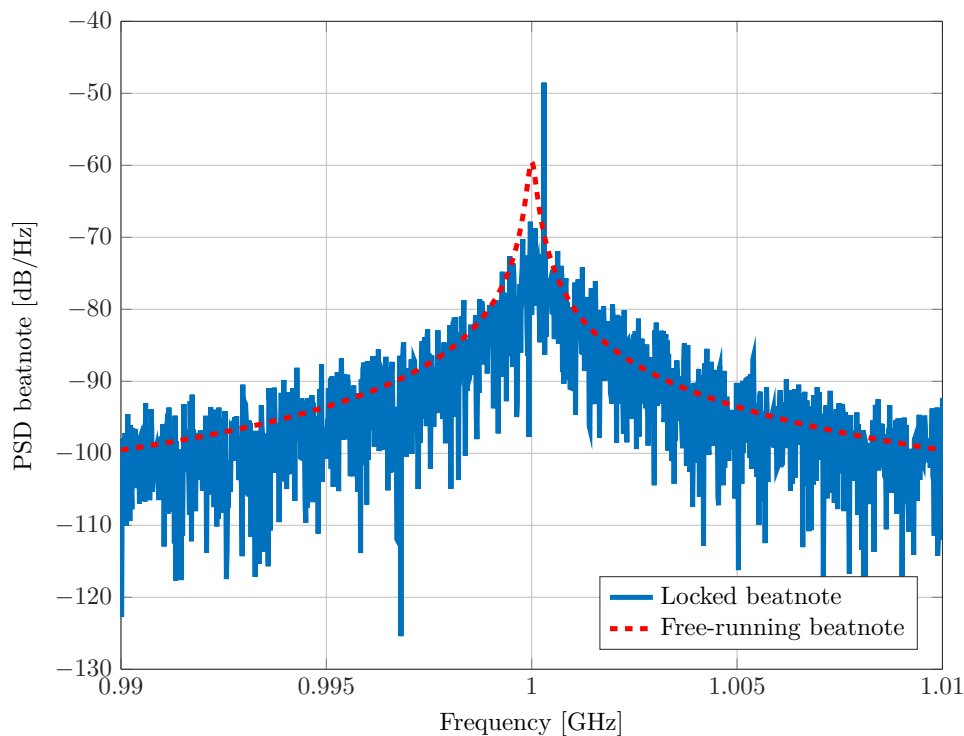
Table 5.4: Loop parameters for system 2

The sizing of the components in the schematic of Figure 5.17 is based on the design parameters of Table 5.4, and on the ADS simulations to create a stable, and linear circuit. The resulting resistor, and capacitor values are listed in Table 5.5.

C_{LP}	R_A	R_B	R_{pot}	R_E
330 pF	100 Ω	560 Ω	1 k Ω	25 Ω

Table 5.5: Resistor and capacitor values of system 2

The tunable gain can take values in the interval $[0; 0.0056]$, which corresponds to a $G_{tot} \in [0; 7.75 \times 10^8]$. The maximal possible gain is even higher than the value calculated in Table 5.4, this is useful in the case that K_f is lower than anticipated. The SIMULINK OPLL testbench of Figure 4.34 is run with the parameters of Table 5.4, the resulting PSD of the beatnote in lock is plotted in Figure 5.19. The improved lineshape of the simulated beatnote PSD confirms that a better performance can be achieved if the slave laser FM cross-over frequency would indeed be as high as $f_{FM} = 10$ MHz. System 2 could make phase locking more easy, provided that loop stability allows it.

Figure 5.19: System 2 OPLL simulation result, $f_{FM} = 10$ MHz

The circuit is simulated in ADS, this time the bias current functionality of the output stage is not considered because no slave laser is available without a dedicated bias input port. The schematic is visible in Figure 5.20. Figures 5.21, and 5.22 plot the results from the AC, and transient simulation respectively. The stability of each op-amp feedback system is ok.

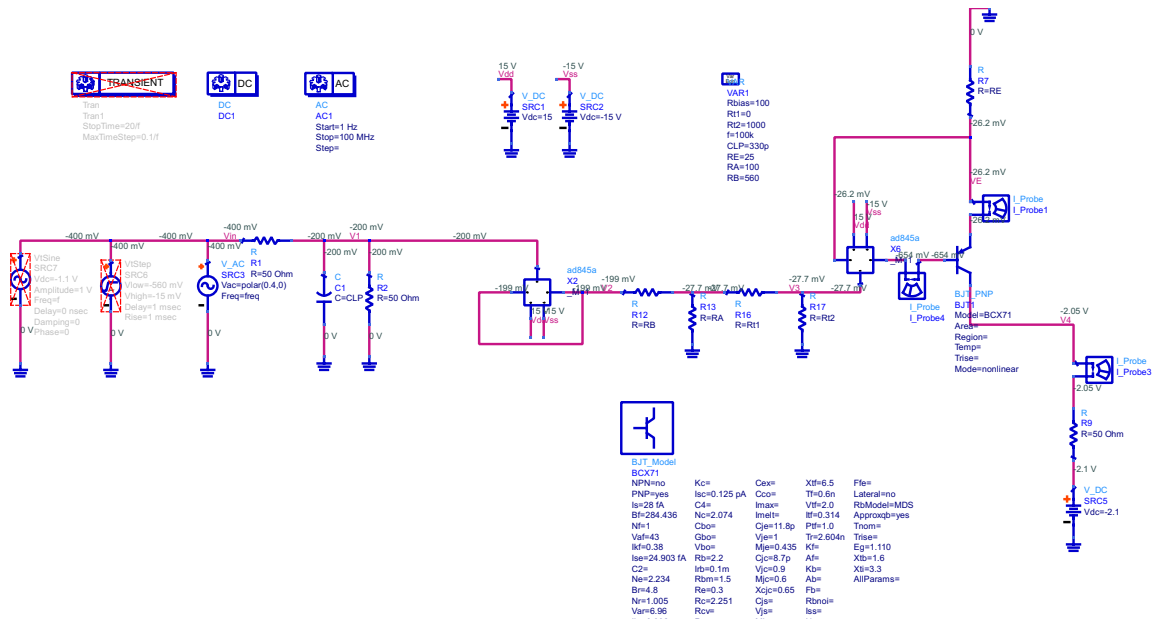


Figure 5.20: ADS system 2 schematic

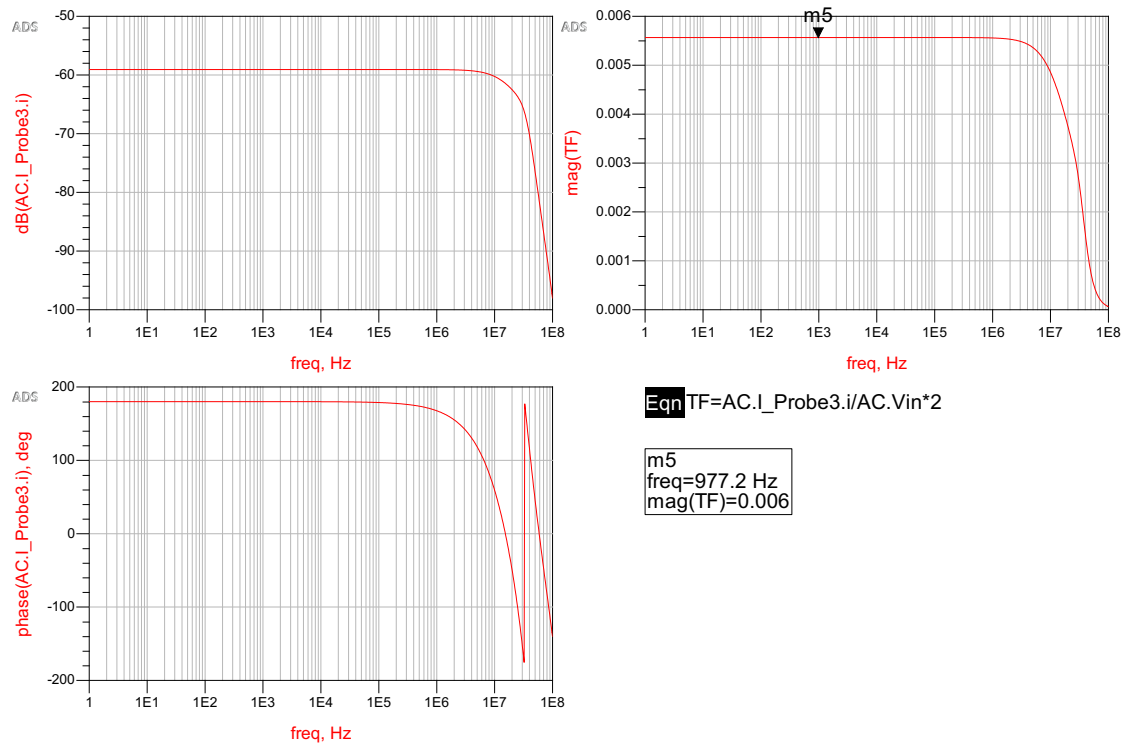


Figure 5.21: ADS system 2 AC simulation result

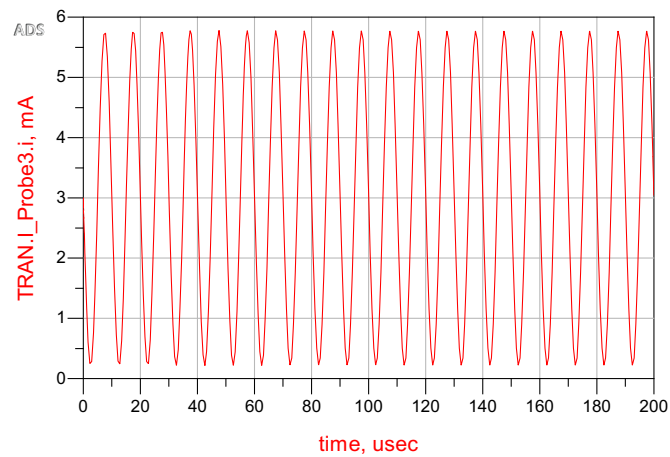
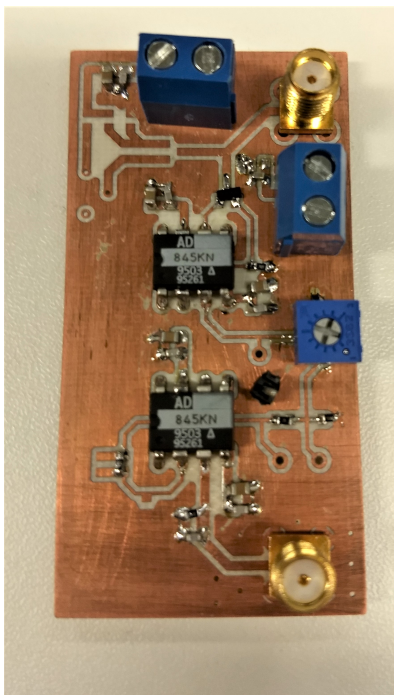


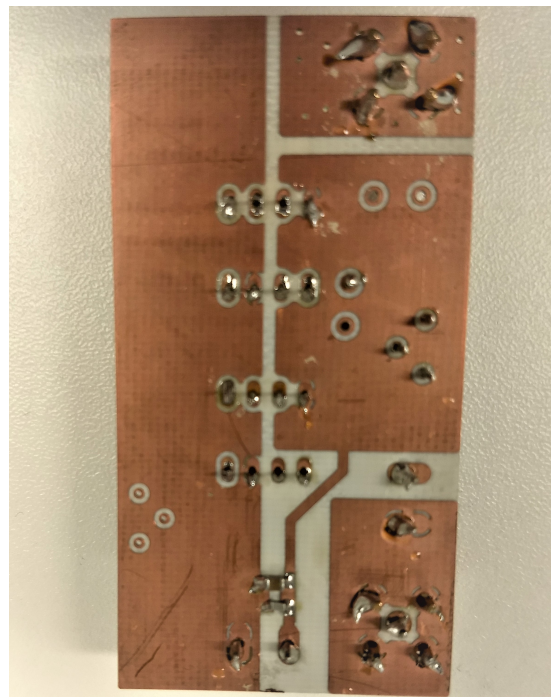
Figure 5.22: ADS system 2 transient simulation result

Printed Circuit Board

The PCB lay-out for system 1 was fabricated twice. The second board can be used for the circuit of system 2 by not soldering some components. Pictures of the finished PCB are visible in Figure 5.23.



(a) PCB top side



(b) PCB bottom side

Figure 5.23: Printed circuit board of system 2

PCB Measurements

The correct behavior of the electronic part of the loop is tested with the measurement setup of Figure 5.15. Figure 5.24 confirms the correct behavior of the XOR-gate, low-pass filter, and tunable gain. The maximal gain between the XOR-gate output voltage $V_{XOR} \in [-560 \text{ mV}; -15 \text{ mV}]$, and the output current of the PCB $I_L \in [-1.5 \text{ mA}; 1.5 \text{ mA}]$ is given by $G_{max} = \frac{0.003}{0.545} = 0.0055$, which is a very good match with the designed for maximal gain $G_{max} = 0.0056$.

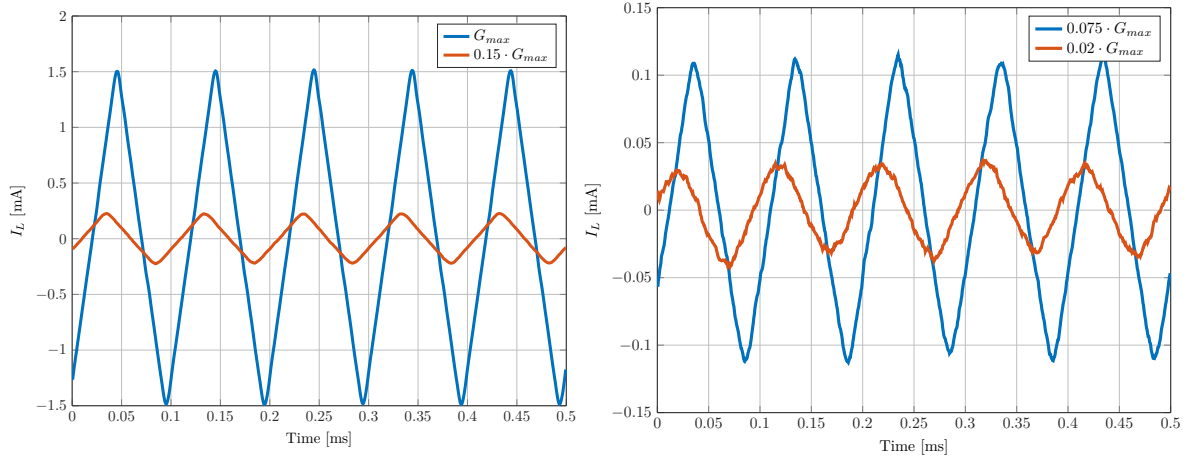


Figure 5.24: $|f_{RF1} - f_{RF2}| = 10 \text{ kHz}$, 4 different gain values

5.3 Complete Setup

All the components of the OPLL are now combined into a complete system. The schematic representation of the setup is visible in Figure 5.25. The loop bandwidth of the designed system 1 is below 1 MHz, the exact value depends on the tunable gain. This low bandwidth will severely limit the capture range of the loop. System 2 potentially has a loop bandwidth up to almost 5 MHz depending on the tunable gain, and loop stability.

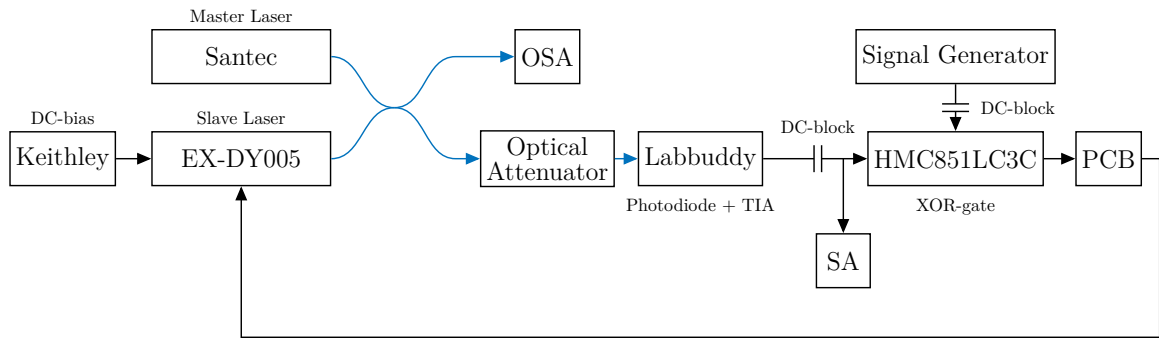


Figure 5.25: Complete OPLL system setup block diagram

First, the tunable master laser frequency is set to overlap with the slave laser with the help of the optical spectrum analyzer (OSA). Secondly, the electrical spectrum analyzer (SA) behind the Labbuddy is used to look at the spectrum of the beatnote. The Santec master laser is tuned further, until the beatnote frequency is well below 6 GHz which is the maximum frequency of the signal generator. Thirdly, the signal generator is set at the measured free-running beatnote frequency. The resulting offset frequency between the signal generator and the beatnote should be within the capture range of the loop, which depends on the small loop bandwidth. Finally, the XOR-gate, and PCB are connected to close the loop.

Changing the RF-offset frequency towards the free-running beatnote can be done with a much finer resolution than tuning the beatnote frequency with the Santec. The goal of this setup is merely to achieve phase lock and verify the design methodology based on the SIMULINK simulation tool of Chapter 4. When phase lock is achieved, the beatnote frequency visible on the SA should follow the RF frequency of the signal generator along the lock range of the loop. Also a finer spectral line is expected as the OPLL in lock removes reduces the phase noise of the beatnote.

The measurement of the complete setup will be discussed in Chapter 6. Pictures of the complete setup are visible in Figure 5.26.



Figure 5.26: Complete OPLL system measurement setup

6

System Measurements

In Chapter 5, the complete OPLL system was built with discrete components. The measurement setup is shown in the block diagram of Figure 5.25, and the pictures are shown in Figure 5.26. Unfortunately, the OPLL setup was unable to achieve phase lock. The observations, problems, and potential solutions are elaborated in this chapter.

6.1 Observations

The behavior of the loop is monitored by the spectrum analyzer (SA), measuring the beatnote before the XOR-gate, illustrated in Figure 5.25. When phase lock is achieved, the beatnote frequency should be fixed, and equal to the signal generator RF frequency. However, the spectrum on the SA is very variable, jumping around over several hundreds of MHz. The frequency jumps look discrete on the SA, even at very high video bandwidth settings. Either the beatnote between the two lasers makes discrete jumps, or the frequency noise of the beatnote is very pronounced, and much higher than anticipated.

6.2 Problem Investigation

The problem is that the free-running beatnote is not stable enough for the low bandwidth OPLL system 1 designed in Chapter 5. Two courses of action are taken in order to solve this issue. Firstly, the first order loop system 2 is designed with the possibility of a much larger gain. If stability allows it (depending on the cross-over frequency f_{FM} between thermal and electronic FM of the slave laser), the loop bandwidth could be improved up to several MHz. Secondly, measures are taken in order to improve the spectrum of the beatnote.

6.2.1 System 1 and 2

The design of both the second order, and first order loops is discussed in sections system 1, and system 2 of Chapter 5 respectively. The first design was based on the assumption that the stability limiting factor $f_{FM} = 1$ MHz. However, this FM cross-over frequency is unknown for the EX-DY7005 DFB slave laser, it could be as high as $f_{FM} = 10$ MHz, which would allow a much larger stable loop bandwidth. System 2 was designed to have the possibility of a potentially larger bandwidth. Unfortunately, also the second system was unable to achieve phase lock. Is the possible bandwidth still too low for the amount of frequency noise in the beatnote, or does the system become unstable for larger gain values? These questions could not be answered based on the spectrum of the beatnote. There was no visible difference in the spectrum for all the different gain values, and no difference between system 1, and system 2. The correct behavior of the PCB's was verified by the measurements discussed in Chapter 5 using the measurement setup of Figure 5.15.

6.2.2 Master Laser

In order to locate the cause of the frequency drifting in the beatnote spectrum, different master lasers are tried: the Santec tunable laser, the Photonetics Tunics 1550, the Yenista Optics tunics-T100s-HP, and a second EX-DY7005 DFB laser. From all these options, using the EX-DY7005 DFB laser both as master, and slave laser gave the most noisy and variable spectrum. This finding already indicates that the used slave laser has a lot of frequency noise.

6.2.3 Slave Laser Temperature Control

On the spectrum analyzer, it seemed like the beatnote made discrete frequency jumps. One investigated theory was the following. The slave laser is temperature controlled by a digital thermo electronic cooler (TEC) controller. Maybe, the digital feedback circuit controlling the

laser temperature is responsible for the jumps in frequency. An InP laser, like the EX-DY7005, typically has a thermal frequency dependence of $10 \text{ GHz}/^\circ\text{C}$. If the digital TEC controller adjusts the temperature in steps of 0.01°C , this would result in frequency steps of 100 MHz . In order to eliminate the frequency jumps, the TEC on the laser was controlled with an external current source instead of the digital feedback loop. A constant current is applied to the TEC, the thermal equilibrium will change slowly while the laser is working. However, this thermal frequency drift was very slow, and the frequency jumping remained in the beatnote spectrum, ruling out that the digital TEC control was responsible.

6.2.4 Slave Laser Bias Current

Initially, the slave laser was biased using a digital Keithley 2450 current source. The spectrum of this current source, showed some unwanted low frequency components. The spectrum of the current source could be improved by connecting an external bias-T (with the capacitor side terminated with 50Ω) after the source. However, this had no measurable effect on the beatnote spectrum. The slave laser was also biased with an older analog source, without improvements on the beatnote spectrum.

6.2.5 Beatnote Measurement

The free-running beatnote of the slave laser, and master laser is measured in the time domain according to the setup in Figure 6.1. The setup is equal to the optical part of the OPLL system of Figure 5.25. The measurement is performed for different combinations of lasers, to find out exactly what causes the strange beatnote spectrum in the loop.

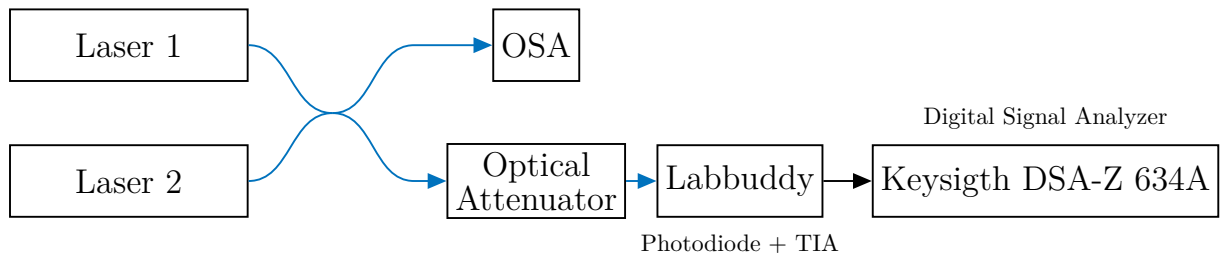


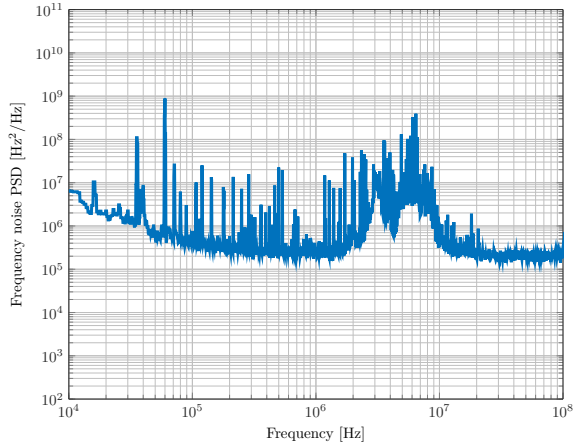
Figure 6.1: Beatnote of two lasers measurement setup

From the time domain traces, it is possible to extract the frequency noise power spectral density. First, the quadrature components are generated for each time trace and divided by each other to eliminate the amplitude information. Secondly, the arctangent is taken from the result to obtain the instantaneous phase information, illustrated by equation (6.1). Thirdly, the carrier

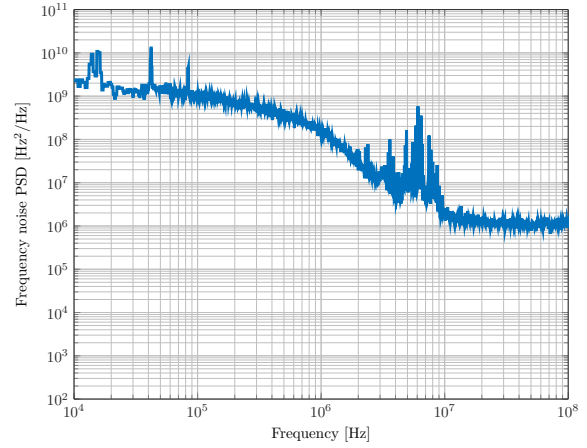
frequency is removed from the instantaneous phase by subtracting the slope of the linear fit on the instantaneous phase. Finally, calculating the power spectral density results in the phase noise PSD, the frequency noise PSD is then equal to $S_f(f) = f^2 S_\phi(f)$. To get a better approximation for the frequency noise PSD, the PSD of multiple hanning windows of the total trace are averaged. A more detailed description of can be found in [21].

$$\arctan\left(\frac{Q(t)}{I(t)}\right) = \arctan\left(\frac{V_0(t)\sin(\theta(t))}{V_0(t)\cos(\theta(t))}\right) = \theta(t) = \omega_{beat}t + \phi(t) \quad (6.1)$$

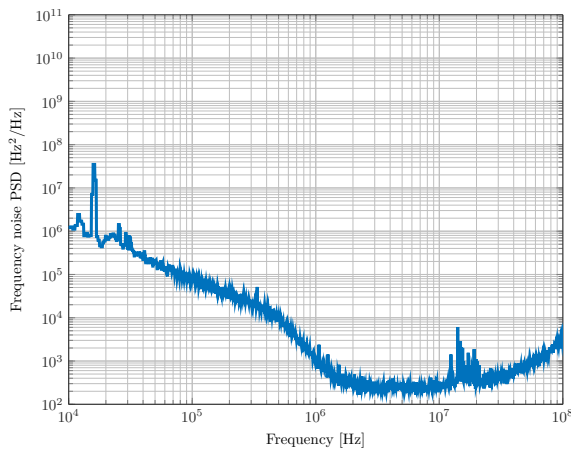
The frequency noise PSD is calculated for the beatnote of multiple laser pairs. The results are shown in Figure 6.2.



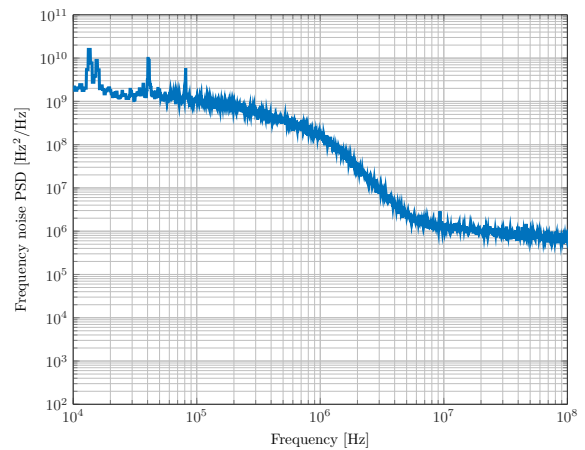
(a) $S_f(f)$ beatnote between Santec and EX-DY7005 DFB lasers



(b) $S_f(f)$ beatnote between Syntune and EX-DY7005 DFB lasers



(c) $S_f(f)$ beatnote between two different Santec lasers



(d) $S_f(f)$ beatnote between Syntune and Santec lasers

Figure 6.2: Calculated frequency noise PSD $S_f(f)$ from different beatnote measurements

By looking at the frequency noise on the free-running beatnote of two Santec tunable lasers in Figure 6.2c, it is clear that the Santec tunable laser has the lowest amount of frequency noise of all the measured lasers. Figures 6.2b, and 6.2d are very similar, the common divisor between the two is the Syntune laser, and is responsible for the large amount of frequency noise. The frequency noise PSD of the beatnote between the EX-DY7005 DFB slave laser and Santec master laser used in the OPLL setup is visible in Figure 6.2a. When comparing with the beatnote between two Santecs, it is clear that the used slave laser has a lot more frequency noise than the Santec master laser.

An estimation of the beatnote linewidth can be made based on the frequency noise PSD. This estimation method is described, and experimentally verified in literature [14, 22]. The method is based on the idea that the frequency noise at lower frequencies contributes to the linewidth, while the contribution from high frequencies, influences the tails of the lineshape. The estimate for the FWHM is given by equation (6.2), where A is the frequency noise power contained in the part of the PSD where $S_f(f) > \frac{8 \ln(2)f}{\pi^2}$. This is described by equation (6.3), where $\mathcal{H}(x)$ is the heaviside function. A detailed description of the method, and a derivation of the formulas can be found in [14].

$$\text{FWHM}_{est} = \sqrt{8 \ln(2) A} \quad (6.2)$$

$$A = \int_{1/T_0}^{\infty} \mathcal{H}(S_f(f) - \frac{8 \ln(2)f}{\pi^2}) S_f(f) df \quad (6.3)$$

The linewidth is estimated for the beatnote measurements of all the laser combinations, the results are tabulated in Table 6.1.

	Santec/Santec	Santec/EX-DY7005	Santec/Syntune	Syntune/EX-DY7005
FWHM_{est}	566 kHz	2.46 MHz	59.1 MHz	62.7 MHz

Table 6.1: Estimated linewidth of laser beatnote measurements

The free-running beatnote used in the OPLL system experiment has a linewidth of $\text{FWHM}_{est} = 2.46$ MHz. System 1 has a designed loop bandwidth of 605.5 kHz, which is a lot smaller than the linewidth of the free-running beatnote. This explains why system 1 is unable to achieve phaselock, within the loop bandwidth, the laser behaves almost like a LED. System 2 could have a bandwidth larger than 2.46 MHz, but only if the FM cross-over frequency f_{FM} is high enough to maintain a stable loop. The FM response of the used slave laser is still unknown, so no conclusion can be drawn for the failure of the first order OPLL system 2. However, the high amounts of frequency noise in the slave laser, will definitely play a role in the problem.

6.3 Proposed solution

The obvious solution for the problems investigated in the previous section, is using a different slave laser that can be frequency modulated with the input current, and with less frequency noise. Unfortunately, there is no substitute slave laser available at the research lab. Another option to consider is designing a new loop employing one of the loop bandwidth enhancement techniques discussed in Chapter 5. This second option was not tried because there was no time left.

7

Conclusion

In this thesis, a careful analysis of optical phase locked loops with single-electrode DFB lasers was performed. Laser characterization measurements for the extraction of the influential parameters like the FM response, and the frequency noise were discussed. A SIMULINK simulation model was created to investigate, and compare the behavior of different loop architectures. The model was used to build a basic OPLL system with a DFB laser. However, unsuccessful due to the large amount of frequency noise in the available DFB laser. The linewidth of the beatnote was measured to be $\text{FWHM} = 2.46 \text{ MHz}$, while the loop bandwidth is limited by the phase reversal in the FM response of the DFB laser.

Semiconductor lasers, like the DFB, are very compact, and allow for compact loop systems. This way propagation delays can be minimized, and bandwidth maximized. However, DFB laser OPLL's pose a bandwidth limitation due to their current-frequency modulation response. Simple first order, and second order loops have been investigated, and optimized for maximal performance. The second order loop shows better performance in the case of increased frequency noise at lower frequencies (like $\frac{1}{f}$ frequency noise), and has advantages such as increased lock range, and zero remaining phase error when a frequency step is applied. The performance of both architectures was insufficient to phase lock the available DFB laser in an actual test system. Due to the estimated low thermal crossover frequency, only lasers with very narrow linewidth have the potential to lock. A more complicated double loop architecture was designed and sim-

ulated, showing tremendous improvement over the single loop architectures.

Performant OPLL's with DFB lasers should be possible with double loop architectures. However this increase in loop complexity is in contrast with the primary reason to use DFB lasers, being the simplicity of fabrication, and simple FM with the input current. Instead of working around the phase reversal in the FM response, a better solution is using semiconductor lasers without this stability limiting factor. DBR lasers can be frequency modulated by thermally modulating the cavity length instead of the laser active region. Special design methods can also be employed to fabricate DFB lasers without FM phase reversal [23].

7.1 Future Work

The linewidth of the DFB lasers on the PIC was measured to be around 100 kHz with the delayed-self-heterodyne measurement technique. This result could be verified with the technique discussed in section 6.2.5. Provided that the frequency noise of the lasers on the PIC is indeed very low, the OPLL setup can be tried again with a DFB slave laser on the PIC. Also the loop delays will be reduced considerably in comparison to the setup with discrete components in Chapter 5. Once a working OPLL experiment has been achieved, more advanced architectures (like the double loop architecture) can be explored, and laser structures can be optimized for OPLL applications. Fine linewidths, and a favorable FM response should be the primary specifications.

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